KNOWLEDGE INSTITUTE OF TECHNOLOGY, SALEM

(An Autonomous Institution)

Approved by AICTE, Affiliated to Anna University, Chennai.
Accredited by NBA (CSE, ECE, EEE & MECH), Accredited by NAAC with „A“ Grade
KIOT Campus, Kakapalayam – 637 504. Salem Dt., Tamil Nadu, India.

M.E. / M.Tech. Regulations 2023

M.E. – Power Electronics and Drives

Curriculum and Syllabi

(For the Students Admitted from the Academic Year 2023 – 2024 onwards)

Version: 1 .0	**Date:** 15.12.2025

KNOWLEDGE INSTITUTE OF TECHNOLOGY (AUTONOMOUS), SALEM -637504

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Accredited by NAAC and NBA (B.E.: Mech., ECE, EEE & CSE)

Website: www.kiot.ac.in

M.E. / M.Tech. / M.C.A. REGULATIONS 2023 (R 2023)

CHOICE BASED CREDIT SYSTEM AND OUTCOME BASED EDUCATION

M.E. POWER ELECTRONICS AND DRIVES

VISION OF THE INSTITUTE

- To be a world-class institution to impart value and need based professional education to the aspiring youth and carving them into disciplined world class professional who have the quest for excellence, achievement orientation and social responsibilities.

MISSION OF THE INSTITUTE

A	To promote academic growth by offering state-of-art undergraduate, postgraduate and doctoral programs and to generate new knowledge by engaging in cutting – edge research
B	To nurture talent, Innovation, entrepreneurship, all-round personality and value system among thestudents and to foster competitiveness among students
C	To undertake collaborative projects which offer opportunities for long-term interaction with academia and industry for creating a sustainable world.
D	To pursue global standards of excellence in all our endeavors namely teaching, research,consultancy, continuing education and support functions

VISION OF THE DEPARTMENT

To produce technically competent Electrical and Electronics Engineers having exemplary skills with ethical and social values.

MISSION OF THE DEPARTMENT

M1	To provide state-of-the art facilities in Electrical and Electronics Engineering for improving the learning environment and research activities
M2	To continuously enrich the knowledge and skill of students towards the employment and creation of innovative products for society
M3	To develop ethical, social-valued and entrepreneurship skilled Electrical and Electronics Engineers

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO 1	To prepare the students for successful career in power electronic industry, research and teaching institutions.
PEO 2	To analyze, design and develop the power electronic converter and drive systems.
PEO 3	To develop the ability to analyze the dynamics in power electronic converter, drive systems and design various controllers to meet the performance criteria.

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PROGRAM OUTCOMES (POs)	
PO 1	An ability to independently carry out research/investigation and development work to solve practical problems
PO 2	An ability to write and present a substantial technical report/document.
PO 3	Students should be able to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.
PO 4	Apply knowledge of basic science and engineering in design and testing of power electronic systems and drives.
PO 5	Interact with Industry in a professional and ethical manner to meet the requirements of societal needs and to contribute sustainable development of the society.
PO 6	Implement cost effective and cutting-edge technologies in power electronics and drives system.

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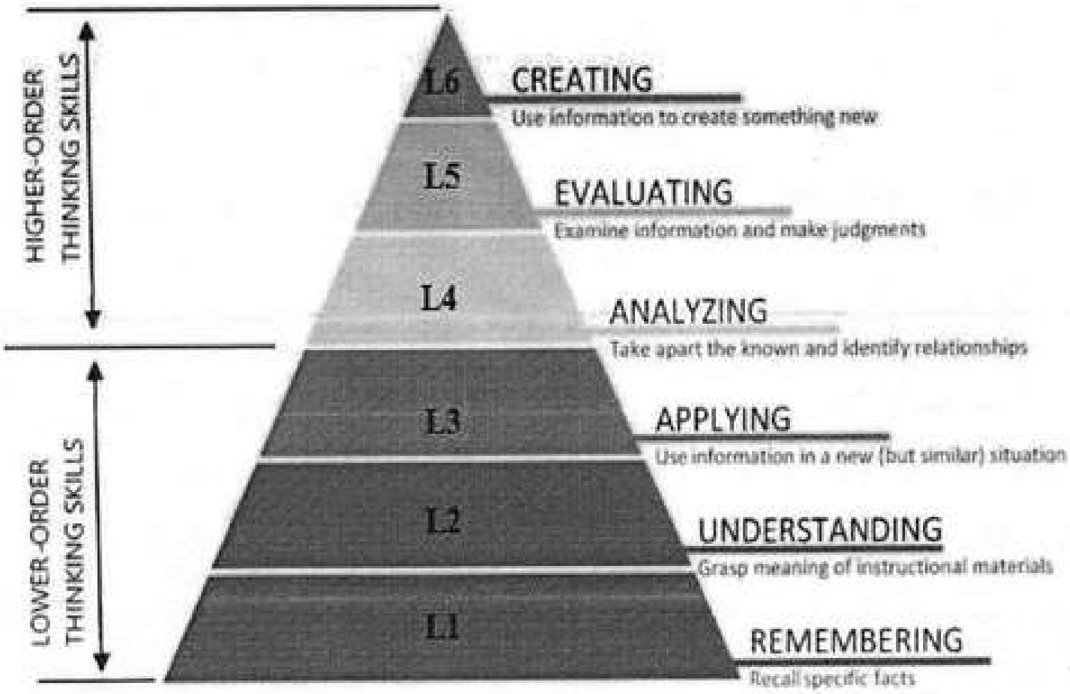
BLOOM'S TAXONOMY LEVELS AND ACTION VERBS

(A) BLOOM'S TAXONOMY LEVELS (BTL):

Bloom's Taxonomy (BT) is based on the belief that learners must first acquire basic foundational knowledge about a subject before progressing to more complex types of thinking, such as analysis and evaluation. Bloom's Taxonomy levels help faculty to guide students through the learning process, from fundamental remembering and understanding to more complex evaluating and creating.



At KIOT, Curriculum Design, Delivery and Assessment (CDDA) are carried out based on the Blooms' Taxonomy Levels (BTL). Its organized set of objectives helps teachers to plan and deliver appropriate instruction, design valid assessment tasks and schemes, and ensure that instruction and assessment are aligned with the objectives.

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(B) BLOOM'S TAXONOMY ACTION VERBS

I. Remembering	II. Understanding	III. Applying	IV. Analyzing	V. Evaluating	VI. Creating
Exhibit memory of previously learned material by recalling facts, terms, basic concepts, and answers.	Demonstrate understanding of facts and ideas by organizing, comparing, interpreting, giving descriptions, and stating main ideas.	Solve problems to new situations by applying acquired knowledge, facts, techniques and rules in a different way.	Examine and break information into parts by identifying motives or causes. Make inferences and find evidence to support generalizations.	Present and defend opinions by making judgments about information, validity of ideas, or quality of work based on a set of criteria.	Compile information together in a different way by combining elements in a new pattern or proposing new Solutions.
• Choose	• Classify	• Apply	• Analyze	• Agree	• Adapt
• Define	• Compare	• Build	• Assume	• Appraise	• Build
• Find	• Contrast	• Choose	• Categorize	• Assess	• Change
• How	• Demonstrate	• Construct	• Classify	• Award	• Choose
• Label	• Explain	• Develop	• Compare	• Choose	• Combine
• List	• Extend	• Experiment with	• Conclusion	• Compare	• Compile
• Match	• Illustrate	• Identify	• Contrast	• Conclude	• Compose
• Name	• Infer	• Interview	• Discover	• Criteria	• Construct
• Omit	• Interpret	• Make use of	• Dissect	• Criticize	• Create
• Recall	• Outline	• Model	• Distinguish	• Decide	• Delete
• Relate	• Relate	• Organize	• Divide	• Deduct	• Design
• Select	• Rephrase	• Plan	• Examine	• Defend	• Develop
• Show	• Show	• Select	• Function	• Determine	• Discuss
• Spell	• Summarize	• Solve	• Inference	• Disprove	• Elaborate
• Tell	• Translate	• Utilize	• Inspect	• Estimate	• Estimate
• What			• List	• Evaluate	• Formulate
• When			• Motive	• Explain	• Happen
• Where			• Relationships	• Importance	• Imagine
• Which			• Simplify	• Influence	• Improve
• Who			• Survey	• Interpret	• Invent
• Why			• Take part in	• Judge	• Make up
			• Test for	• Justify	• Maximize
			• Theme	• Mark	• Minimize
				• Measure	• Modify
				• Opinion	• Original
				• Perceive	• Originate
				• Prioritize	• Plan
				• Prove	• Predict
				• Rate	• Propose
				• Recommend	• Solution
				• Rule on	• Solve
				• Select	• Suppose
				• Support	• Test
				• Value	• Theory

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KNOWLEDGE INSTITUTE OF TECHNOLOGY (AUTONOMOUS), SALEM - 637504											
M.E. POWER ELECTRONICS AND DRIVES									Version: 1.0		
Courses of Study and Scheme of Assessment (Regulations 2023)									Date: 15.12.2025		
Sl. No.	Course Code	Course Title	Periods / Week						Maximum Marks		
			CAT	CP	L	T	P	C	IA	ESE	Total
SEMESTER I											
THEORY											
1	ME23MA106	Applied Mathematics for Power Electronics Engineers	FC	4	3	1	0	4	40	60	100
2	UX23RM201	Research Methodology and IPR	RM	3	2	1	0	3	40	60	100
3	ME23PE301	Analysis of Electrical Machines	PC	4	3	1	0	4	40	60	100
4	ME23PE302	Analysis of Power Converters	PC	4	3	1	0	4	40	60	100
5	ME23PE303	Modeling and Design of SMPS	PC	3	3	0	0	3	40	60	100
PRACTICAL											
6	ME23PE304	Power Converters Laboratory	PC	4	0	0	4	2	60	40	100
7	ME23PE305	Analog and Digital Controllers Laboratory	PC	4	0	0	4	2	60	40	100
EMPLOYABILITY ENHANCEMENT											
8	UX23PT801	Technical Seminar / Case Study Presentation	EEC	2	0	0	2	NC	100	-	100
		Total		**28**	**14**	**4**	**10**	**22**	**420**	**380**	**800**
SEMESTER II											
THEORY											
1	ME23PE306	Special Electrical Machines	PC	3	2	1	0	3	40	60	100
2	ME23PE307	Electric Vehicles and Power Management	PC	4	3	1	0	4	40	60	100
3	ME23PE4XX	Professional Elective - I	PE	3	3	0	0	3	40	60	100
4	ME23PE4XX	Professional Elective - II	PE	3	3	0	0	3	40	60	100
5	ME23XX5XX	Open Elective – I	OE	3	3	0	0	3	40	60	100
6	UX23MC701	Universal Human Values and Ethics	MC	3	1	0	2	NC	100	-	100
7	UX23AC7XX	Audit course	AC	2	2	0	0	NC	100	-	100
PRACTICAL											
8	ME23PE308	Power Electronics and Drives Laboratory	PC	4	0	0	4	2	60	40	100
9	ME23PE309	Design Laboratory for Power Electronics Systems	PC	4	0	0	4	2	60	40	100
EMPLOYABILITY ENHANCEMENT											
10	UX23PT802	Research Paper Review and Presentation	EEC	2	0	0	2	1	100	-	100
		Total		**31**	**17**	**2**	**12**	**21**	**620**	**380**	**1000**

w.e.f. 22/12/2025
Passed in BoS Meeting held on 06/12/2025
Approved in Academic Council Meeting held on 15/12/2025

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M.E. POWER ELECTRONICS AND DRIVES												
Courses of Study and Scheme of Assessment (Regulations 2023)												
Sl. No.	**Course Code**	**Course Title**	**Periods / Week**						**Maximum Marks**			
			CAT	**CP**	**L**	**T**	**P**	**C**	**IA**	**ESE**	**Total**	
SEMESTER III												
THEORY												
1	ME23PE310	Analysis of Electrical Drives	PC	4	3	1	0	4	40	60	100	
2	ME23PE4XX	Professional Elective - III	PE	3	3	0	0	3	40	60	100	
3	ME23PE4XX	Professional Elective - IV	PE	3	3	0	0	3	40	60	100	
4	ME23XX5XX	Open Elective – II	OE	3	3	0	0	3	40	60	100	
PRACTICAL												
5	ME23PE601	Project Work – Phase I	PW	12	0	0	12	6	60	40	100	
Total				**25**	**12**	**1**	**12**	**19**	**220**	**280**	**500**	
SEMESTER IV												
PRACTICAL												
1	ME23PE602	Project Work – Phase II	PW	24	0	0	24	12	60	40	100	
Total				**24**	**0**	**0**	**24**	**12**	**60**	**40**	**100**	
Total Number of Credits: 74												

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PROFESSIONAL ELECTIVE COURSES: VERTICALS

Professional Elective	Vertical 1 Power Converters	Vertical 2 Power Electronics for Power Systems	Vertical 3 Intelligent Systems	Vertical 4 Embedded Systems
Semester 2 (Elective 1&2)	Power Semiconductor Devices	Energy Storage Technologies	Soft Computing Techniques	System Design Using Microcontroller
	Control of Power Electronic Circuits	Wind Energy Conversion System	Machine Learning and Deep Learning	DSP Based System Design
	Modern Rectifiers and Resonant Converters	Power Electronics for Renewable Energy Systems	Optimization Techniques	MEMS Design: Sensors and Actuators
	HVDC and FACTS	-	-	IoT for Smart Systems
Semester 3 (Elective 3&4)	Nonlinear Dynamics for Power Electronics Circuits	Renewable Energy Technology	Python Programming for Machine Learning	Automotive Embedded System
	Advanced Power Converters	Grid Integration of Renewable Energy Sources	AI for Power Electronics	Embedded Control Systems
	High Power Converters for AC Drives	Distributed Generation and Micro Grid	Intelligent Control and Automation	FPGA based Controllers for Power Converters
	-	Power Quality	-	-
	-	Energy Management and Auditing	-	-

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PROFESSIONAL ELECTIVE COURSES: VERTICALS

VERTICAL 1: POWER CONVERTERS

Sl. No.	Course Code	Course Title	Periods / Week						Maximum Marks		
			CAT	CP	L	T	P	C	IA	ESE	Total
1	ME23PE401	Power Semiconductor Devices	PE	3	3	0	0	3	40	60	100
2	ME23PE402	Control of Power Electronic Circuits	PE	3	3	0	0	3	40	60	100
3	ME23PE403	Modern Rectifiers and Resonant Converters	PE	3	3	0	0	3	40	60	100
4	ME23PE404	HVDC and FACTS	PE	3	3	0	0	3	40	60	100
5	ME23PE415	Nonlinear Dynamics for Power Electronics Circuits	PE	3	3	0	0	3	40	60	100
6	ME23PE416	Advanced Power Converters	PE	3	3	0	0	3	40	60	100
7	ME23PE417	High Power Converters for AC Drives	PE	3	3	0	0	3	40	60	100

VERTICAL 2: POWER ELECTRONICS FOR POWER SYSTEMS

Sl. No.	Course Code	Course Title	Periods / Week						Maximum Marks		
			CAT	CP	L	T	P	C	IA	ESE	Total
1	ME23PE405	Energy Storage Technologies	PE	3	3	0	0	3	40	60	100
2	ME23PE406	Wind Energy Conversion System	PE	3	3	0	0	3	40	60	100
3	ME23PE407	Power Electronics for Renewable Energy Systems	PE	3	3	0	0	3	40	60	100
4	ME23PE418	Renewable Energy Technology	PE	3	3	0	0	3	40	60	100
5	ME23PE419	Grid Integration of Renewable Energy Sources	PE	3	3	0	0	3	40	60	100
6	ME23PE420	Distributed Generation and Micro Grid	PE	3	3	0	0	3	40	60	100
7	ME23PE421	Power Quality	PE	3	3	0	0	3	40	60	100
8	ME23PE422	Energy Management and Auditing	PE	3	3	0	0	3	40	60	100

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VERTICAL 3: INTELLIGENT SYSTEMS

Sl. No.	Course Code	Course Title	Periods / Week						Maximum Marks		
			CAT	CP	L	T	P	C	IA	ESE	Total
1	ME23PE408	Soft Computing Techniques	PE	3	3	0	0	3	40	60	100
2	ME23PE409	Machine Learning and Deep Learning	PE	3	3	0	0	3	40	60	100
3	ME23PE410	Optimization Techniques	PE	3	3	0	0	3	40	60	100
4	ME23PE423	Python Programming for Machine Learning	PE	3	3	0	0	3	40	60	100
5	ME23PE424	AI for Power Electronics	PE	3	3	0	0	3	40	60	100
6	ME23PE425	Intelligent Control and Automation	PE	3	3	0	0	3	40	60	100

VERTICAL 4: EMBEDDED SYSTEMS

Sl. No.	Course Code	Course Title	Periods / Week						Maximum Marks		
			CAT	CP	L	T	P	C	IA	ESE	Total
1	ME23PE411	System Design Using Microcontroller	PE	3	3	0	0	3	40	60	100
2	ME23PE412	DSP Based System Design	PE	3	3	0	0	3	40	60	100
3	ME23PE413	MEMS Design: Sensors and Actuators	PE	3	3	0	0	3	40	60	100
4	ME23PE414	IoT for Smart Systems	PE	3	3	0	0	3	40	60	100
5	ME23PE426	Automotive Embedded System	PE	3	3	0	0	3	40	60	100
6	ME23PE427	Embedded Control Systems	PE	3	3	0	0	3	40	60	100
7	ME23PE428	FPGA based Controllers for Power Converters	PE	3	3	0	0	3	40	60	100

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OPEN ELECTIVES

Sl. No.	Course Code	Course Title	Offered by	Periods / Week						Maximum Marks		
				CAT	CP	L	T	P	C	IA	ESE	Total
Except M.E. Industrial Safety Engineering												
1	ME23IS501	Environmental Safety	MECH	OE	3	3	0	0	3	40	60	100
2	ME23IS502	Electrical Safety	MECH	OE	3	3	0	0	3	40	60	100
3	ME23IS503	Safety in Engineering Industry	MECH	OE	3	3	0	0	3	40	60	100
4	ME23IS504	Design of Experiments	MECH	OE	3	3	0	0	3	40	60	100
5	ME23IS505	Circular Economy	MECH	OE	3	3	0	0	3	40	60	100
Except M.E. Automotive Electronics												
6	ME23AE501	Fundamentals of Automotive Electronics	ECE	OE	3	3	0	0	3	40	60	100
7	ME23AE502	Vehicle Sensors and Electronic Control Systems	ECE	OE	3	3	0	0	3	40	60	100
8	ME23AE503	Automotive Embedded Systems	ECE	OE	3	3	0	0	3	40	60	100
9	ME23AE504	Automotive Communication Networks	ECE	OE	3	3	0	0	3	40	60	100
10	ME23AE505	Introduction to Electric and Hybrid Vehicle Electronics	ECE	OE	3	3	0	0	3	40	60	100
Except M.E. Power Electronics and Drives												
11	ME23PE501	Electric Vehicle Technology	EEE	OE	3	3	0	0	3	40	60	100
12	ME23PE502	Energy Conservation and Management	EEE	OE	3	3	0	0	3	40	60	100
13	ME23PE503	Smart Grid	EEE	OE	3	3	0	0	3	40	60	100
14	ME23PE504	Solar PV and Energy Storage Systems	EEE	OE	3	3	0	0	3	40	60	100
Except M.E. Software Engineering												
15	ME23SE501	Blockchain Technologies	CSE	OE	3	3	0	0	3	40	60	100
16	ME23SE502	Deep Learning	CSE	OE	3	3	0	0	3	40	60	100
17	ME23SE503	Full Stack Web Application Development	CSE	OE	3	3	0	0	3	40	60	100
18	ME23SE504	Principles of Multimedia	CSE	OE	3	3	0	0	3	40	60	100
19	ME23SE505	Cyber Physical Systems	CSE	OE	3	3	0	0	3	40	60	100
AUDIT COURSES												
1	UX23AC701	Industrial Safety Engineering	MECH	AC	2	2	0	0	0	100	-	100
2	UX23AC702	Sustainability Engineering	MECH	AC	2	2	0	0	0	100	-	100
3	UX23AC703	Energy Audit	EEE	AC	2	2	0	0	0	100	-	100
4	UX23AC704	Design Thinking	CSE	AC	2	2	0	0	0	100	-	100
5	UX23AC705	Business Analytics	CSE	AC	2	2	0	0	0	100	-	100

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SEMESTER-WISE CREDITS DISTRIBUTION

SUMMARY							
Sl. No.	Course Category	Credits per Semester				Credits	Credit %
		I	II	III	IV		
1	FC	4	-	-	-	4	5
2	RM	3	-	-	-	3	4
3	PC	15	11	4	-	30	41
4	PE	-	6	6	-	12	16
5	OE	-	3	3	-	6	8
6	PW	-	-	6	12	18	24
7	MC/AC	-	✓	-	-	0	0
8	EEC	✓	1	-	-	1	1
	Total	22	21	19	12	74	100

NOMENCLATURE

CAT	Category of Course	FC	Foundation Courses	MC/AC	Mandatory Courses/ Audit Courses
CP	Contact Periods	RM	Research methodology and IPR courses	EEC	Employability Enhancement Courses
L	Lecture Hours	PC	Professional Core Courses	IA	Internal Assessment
T	Tutorial Hours	PE	Professional Elective Courses	ESE	Semester End Examination
P	Practical Hours	OE	Open Elective Courses	PW	Project Work Courses
C	Credits				

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ME23MA106	APPLIED MATHEMATICS FOR POWER ELECTRONICS ENGINEERS	CP	L	T	P	C
		4	3	1	0	4
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:

1.	To develop the ability to apply the concepts of matrix theory in Electrical Engineering problems.
2.	To familiarize the students in the field of differential equations to solve boundary value problems associated with engineering applications.
3.	To develop the ability among the students to solve problems using Laplace transform associated with engineering applications.
4.	To introduce the effective mathematical tools for the solutions of partial differential equations that model several physical processes and to develop Z transform techniques for discrete time systems.
5.	To develop the ability among the students to solve problems using Fourier series associated with engineering applications.

UNIT–I	**MATRIX THEORY**	9+3
	The Cholesky decomposition - Generalized Eigenvectors - Canonical basis -QR factorization – Singular value decomposition - Pseudo inverses - Least square approximation.	
UNIT–II	**CALCULUS OF VARIATIONS**	9+3
	Concept of variations and its properties - Euler's theorem - Functional dependent on first and higher order of derivatives - Functionals dependent on functions of several independent variables - Variational problems with moving boundaries - Isoperimetric problems - Direct methods : Rayleigh Ritz method and Kantorovich problems .	
UNIT–III	**LAPLACE TRANSFORM TECHNIQUES FOR PARTIAL DIFFERENTIAL EQUATIONS**	9+3
	Definitions - Properties - Transform error function - Bessel's function – Dirac Delta function – Unit step function - Convolution theorem – Inverse Laplace transform - Complex inversion formula - Solutions to partial differential equations : Heat and Wave equations	
UNIT – IV	**Z - TRANSFORM TECNIQUES FOR PARTIAL DIFFERENTIAL EQUATIONS**	9+3
	Z-transforms - Elementary properties – Convergence of Z-transforms – Initial and final value theorems - Inverse Z - transform (using partial fraction and residues) - Convolution theorem - Formation of difference equations – Solution of difference equations using Z – transforms.	
UNIT–V	**FOURIER SERIES**	9+3
	Fourier Trigonometric series : Periodic function as power signals – Convergence of series - Even and odd functions : Cosine and sine series – Non periodic function - Extension to other intervals - Power signals : Exponential Fourier series - Parseval's theorem and power spectrum ;	

	Eigenvalue problems and orthogonal functions - Regular Sturm –Liouville systems - Generalized Fourier series.	
	Total(LT) : 60 Periods	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Apply the concepts of matrix theory in Electrical Engineering problems.	L3 – Apply
CO2	Solve boundary value problems associated with engineering applications.	L3 – Apply
CO3	Solve problems using Laplace transform associated with Engineering	L3 – Apply
CO4	Use the effective mathematical tools for the solutions of partial differential equations by using Z transform techniques for discrete time systems.	L3 – Apply
CO5	Solve problems using Fourier series associated with engineering applications.	L3 – Apply
	REFERENCE BOOKS:	
1.	Richard Bronson, MATRIX OPERATION , Schaum's outline series, Second Edition, McGraw Hill, New Delhi , 2011.	
2.	Elsgolc. L.D., " CALCULUS OF VARIATIONS " , Dover Publications Inc., New York, 2007.	
3.	SankaraRao. K , INTRODUCTION TO PARTIAL DIFFERENTIAL EQUATIONS , Prentice Hall of India Pvt . Ltd, New Delhi , 1997.	
4.	Grewal.B.S., "Higher Engineering Mathematics", Khanna Publishers, New Delhi, 44th Edition , 2018.	

Mapping of Cos with Pos and PSOs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3		2	2		1
CO2	3		2	2		1
CO3	3		2	3		1
CO4	3		2	3		1
CO5	3		2	3		1
Avg.	**3.0**		**2.0**	**2.6**		**1.0**

1-Low,2-Medium,3-High

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w.e.f. 13/10/2025
Passed in BoS Meeting held on 18/09/2025
Approved in Academic Council Meeting held on 11/10/2025

UX23RM201	RESEARCH METHODOLOGY AND IPR	CP	L	T	P	C
		3	2	1	0	3
Programme & Branch	**Common to all M.E (SE,AE,PED,ISE) and M.C.A**	**Version: 1.0**				

Course Objectives:

1.	To analyze the significance of research and formulate well-defined research questions.
2.	To apply appropriate research methods and critically evaluate research articles.
3.	To create well-structured research papers and utilize research tools proficiently.
4.	To produce effective technical reports and deliver impactful presentations.
5.	To understand forms of intellectual property and analyze their implications on technological research and international cooperation.

UNIT–I	**CONCEPT OF RESEARCH**	**6+3**
	Meaning and Significance of Research -Skills, Habits and Attitudes for Research -Time Management -Status of Research in India -Why, How, and What a Research is? -Types and Process of Research -Outcome of Research -Sources of Research Problem -Characteristics of a Good Research Problem -Errors in Selecting a Research Problem -Importance of Keywords -Literature Collection - Analysis -Citation Study - Gap Analysis -Problem Formulation Techniques.	
UNIT–II	**RESEARCH METHODS AND JOURNALS**	**6+3**
	Interdisciplinary Research -Need for Experimental Investigations -Data Collection Methods -Appropriate Choice of Algorithms / Methodologies / Methods -Measurement and Result Analysis -Investigation of Solutions for Research Problem -Interpretation -Research Limitations -Journals in Science/Engineering -Indexing and Impact factor of Journals -Citations-h Index - i10 Index -Journal Policies -How to Read a Published Paper -Ethical Issues Related to Publishing- Plagiarism and Self-Plagiarism.	
UNIT–III	**PAPER WRITING AND RESEARCH TOOLS**	**6+3**
	Types of Research Papers - Original Article/Review Paper/Short Communication/Case Study-When and Where to Publish? - Journal Selection Methods -Layout of a Research Paper -Guidelines for Submitting the Research Paper -Review Process - Addressing Reviewer Comments - Use of tools / Techniques for Research -Hands-on Training related to Reference Management Software - EndNote - Introduction to Origin, SPSS, etc. -Software for Detection of Plagiarism.	
UNIT – IV	**EFFECTIVE TECHNICAL THESIS WRITING/PRESENTATION**	**6+3**
	How to Write a Report - Language and Style -Format of Project Report - Use of Quotations -Method of Transcription Special Elements -Title Page - Abstract - Table of Contents - Headings and Sub-Headings -Footnotes - Tables and Figures - Appendix - Bibliography etc. -Different Reference Formats -Presentation using PPTs.	
UNIT – V	**NATURE OF INTELLECTUAL PROPERTY**	**6+3**
	Patents - Designs - Trade and Copyright - Process of Patenting and Development - Technological research - innovation - patenting - Development International Scenario - International Cooperation on Intellectual Property - Procedure for Grants of Patents.	
	Total(LT)	**45 Periods**

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Illustrate the importance and objectives of research in contributing to knowledge and solving real-world problems.	L2 - Understand
CO2	Experiment with data collection techniques, choosing fitting approaches to ensure sound research framework and methodology.	L3 - Apply
CO3	Utilize research & analytic tools for enhancing the research publication	L2 - Understand
CO4	Apply knowledge to produce presentations and technical reports that effectively communicate research findings.	L3 - Apply
CO5	Explain types of intellectual property and comprehend patenting as essential for safeguarding innovation and creativity.	L2 - Understand
	REFERENCE BOOKS:	
1.	Cooper, Donald R.; Schindler, Pamela S.; Sharma, J. K., Business Research Methods, 12th Edition, Tata McGraw Hill Education, 2012.	
2.	DePoy, Elizabeth; Gitlin, Laura N., Introduction to Research: Understanding and Applying Multiple Strategies, 6th Edition, Elsevier Health Sciences, 2015.	
3.	Walliman, Nicholas, Research Methods: The Basics, 3rd Edition, Routledge, 2017.	
4.	Bettig, Ronald V., Copyrighting Culture: The Political Economy of Intellectual Property, 1st Edition, Routledge, 2018.	
5.	The Institute of Company Secretaries of India, Statutory body under an Act of parliament, "Professional Programme Intellectual Property Rights, Law and practice", September 2013.	

Mapping of COs with POs						
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3	3				
CO2	3	3				
CO3	3	3				
CO4	3	3			1	
CO5	3	3			1	
Avg.	**3**	**3**			**1**	
1-Low,2-Medium,3-High						

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ME23PE301	ANALYSIS OF ELECTRICAL MACHINES	CP	L	T	P	C
		4	3	1	0	4
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:		
1.	To understand the principles of electromechanical energy conversion in electrical machines and to know the dynamic characteristics of DC motors	
2.	To study the concepts related with AC machines, magnetic noise and harmonics in rotating electrical machines.	
3.	To interpret the principles of reference frame theory	
4.	To study the principles of three phase, doubly fed and 'n' phase induction machines in machine variables and reference variables.	
5.	To understand the principles of three phase, synchronous machine in machine variables and reference variables.	
UNIT–I	**ELECTROMECHANICAL ENERGY CONVERSION AND DC MACHINES**	**9+3**
	Magnetic circuits, permanent magnet, Energy conservation - stored magnetic energy, co-energy - force and torque in singly and doubly excited systems – Elementary DC machine and analysis of steady state operation - Voltage and torque equations – dynamic characteristics - DC motors – Time domain block diagrams - solution of dynamic characteristic by Laplace transformation	
UNIT–II	**AC MACHINES -CONCEPTS**	**9+3**
	Distributed Windings - Winding Functions - Air-Gap Magnetomotive Force -Rotating MMF - Flux Linkage and Inductance -Resistance - Voltage and Flux Linkage Equations for Distributed Winding Machines-magnetic noise and harmonics in rotating electrical machines, Modeling of 'n' phase machines.	
UNIT–III	**REFERENCE FRAME THEORY**	**9+3**
	Historical background – phase transformation and commutator transformation – transformation of variables from stationary to arbitrary reference frame – transformation of balanced set-variables observed from several frames of reference.	
UNIT – IV	**INDUCTION MACHINES**	**9+3**
	Three phase induction machine and doubly fed induction machine-equivalent circuit and analysis of steady state operation – free acceleration characteristics – voltage and torque equations in machine variables and arbitrary reference frame variables – analysis of dynamic performance for load torque variations- Transformation theory for 'n' phase induction machine.	
UNIT – V	**SYNCHRONOUS MACHINES**	**9+3**
	Three phase synchronous machine and analysis of steady state operation - voltage and torque equations in machine variables and rotor reference frame variables (Park's equations) – analysis of dynamic performance for load torque variations –Krons primitive machine	
	Total(LT)	**60 Periods**

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Understand the principles of electromechanical energy conversion and characteristics of DC motors.	L2 - Understand
CO2	Understand the concepts related with AC machines and modeling of 'n' phase machines.	L2 - Understand
CO3	Apply procedures to develop induction machine models in both machine variable form and reference variable forms.	L3 - Apply
CO4	Apply the procedures to develop synchronous machine models in machine variables form and reference variable form.	L3 - Apply
CO5	Understand the principles of electromechanical energy conversion and characteristics of DC motors.	L2 - Understand
	REFERENCE BOOKS:	
1.	Stephen D. Umans, "Fitzgerald & Kingsley's Electric Machinery", Tata McGraw Hill, 7th Edition, 2020.	
2.	Bogdan M. Wilamowski, J. David Irwin, The Industrial Electronics Handbook, Power Electronics and Motor Drives, CRC Press, 2nd Edition, 2011.	
3.	Paul C. Krause, Oleg Wasynczuk, Scott D. Sudhoff, Steven D. Pekarek, "Analysis of Electric Machinery and Drive Systems", Wiley-IEEE Press, 3rd Edition, 2013.	
4.	R. Krishnan, Electric Motor & Drives: Modeling, Analysis and Control, Pearson Education, 1st Imprint, 2015.	
5.	R.Ramanujam, Modeling and Analysis of Electrical Machines, I.k.International Publishing House Pvt.Ltd, 2018.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3		3	3	1	2
CO2	3		3	3	1	2
CO3	3		3	3	1	2
CO4	3		3	3	1	2
CO5	3		3	3	1	2
Avg.	**3**		**3**	**3**	**1**	**2**

1-Low,2-Medium,3-High

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ME23PE302	ANALYSIS OF POWER CONVERTERS	CP	L	T	P	C
		4	3	1	0	4
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:

1.	To provide the mathematical fundamentals necessary for deep understanding of power converter operating modes.
2.	To introduce the electrical circuit concepts behind the different working modes of power converters so as to enable deep understanding of their operation.
3.	To impart required skills to formulate and design inverters for generic load and for machine loads.
4.	To equip with required skills to derive the criteria for the design of power converters starting from basic fundamentals.
5.	To inculcate knowledge to perform analysis and comprehend the various operating modes of different configurations of power converters

UNIT–I	**SINGLE PHASE AC-DC CONVERTER**	**9+3**
	Static Characteristics of power diode, SCR and GTO, half controlled and fully controlled converters with R-L, R-L-E loads and freewheeling diodes – continuous and discontinuous modes of operation - inverter operation and its limit –Sequence control of converters – performance parameters – effect of source impedance and overlap-reactive power and power balance in converter circuit.	
UNIT–II	**THREE PHASE AC-DC CONVERTER**	**9+3**
	Half controlled and fully controlled converters with R, R-L, R-L-E loads and freewheeling diodes – inverter operation and its limit – performance parameters – effect of source impedance and overlap - 12 pulse converter –Applications - Excitation system, DC drive system.	
UNIT–III	**SINGLE PHASE INVERTERS**	**9+3**
	Introduction to self-commutated switches : MOSFET and IGBT - Principle of operation of half and full bridge inverters – Performance parameters – Voltage control of single phase inverters using various PWM techniques – various harmonic elimination techniques – Design of UPS - VSR operation.	
UNIT – IV	**THREE PHASE INVERTERS**	**9+3**
	180 degree and 120 degree conduction mode inverters with star and delta connected loads – voltage control of three phase inverters: single, multi pulse, sinusoidal, space vector modulation techniques – VSR operation-Application – Induction heating, AC drive system – Current source inverters.	
UNIT – V	**MODERN INVERTERS**	**9+3**
	Multilevel concept – diode clamped – flying capacitor – cascaded type multilevel inverters - Comparison of multilevel inverters - application of multilevel inverters – PWM techniques for MLI – Single phase &Three phase Impedance source inverters – Filters.	
	Total(LT)	**60 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester	

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	Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the operation and characteristics of power semiconductor devices used in converters and inverters.	L2 - Understand
CO2	Analyze single-phase and three-phase AC–DC converters with different loads under various operating modes.	L2 - Analyze
CO3	Design and simulate phase-controlled rectifiers for generic load and machine loads.	L3 - Apply
CO4	Design and simulate switched mode inverters for generic load and for machine loads.	L3 - Apply
CO5	Apply PWM and harmonic reduction techniques to improve the performance of power electronic converters and inverters.	L3 - Apply
	REFERENCE BOOKS:	
1.	Rashid M.H., "Power Electronics Circuits, Devices and Applications ", Pearson, 4th Edition, 10th Impression 2021.	
2.	Jai P. Agrawal, "Power Electronics System Theory and Design", Pearson Education, 1st Edition, 2015	
3.	Philip T. Krein, "Elements of Power Electronics" Indian edition Oxford University Press, 2nd Edition, 2022	
4.	Ned Mohan, T.M.Undeland and W.P.Robbins, "Power Electronics: converters, Application and design", 3rd edition Wiley, 2007.	
5.	Bimal.K.Bose "Modern Power Electronics and AC Drives", Pearson Education, 2nd Edition, 2003	
6.	P.C.Sen, "Modern Power Electronics", S.Chand Publishing, 2005.	

Mapping of COs with POs

COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3		3	3	1	2
CO2	3		3	3	1	2
CO3	3		3	3	1	2
CO4	3		3	3	1	2
CO5	3		3	3	1	2
Avg.	3		3	3	1	2

1-Low,2-Medium,3-High

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ME23PE303	MODELING AND DESIGN OF SMPS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:		
1.	To inculcate knowledge on steady state analysis of Non-Isolated DC-DC converter	
2.	To perform steady state analysis of Isolated DC-DC converter	
3.	To educate on different converter dynamics	
4.	To impart knowledge on the design of controllers for DC-DC converters	
5.	To familiarize the design magnetics for SMPS applications	
UNIT–I	**ANALYSIS OF NON-ISOLATED DC-DC CONVERTERS**	**9**
	Buck, Boost, Buck- Boost and Cuk converters: Principles of operation – Continuous conduction mode– Concepts of volt-sec balance and charge balance – Analysis and design based on steady- state relationships – Introduction to discontinuous conduction mode - SEPIC topology - design examples - Applications to Battery operated vehicle, PV system.	
UNIT–II	**ANALYSIS OF ISOLATED DC-DC CONVERTERS**	**9**
	Introduction - classification- forward- flyback- pushpull – half bridge – full bridge topologies- design of SMPS - Applications to Battery operated vehicle	
UNIT–III	**CONVERTER DYNAMICS**	**9**
	AC equivalent circuit analysis – State space averaging – Circuit averaging – Averaged switch modeling – Transfer function model for buck, boost, buck-boost and cuk converters – Input filters.	
UNIT – IV	**CONTROLLER DESIGN**	**9**
	Review of P, PI, and PID control concepts – gain margin and phase margin – Bode plot based analysis – Design of controller for buck, boost, buck-boost and cuk converters	
UNIT – V	**DESIGN OF MAGNETICS**	**9**
	Basic magnetic theory revision – Inductor design – Design of mutual inductance – Design of transformer for isolated topologies – Ferrite core table and selection of area product – wire table – selection of wire gauge	
	Total(L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**

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CO1	Understand the operation and design principles of non-isolated DC–DC converters.	L2 - Understand
CO2	Understand the operation and basic design of isolated DC–DC converters with different topologies.	L2 - Understand
CO3	Derive transfer function models for DC–DC converters using small-signal analysis.	L3 - Apply
CO4	Apply controller design methods to improve the performance and stability of DC–DC converters.	L3 - Apply
CO5	Apply magnetic design principles to calculate inductors and transformers for SMPS applications.	L3 - Apply
	REFERENCE BOOKS:	
1.	John G. Kassakian, Martin F. Schlecht, George C. Verghese, "Principles of Power Electronics", Pearson, 2010.	
2.	Simon Ang and Alejandra Oliva, "Power-Switching Converters", CRC press, 3rd edition, 2011.	
3.	Philip T. Krein, "Elements of Power Electronics" Indian edition Oxford University Press, 2nd Edition, 2022	
4.	Ned Mohan, "Power Electronics: A first course", Wiley, 1st edition, 2011.	
5.	Issa Batarseh, Ahmad Harb, "Power Electronics- Circuit Analysis and Design, 2nd edition, 2018.	
6.	V.Ramanarayanan, "Course material on Switched mode power conversion", Indian Institute of Science, 2007.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3		3	3	1	2
CO2	3		3	3	1	2
CO3	3		3	3	1	2
CO4	3		3	3	1	2
CO5	3		3	3	1	2
Avg.	**3**		**3**	**3**	**1**	2

1-Low,2-Medium,3-High

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ME23PE304	POWER CONVERTERS LABORATORY	CP	L	T	P	C
		4	0	0	4	2
Programme & Branch	**M.E. – POWER ELECTRONICS AND DRIVES**	**Version: 1.0**				

Course Objectives:		
1.	To provide the basic understanding of the dynamic behavior of the power electronic switches	
2.	To make the students familiar with the digital processors used in generation of gate pulses for the power electronic switches	
3.	To make the students acquire knowledge on the design of power electronic circuits and implementing the same using simulation tools	
4.	To facilitate the students to design gate drive circuits for power converters	
5.	To provide the fundamentals of DC-AC power converter topologies and analyze the harmonics.	
	LIST OF EXPERIMENTS	
1.	Study of switching characteristics of Power MOSFET & IGBT.	
2.	Circuit Simulation of Three-phase semi-converter with R,RL& RLE load.	
3.	Circuit Simulation of Three-phase fully controlled converter with R, RL & RLE load.	
4.	Circuit Simulation of Three-phase Voltage Source Inverter in 180 and 120 degree mode of conduction.	
5.	Circuit simulation of Three-phase PWM inverter and study of spectrum analysis for various modulation indices.	
6.	Simulation of Four quadrant operation of DC Chopper.	
7.	Generation of Gating pulse for converter and inverter.	
8.	Simulation of a single-phase Z-source inverter with R load.	
9.	Simulation of three-phase AC voltage Controller with R load.	
10.	Simulation of a five-level cascaded multilevel inverter with R load.	
11.	Simulation of a Flyback DC-DC converter	
	Total (P)	**60 Periods**
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Understand the switching behaviour of Power Electronic Switches.	L2-Understand
CO2	Simulate AC–DC converters (semi-controlled and fully controlled) under different load conditions.	L3-Apply
CO3	Apply microcontrollers/Arduino to generate gating pulses for DC–DC and inverter circuits.	L3-Apply
CO4	Design and simulate various topologies of inverters and analyze their harmonic spectrum.	L3-Apply
CO5	Design the gate drive power converter circuits and Analyze the three-phase controlled rectifiers and isolated DC-DC converters for designing the power supplies.	L3-Apply

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Mapping of COs with POs						
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	2		3	3	2	2
CO2	2		3	3	2	2
CO3	2		3	3	2	2
CO4	2		3	3	2	2
CO5	2		3	3	2	2
Avg.	**2**		**3**	**3**	**2**	**2**
1-Low,2-Medium,3-High						

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ME23PE305	ANALOG AND DIGITAL CONTROLLERS LABORATORY	CP	L	T	P	C
		4	0	0	4	2
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:

1.	To understand the concepts related with analog and digital controllers.
2.	To design and understand the op-amp circuits and microcontroller circuits for power electronics.
3.	To study and design the driving circuits, sensing circuits, protection circuits for power converters.
4.	To design and select the appropriate digital controller for power converters along with control strategy.
5.	To analyze and implement closed-loop control techniques for power electronic converters using analog and digital platforms.

	LIST OF EXPERIMENTS	
1.	Amplifiers and buffer design and verification by using Opamp	
2.	Filter design and verification by using Opamp	
3.	ON/OFF controller design and verification by using analog circuits	
4.	Design of Driver Circuit using IR2110	
5.	Waveform generation by using look up table	
6.	Generation of PWM gate pulses with duty cycle control using PWM peripheral of microcontroller (TI-C2000 family/ PIC18)	
7.	Duty cycle control from IDE	
8.	Duty Cycle control using a POT connected to ADC peripheral in a standalone mode	
9.	Generation of Sine-PWM pulses for a single and three phase Voltage Source Inverter with control of modulation index using PWM peripheral of microcontroller (TI C2000 family/PIC 18)	
10.	Design and testing of signal conditioning circuit to interface voltage/current sensor with microcontroller (TI-C2000 family/ PIC18)	
11.	Interface Hall effect voltage and current sensor with microcontroller and display the current waveform in the IDE and validate with actual waveform in DSO	
12.	Design of closed loop P, I and PI controllers using OP-AMP	
13.	Design of closed loop P, I and PI controllers using TI-C2000 family/ PIC18	
	Total (P)	**60 Periods**
	Course Outcomes: Upon completion of this course, the students will be able to:	**BLOOM'S Taxonomy**
CO1	Identify suitable analog and digital controllers for various power converter designs	L2-Understand
CO2	Understand the need for gate driver, sensing, and protection circuits in power converters.	L2-Understand
CO3	Apply different control strategies using analog and digital controllers for converter design.	L3-Apply

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CO4	Design and test appropriate driving and protection circuits for reliable converter operation.	L3-Apply
CO5	Design analog and digital controller-based solutions for real-time power electronic applications.	L3-Apply

Mapping of COs with POs						
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	2		3	3	2	2
CO2	2		3	3	2	2
CO3	2		3	3	2	2
CO4	2		3	3	2	2
CO5	2		3	3	2	2
Avg.	**2**		**3**	**3**	**2**	**2**
1-Low,2-Medium,3-High						

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UX23PT801	TECHNICAL SEMINAR / CASE STUDY PRESENTATION	CP	L	T	P	C
		2	0	0	2	NC
Programme & Branch	Common to all M.E (SE,AE,PED,ISE)	Version: 1.0				

Course Objectives:

1	To encourage the students to study advanced engineering developments
2	To prepare and present the technical and case study reports

Method of Evaluation:

The students need to identify an area of interest or topic in their programme of study or case study and prepare a 5-10 page report and a presentation. Based on the report and presentation, the course is evaluated for 100 marks. Minimum 50 marks is essential to pass. In case a student fails, he has to make such presentation in the subsequent semesters. The evaluation guidelines will be issued by the Head of the Department before the commencements of the course. The objectives are improving literature searching capabilities, comprehension and ability to write reports and to make presentations. It is assessed in Internal Assessment mode only and no End Semester Examination.

Total (P) : 30 PERIODS

Course Outcomes: Upon completion of this course the students will be able to:		BLOOM'S Taxonomy
CO1	Perform the review and present technological developments in their field	L3 - Apply
CO2	Interpret the case study report and make a decision	L3 - Apply

Mapping of COs with POs						
CO	PO1	PO2	PO3	PO4	PO5	PO6
1		3				
2		3				
Avg		**3**				
1-Low, 2 -Medium, 3-High.						

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ME23PE306	SPECIAL ELECTRICAL MACHINES	CP	L	T	P	C
		3	2	1	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:	
1.	To understand construction and operation of stepper motors and their control.
2.	To learn principles, control, and applications of switched reluctance motors.
3.	To understand operation and control of permanent magnet brushless DC motors.
4.	To learn principles and control of permanent magnet synchronous motors.
5.	To understand axial flux machines and their applications.

UNIT–I	**STEPPER MOTORS**	**6+3**
	Constructional Features – Principle of Operation – Types – Torque Predictions – Linear and Non-linear Analysis – Characteristics – Drive Circuits – Closed Loop Control –Applications .	
UNIT–II	**SWITCHED RELUCTANCE MOTORS**	**6+3**
	Constructional Features – Principle of Operation - Torque Prediction – Characteristics - Power Controllers - Control of SRM drive - Speed Control - Current Control - Design Procedures - Sensorless Operation of SRM - Current Sensing - Rotor Position Measurement and Estimation Methods - Sensorless Rotor Position Estimation - Inductance based Estimation – Applications .	
UNIT–III	**PERMANENT MAGNET BRUSHLESS DC MOTORS**	**6+3**
	Fundamentals of Permanent Magnets - Types - Principle of Operation - Magnetic Circuit Analysis EMF and Torque Equations - Characteristics - Controller Design - Transfer Function of Machine, Load and Inverter- Current and Speed Controller .	
UNIT – IV	**PERMANENT MAGNET SYNCHROUNOUS MOTORS**	**6+3**
	Permanent Magnet AC Machines - Configurations - PMSM - Principle of Operation – EMF and Torque Equations - Phasor Diagram - Torque Speed Characteristics – Modeling and Small Signal Equations - Evaluation of Control Characteristics - Design of Current and Speed Controllers - Constructional Features - Operating Principle and Characteristics of Synchronous Reluctance Motor .	
UNIT – V	**AXIAL FLUX MACHINES**	**6+3**
	Axial Flux Permanent Magnet Machines - Comparison with Radial Flux Machines - Development – Geometries - Principle of Operation - Torque Production - Applications - Axial flux Switched Reluctance Machine - Topologies and Structures - Operating Principles - Output Equation - Applications .	
	Total(LT)	**45 Periods**

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the construction, operation, and control of stepper motors.	L2 - Understand
CO2	Describe the working principles and control methods of switched reluctance motors.	L2 - Understand
CO3	Apply basic control concepts for permanent magnet brushless DC motors to find transfer function.	L3 - Apply
CO4	Explain the operation and control characteristics of permanent magnet synchronous motors.	L2 - Understand
CO5	Describe axial flux machines and their applications.	L2 - Understand
	REFERENCE BOOKS:	
1.	Jacek F. Gieras, Dr. Rong-Jie Wang, Professor Maarten J. Kamper - Axial Flux Permanent Magnet Brushless Machines-Springer Netherlands, 2008.	
2.	Bilgin, Berker_ Emadi, Ali_ Jiang, James Weisheng - Switched reluctance motor drives: fundamentals to applications-CRC, 2019.	
3.	Ramu Krishnan - Permanent Magnet Synchronous and Brushless DC Motor Drives -CRC Press, Marcel Applications -CRC Press, 2001.	
4.	T.Kenjo, ' Stepping motors and their microprocessor controls', Oxford University press, New Delhi, 2000 Dekker, 2009.	
5.	T.J.E. Miller, 'Brushless magnet and Reluctance motor drives', Clarendon press, London, 1989.	

Mapping of COs with POs						
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3	1	3	2	2	2
CO2	3	1	3	2	2	2
CO3	3	1	3	2	2	2
CO4	3	1	3	2	2	2
CO5	3	1	3	2	2	2
Avg.	3	1	3	2	2	2
1-Low,2-Medium,3-High						

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ME23PE307	ELECTRIC VEHICLES AND POWER MANAGEMENT	CP	L	T	P	C
		4	3	1	0	4
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				
Course Objectives:						
1.	To understand electric and hybrid vehicles and basic vehicle mechanics.					
2.	To learn EV architecture and powertrain components.					
3.	To understand power electronics and motor drives used in electric vehicles.					
4.	To learn battery energy storage systems for electric vehicles.					
5.	To understand alternative energy storage systems for transportation.					
UNIT–I	ELECTRIC VEHICLES AND VEHICLE MECHANICS	9+3				
	Electric Vehicles - Hybrid Electric Vehicles - Engine Ratings - Comparisons of EV with Internal Combustion Engine Vehicles - Fundamentals of Vehicle Mechanics .					
UNIT–II	ARCHITECTURE OF EV's AND POWER TRAIN COMPONENTS	9+3				
	Architecture of EV's and HEV's – Plug-in Hybrid Electric Vehicles - Power Train Components and Sizing - Gears -Clutches - Transmission and Brakes .					
UNIT–III	POWER ELECTRONICS AND MOTOR DRIVES	9+3				
	Electric Drive Components – Power Electronic Switches - Four Quadrant Operation of DC Drives – Induction Motor and Permanent Magnet Synchronous Motor-based Vector Control Operation – Switched Reluctance Motor Drives - EV Motor sizing .					
UNIT – IV	BATTERY ENERGY STORAGE SYSTEM	9+3				
	Battery Basics - Different Types - Battery Parameters - Battery Life & Safety Impacts - Battery Modeling - Design of Battery for Large Vehicles .					
UNIT – V	ALTERNATIVE ENERGY STORAGE SYSTEMS	9+3				
	Introduction to Fuel Cell – Types - Operation and Characteristics - Proton Exchange Membrane - Fuel Cell for E-Mobility – Hydrogen Storage Systems – Super Capacitors for Transportation Applications .					
	Total(LT)	60 Periods				

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain electric vehicle concepts and fundamentals of vehicle mechanics.	L2 - Understand
CO2	Describe EV architecture and powertrain components.	L2 - Understand
CO3	Apply basic power electronics and motor drive concepts in EV systems.	L3 - Apply
CO4	Explain battery characteristics, modeling, and design aspects for EVs.	L2 - Understand
CO5	Describe fuel cells, supercapacitors, and alternative energy storage systems.	L2 - Understand
	REFERENCE BOOKS:	
1.	Iqbal Hussain, "Electric and Hybrid Vehicles: Design Fundamentals, Second Edition" CRC Press, Taylor & Francis Group, Second Edition, 2011.	
2.	Ali Emadi, Mehrdad Ehsani, John M.Miller, "Vehicular Electric Power Systems", Special Indian Edition, Marcel dekker, Inc 2010.	
3.	Mehrdad Ehsani, YiminGao, Sebastian E. Gay, Ali Emadi, 'Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design', CRC Press, 2004.	
4.	C.C. Chan and K.T. Chau, 'Modern Electric Vehicle Technology', OXFORD University Press, 2001.	
5.	Wie Liu, "Hybrid Electric Vehicle System Modeling and Control",Second Edition, John Wiley & Sons, 2017.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3		3	2	3	2
CO2	3		3	2	3	2
CO3	3		3	2	3	2
CO4	3		3	2	3	2
CO5	3		3	2	3	2
Avg.	3		3	2	3	2

1-Low,2-Medium,3-High

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UX23MC701	UNIVERSAL HUMAN VALUES AND ETHICS	CP	L	T	P	C
		3	1	0	2	NC
Programme & Branch	COMMON TO M.E. (SE,AE,PED,ISE), M.C.A & MBA (IEV)	Version: 1.1				

Course Objectives:

1.	To understand the concept of Universal Human Values.
2.	To discuss theoretical and practical implications of UHV.
3.	To relate the use of harmony in the family and society.
4.	To classify the harmony in the nature methods.
5.	To construct effective human values in personal and professional in life.

UNIT–I	**INTRODUCTION TO VALUE EDUCATION**	**3+6**
	Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education) - Understanding Value Education - Sharing about Oneself - Self-exploration as the Process for Value Education - Continuous Happiness and Prosperity – the Basic Human Aspirations - Exploring Human Consciousness - Happiness and Prosperity – Current Scenario - Method to Fulfil the Basic Human Aspirations - Exploring Natural Acceptance . * **Experiential Learning**: Practice sessions to discuss natural acceptance in human being.	
UNIT–II	**HARMONY IN THE HUMAN BEING**	**3+6**
	Understanding Human being as the Co-existence of the Self and the Body - Distinguishing between the Needs of the Self and the Body - Exploring the difference of Needs of Self and Body - The Body as an Instrument of the Self - Understanding Harmony in the Self - Exploring Sources of Imagination in the Self- Harmony of the Self with the Body - Programme to ensure self-regulation and Health - Exploring Harmony of Self with the Body . * **Experiential Learning**: Practice sessions to discuss the awareness of self, help in managing stress and anxiety.	
UNIT–III	**HARMONY IN THE FAMILY AND SOCIETY**	**3+6**
	Harmony in the Family – the Basic Unit of Human Interaction - 'Trust' – the Foundational Value in Relationship - Exploring the Feeling of Trust - 'Respect' – as the Right Evaluation - Exploring the Feeling of Respect - Other Feelings , Justice in Human-to-Human Relationship - Understanding Harmony in the Society - Vision for the Universal Human Order - Exploring Systems to fulfil Human Goal. * **Experiential Learning**: Case Studies on Family and societal issues.	

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UNIT – IV	**HARMONY IN THE NATURE/EXISTENCE**	3+6
	Understanding Harmony in the Nature – Interconnectedness , self-regulation and Mutual Fulfilment among the Four Orders of Nature - Exploring the Four Orders of Nature - Realizing Existence as Co-existence at All Levels - The Holistic Perception of Harmony in Existence - Exploring Co-existence in Existence. * **Experiential Learning:** Case Study to discuss human being as cause of imbalance in nature, pollution, depletion of resources and role of technology.	
UNIT – V	**IMPLICATIONS OF THE HOLISTIC UNDERSTANDING - A LOOK AT PROFESSIONAL ETHICS**	3+6
	Natural Acceptance of Human Values - Definitiveness of (Ethical) Human Conduct - Exploring Ethical Human Conduct - A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order - Competence in Professional Ethics - Exploring Humanistic Models in Education - Holistic Technologies, Production Systems and Management Models -Typical Case Studies - Strategies for Transition towards Value-based Life and Profession - Exploring Steps of Transition towards Universal Human Order. * **Experiential Learning:** Case studies of holistic technologies and Professional ethics.	
	Total (LP) : 45 Periods	
	*** Internal Evaluation only.**	
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the class room teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Interpret the concepts of Universal Human Values.	L2 - Understand
CO2	Summarize both theoretical and practical implications of Universal Human Values.	L2 - Understand
CO3	Build the harmony in family and society.	L3 - Apply
CO4	Practice harmony in all human existence.	L3 - Apply
CO5	Relate human values in both personal and professional life.	L2 - Understand
	REFERENCE BOOKS:	
1.	R.R Gaur, R Sangal, G P Bagaria, A foundation course in Human Values and professional Ethics - Teachers Manual, Excel books, New D...	

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2.	B L Bajpai, 2004, Indian Ethos and Modern Management, New Royal Book Co., Lucknow, Reprinted 2008.	
3.	Frankl, Viktor E. Yes to Life In spite of Everything, Penguin Random House, London, 2019.	
4.	Van Zomeren, M., & Dovidio, J. F. The Oxford Handbook of the Human Essence (Eds.), New York Oxford University Press, 2018.	
5.	R R Gaur, R Asthana, G P Bagaria, A Foundation Course in Human Values and Professional Ethics, 2nd Revised Edition, Excel Books, New Delhi, 2019.	
6.	A.N. Tripathi, Human Values, New Age Intl. Publishers, New Delhi, 2004.	
S.No.	**WEB REFERENCES:**	
1.	https://www.researchgate.net/publication/382495858_Universal_Human_Value	
2.	https://fdp-si.aicte-india.org/UHVII.php	
	VIDEO REFERENCES:	

S.No.	**Publisher**	**Website link**	**Type of Content**
1.	Dr.Rajneesh Arora	https://www.youtube.com/watch?v=SdXIB0k7kIQ&list=PLWDeKF97v9SP7wSlapZcQRrT7OH0ZIGC4&index=2	Video Lecture
2.	Mr.Umesh Jadhav	https://www.youtube.com/watch?v=_g8Hw2APx8Q	Video Lecture

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ME23PE308	POWER ELECTRONICS AND DRIVES LABORATORY	CP	L	T	P	C
		4	0	0	4	2
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:	
1.	To understand closed-loop speed control of DC and BLDC motor drives.
2.	To learn speed control techniques for switched reluctance and stepper motors.
3.	To understand control of induction motor drives using V/F control.
4.	To learn converter-fed motor drive control through simulation.
5.	To understand microcontroller-based motor speed control methods.

	LIST OF EXPERIMENTS	
1.	PID Controller based Closed Loop Speed control of BLDC motor.	
2.	Speed control of SRM motor.	
3.	Speed control of Chopper fed DC motor.	
4.	V/F control of Three-Phase Induction motor.	
5.	Micro controller-based speed control of Stepper motor.	
6.	Simulation of closed loop control of BLDC Motor drive.	
7.	Simulation of Field Oriented Control of Permanent Magnet SRM.	
8.	Simulation of closed loop control of Converter fed DC drive.	
9.	Simulation of VSI fed three phase Induction motor drive.	
	Total (P)	**60 Periods**

	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Apply PID control techniques for closed-loop speed control of motor drives.	L3-Apply
CO2	Apply speed control methods for SRM and stepper motor drives.	L3-Apply
CO3	Apply V/F control technique for three-phase induction motor drives.	L3-Apply
CO4	Apply simulation tools to analyze closed-loop control of converter-fed drives.	L3-Apply
CO5	Apply microcontroller-based methods for motor speed control applications.	L3-Apply

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		3	3	2	2
CO2	2		3	3	2	2
CO3	2		3	3	2	2
CO4	2		3	3	2	2
CO5	2		3	3	2	2
Avg.	**2**		**3**	**3**	**2**	**2**

1-Low,2-Medium,3-High

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ME23PE309	DESIGN LABORATORY FOR POWER ELECTRONICS SYSTEMS	CP	L	T	P	C
		4	0	0	4	2
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:

1.	To understand selection and design of passive and active components for power converters.
2.	To learn the design and testing of isolated power converter circuits.
3.	To learn the design and testing of non-isolated power converter circuits.
4.	To understand performance evaluation and verification of power converter operation.
5.	To develop practical implementation skills through a mini project with applications.

	LIST OF EXPERIMENTS	
1.	Selection and Design of components (Inductor, Capacitor, transformers and devices) for power converters.	
2.	Design and testing of Isolated converter design and verification.	
3.	Design and testing of Non-isolated converter design and verification.	
4.	Mini Project Demonstration with applications.	
	Total (P)	**60 Periods**
	Course Outcomes: Upon completion of this course, the students will be able to:	**BLOOM'S Taxonomy**
CO1	Apply component selection and design methods for power converter circuits.	L3-Apply
CO2	Apply basic design and testing procedures for isolated power converters.	L3-Apply
CO3	Apply basic design and testing procedures for non-isolated power converters.	L3-Apply
CO4	Apply testing methods to verify converter performance and operation.	L3-Apply
CO5	Apply power electronics concepts to demonstrate a mini project application.	L3-Apply

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		3	3	2	2
CO2	2		3	3	2	2
CO3	2		3	3	2	2
CO4	2		3	3	2	2
CO5	2		3	3	2	2
Avg.	**2**		**3**	**3**	**2**	**2**

1–Low,2–Medium,3–High

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UX23PT802	RESEARCH PAPER REVIEW AND PRESENTATION	CP	L	T	P	C
		2	0	0	2	1
Programme & Branch	Common to all M.E (ISE, AE, PED, SE) and M.C.A.	Version : 1.0				

Course Objectives:

1.	To learn scientific paper reading and wiring skills.
2.	To learn the literature review and report wiring skills.
3.	To understand the research gap and formulation of the research problem.

The work involves the following steps:

1 Assigning the faculty supervisor
2 Selecting a subject, narrowing the subject into a topic
3 Stating an objective.
4 Collecting the relevant bibliography (atleast 20 research papers)
5 Studying the papers understanding the authors contributions and critically analyzing each paper.
6 Preparing a 20-25 page literature review report
7 Preparing conclusions based on the literature review report.
8 Writing the Final Review Paper
9 Final Presentation to the review committee

Evaluation method:

A faculty supervisor will be assigned to each student. The supervisor will assign a topic to the student. The student has to review the literature pertaining to the topic, prepare a 20-25 page report and make a presentation. Minimum 20 research papers have to be reviewed out of which 60% have to be in the recent 05 years. The format for the research paper report and guidelines for assessment will be issued by the Head of the Department before the commencement of the course. The evaluation will be carried out based on the research paper report and presentation, and is evaluated for 100 marks. Minimum 50 marks is essential to pass. In case a student fails, he or she has to redo the course in the forthcoming semesters. Assessment is by Internal Assessment mode only no End Semester Examination.

Total : 30 PERIODS

COURSE OUTCOMES: Upon completion of this course the students will be able to:		BLOOM'S Taxonomy
CO1	Write a scientific review paper in their field	L3 - Apply
CO2	Identify the research gap and formulate the research problem	L3 - Apply

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1		3				
CO2		3				
Average		3				

1-Low, 2 -Medium, 3-High.

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ME23PE310	ANALYSIS OF ELECTRICAL DRIVES	CP	L	T	P	C
		4	3	1	0	4
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:	
1.	To understand DC motor fundamentals and mechanical system characteristics.
2.	To learn converter and chopper control techniques for DC motor drives.
3.	To understand closed loop control methods for DC motor drives.
4.	To learn stator controlled induction motor drives using VSI and CSI.
5.	To understand rotor controlled induction motor drive schemes.

UNIT–I	**DC MOTORS FUNDAMENTALS AND MECHANICAL SYSTEMS**	**9+3**
	DC Motor - Types - Induced EMF - Speed-Torque Relations - Speed Control: Armature and Field Speed Control - Ward Leonard Control – Constant Torque and Constant Horse Power Operation - Characteristics of Mechanical System – Dynamic Equations - Components of Torque - Types of Load - Requirements of Drives Characteristics - Stability of Drives – Multi-Quadrant Operation - Drive Elements, Types of Motor Duty and Selection of Motor Rating .	
UNIT–II	**CONVERTER AND CHOPPER CONTROL**	**9+3**
	Principle of Phase Control – Fundamental Relations - Analysis of Series and Separately Excited DC Motor with Single-phase and Three-phase Converters – Performance Parameters and Performance Characteristics - Introduction to Time Ratio Control and Frequency Modulation - Chopper Controlled DC Motor – Performance Analysis - Multi-quadrant Control – Chopper based Implementation of Braking Schemes .	
UNIT–III	**CLOSED LOOP CONTROL**	**9+3**
	Modeling of Drive Elements – Equivalent Circuit, Transfer Function of Self and Separately Excited DC Motors - Linear Transfer Function Model of Power Converters - Sensing and Feeds back Elements – Closed Loop Speed Control – Current and Speed Loops, P, PI and PID Controllers – Response Comparison - Simulation of Converter and Chopper fed DC Drive .	
UNIT – IV	**VSI AND CSI FED STATOR CONTROLLED INDUCTION MOTOR CONTROL**	**9+3**
	AC Voltage Controller – Six Step Inverter Voltage Control -Closed Loop Variable Frequency PWM Inverter Fed Induction Motor (IM) with Braking - CSI fed IM Variable Frequency Motor Drives – Pulse Width Modulation Techniques – Simulation of Closed Loop Operation of Stator Controlled Induction Motor Drives .	

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UNIT – V	ROTOR CONTROLLED INDUCTIONMOTOR DRIVES	9+3
	Static Rotor Resistance Control – Injection of Voltage in the Rotor Circuit – Static Scherbius Drives – Static and Modified Kramer Drives – Sub-Synchronous and Super-Synchronous Speed Operation of Induction Machines – Simulation of Closed Loop Operation of Rotor Controlled Induction Motor Drives .	
	Total(LT)	**60 Periods**

	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain DC motor characteristics, speed control methods, and drive requirements.	L2 - Understand
CO2	Explain converter and chopper control of DC motor drives and their performance.	L2 - Understand
CO3	Apply closed loop control concepts for speed and current control of DC motor drives.	L3 - Apply
CO4	Explain stator controlled induction motor drives using inverter and PWM techniques.	L2 - Understand
CO5	Explain rotor controlled induction motor drives and their speed control methods.	L2 - Understand
	REFERENCE BOOKS:	
1.	Gopal K. Dubey, "Fundamentals of Electrical Drives", 2nd Edition, Narosa Publishing House, 2009.	
2.	Vedam Subramanyam, "Electric Drives: Concepts and Applications", 1st Edition, Tata McGraw-Hill Publishing Company, 2002.	
3.	P. C. Sen, "Thyristor DC Drives", 1st Edition, John Wiley & Sons, 1981.	
4.	Werner Leonhard, "Control of Electrical Drives", 1st Edition, Narosa Publishing House, 1992.	
5.	J. M. D. Murphy, F. G. Turnbull, "Thyristor Control of AC Motors", 1st Edition, Pergamon Press, 1988.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3		3	2	3	2
CO2	3		3	2	3	2
CO3	3		3	2	3	2
CO4	3		3	2	3	2
CO5	3		3	2	3	2
Avg.	3		3	2	3	2

1–Low,2–Medium,3–High

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ME23PE601	PROJECT WORK – PHASE I	CP	L	T	P	C
		12	0	0	12	6
Programme & Branch	**M.E. – POWER ELECTRONICS AND DRIVES**	**Version: 1.0**				

Course Objectives:	
1.	To help students identify real-world engineering problems through literature survey and analysis.
2.	To develop the ability to plan and execute an engineering project independently.
3.	To improve skills in technical report writing and project documentation.
4.	To encourage the application of engineering concepts to practical design or case study problems.
5.	To enhance communication and presentation skills through reviews and viva-voce examination.

COURSE CONTENT		
	Each student will individually work on a project topic related to engineering design or manufacturing applications. The topic may be theoretical or based on a case study and must be selected after a detailed literature survey. The chosen topic should be approved by the Head of the Department and carried out under the guidance of a faculty member with expertise in the selected area. The student must submit a project proposal for approval and clearly define the problem, objectives, and methodology. During the course, three project review meetings will be conducted by the Project Review Committee to evaluate progress and provide guidance and suggestions for improvement. At the end of the semester, the student must submit a detailed Phase-I report, including the problem definition, literature review, and proposed methodology. The final evaluation will include continuous assessment during reviews and a viva-voce examination conducted by a panel of examiners, including one external examiner.	
	Total(P)	**180 Periods**
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Identify and define an engineering problem based on a detailed literature survey.	L2 - Understand
CO2	Analyze existing research and technical information related to the selected project topic.	L4 - Analyze
CO3	Design an appropriate methodology to solve the identified engineering problem.	L3 – Apply
CO4	Develop a technical project report with proper structure and documentation.	L6 - Create
CO5	Present and defend the project work effectively during reviews and viva-voce examination.	L3 - Apply

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Mapping of COs with POs						
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3	3	3	2	3	2
CO2	3	3	3	2	3	2
CO3	3	3	3	2	3	2
CO4	3	3	3	2	3	2
CO5	3	3	3	2	3	2
Avg.	3	3	3	2	3	2
1-Low,2-Medium,3-High						

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ME23PE602	PROJECT WORK – PHASE II	CP	L	T	P	C
		24	0	0	24	12
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:

1.	To implement and complete the approved project work by applying concepts learned.
2.	To develop the ability to implement, test, and validate engineering solutions.
3.	To enhance research skills through technical writing and paper publication.
4.	To improve problem-solving and decision-making skills through continuous project reviews.
5.	To strengthen technical communication skills through report writing and viva-voce examination.

COURSE CONTENT

	This phase is a continuation of the Phase I project and will be carried out on the same topic under the guidance of the same supervisor. The student will continue the work based on the approved methodology and complete the project to the satisfaction of the supervisor and the Project Review Committee. Three project review meetings will be conducted during the semester to monitor progress and provide suggestions for improvement. The student is expected to publish at least one research paper related to the project in a national or international conference. At the end of the course, the student must submit a detailed project report to the Head of the Department. The final evaluation will be based on the project report and a viva-voce examination conducted by a panel of examiners, including one external examiner.	
	Total(P)	**360 Periods**
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Implement the approved project plan and carry out the project work successfully.	L3 - Apply
CO2	Analyze project results and evaluate system performance or outcomes.	L4 - Analyze
CO3	Design and implement improvements based on review committee feedback.	L5 - Evaluate
CO4	Prepare a complete technical report and research paper suitable for publication	L6 - Create
CO5	Present and defend the project work effectively during reviews and viva-voce.	L3 - Apply

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Mapping of COs with POs						
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3	3	3	2	3	2
CO2	3	3	3	2	3	2
CO3	3	3	3	2	3	2
CO4	3	3	3	2	3	2
CO5	3	3	3	2	3	2
Avg.	3	3	3	2	3	2
1-Low,2-Medium,3-High						

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ME23PE401	POWER SEMICONDUCTOR DEVICES	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:	
1.	To understand the fundamentals of power switching devices and wide bandgap technologies.
2.	To learn the operating principles and characteristics of current controlled power devices.
3.	To study the working and behavior of voltage controlled power devices.
4.	To understand methods for device selection, gate driving, and protection circuits.
5.	To learn thermal management and cooling techniques for power electronic devices.

UNIT–I	INTRODUCTION	9
	Overview of Power Switching Devices (L1) – Ideal Switch - Power Handling Capability – SOA - Switching Characteristics – Rating, Features and Brief History of Silicon Carbide - Physical Properties of Silicon Carbide Devices – Unipolar and Bipolar Devices - GaN Technology Overview .	
UNIT–II	CURRENT CONTROLLED DEVICES	9
	BJT's – Switching Characteristics - Negative Temperature Coefficient and Second Breakdown - Thyristors–Static and Transient Characteristics - Series and Parallel Operation - Comparison of BJT and Thyristor – Steady State and Dynamic Models of BJT & Thyristor - Applications - GaN Transistor Electrical Characteristics .	
UNIT–III	VOLTAGE CONTROLLED DEVICES	9
	Power MOSFETs and IGBTs – Principle of Voltage Controlled Devices - Static and Switching Characteristics - Steady State and Dynamic Models of MOSFET and IGBTs - Intelligent Power Modules - Study of Modules like APTGT100TL170G, MSCSM70TAM05TPAG - Integrated Gate Commutated Thyristor - Applications .	
UNIT – IV	DEVICE SELECTION , DRIVING AND PROTECTING CIRCUITS	9
	Device Selection Strategy – Switching Losses – EMI due to Switching - Necessity of Isolation, Pulse Transformer and Optocoupler – Gate Drive Integrated Circuit - Study of Driver IC IRS2110/2113 – Gate Drivers for SCR, MOSFET and IGBTs – Over Voltage, Over Current and Gate Protections - Design of Snubbers Circuit .	
UNIT – V	THERMAL PROTECTION	9
	Heat Transfer (L1) – Conduction, Convection and Radiation - Guidance for Hear Sink Selection – Thermal Resistance and Impedance – Electrical Analogy of Thermal Components - Heat Sink Types and Design – Mounting Types - Switching Loss Calculation for Power Device .	
	Total(L)	45 Periods

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the operation and features of power semiconductor devices.	L2 - Understand
CO2	Describe the characteristics and applications of current controlled devices.	L2 - Understand
CO3	Explain the working and models of voltage controlled devices.	L2 - Understand
CO4	Apply basic criteria to choose devices and use gate driving and protection circuits.	L3 - Apply
CO5	Calculate device losses and use suitable thermal protection methods.	L3 - Apply
	REFERENCE BOOKS:	
1.	Rashid M.H., " Power Electronics Circuits, Devices and Applications ", Pearson, 4th Edition, 10th Impression 2021.	
2.	Mohan, Undeland and Robins, "Power Electronics: Converters Applications and Design, Media Enhanced 3rd Edition, Wiley, 2007	
3.	Tsunenobu Kimoto and James A. Cooper , Fundamentals of Silicon Carbide Technology: Growth, Characterization, Devices, and Applications, First Edition, 2014 John Wiley & Sons Singapore Pte Ltd.	
4.	Alex Lidow, Johan Strydom, Michael de Rooij, David Reusch, GaN Transistors for efficient power conversion, Second Edition, Wiley, 2015	
5.	Biswanath Paul, "Power Electronics", Universities Press 2019	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		3	2	2	2
CO2	1		2	1	3	3
CO3	1		2	1	3	3
CO4	2		3	2	2	1
CO5	2		3	2	2	1
Avg.	1.6		2.6	1.6	2.4	2

1-Low,2-Medium,3-High

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ME23PE402	CONTROL OF POWER ELECTRONIC CIRCUITS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:	
1.	To understand control concepts used in DC–DC converter systems.
2.	To learn linear feedback and state-space based controller techniques.
3.	To understand control methods for AC–DC converters with power factor correction.
4.	To learn the basic principles of sliding mode control for converters.
5.	To understand flatness-based control concepts applied to power converters.

UNIT–I	**INTRODUCTION**	**9**
	Introduction to control of power electronic circuits (L1) - Overview of control objectives - feedback concepts - and basic control structures in power converters - Classification of control strategies .	
UNIT–II	**CONTROLLER DESIGN FOR BASIC DC–DC CONVERTERS**	**9**
	Linear control concepts for DC–DC converters - Small-signal modeling and linearization - Voltage-mode and peak current-mode control Controller design for converters operating in discontinuous conduction mode - Introduction to state-space control concepts .	
UNIT–III	**CONTROLLER DESIGN FOR BASIC AC-DC CONVERTERS**	**9**
	Introduction - Operating Principle of Single-Phase PFCs - Control of PFCs - Designing the Inner Average-Current-Control Loop - Designing the Outer Voltage-Control Loop - Example of Single-Phase PFC Systems .	
UNIT – IV	**SLIDING MODE CONTROL**	**9**
	Introduction - Variable Structure Systems - Control of Single Switch Regulated Systems - Sliding Surfaces - Equivalent Control and the Ideal Sliding Dynamics - Accessibility of the Sliding Surface - Invariance Conditions for Matched Perturbations - Application to Power Converters .	
UNIT – V	**FLATNESS BASED CONTROL**	**9**
	Flatness - Use of the Differential Flatness Property - Controller Development using Flatness - Application to Power Converters .	
	Total(L)	**45 Periods**

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	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain controller structures used in DC-DC converters.	L2 - Understand
CO2	Apply basic linear feedback concepts to controller models.	L3 - Apply
CO3	Explain control loops used in AC-DC PFC converter systems.	L2 - Understand
CO4	Describe the principles and operation of sliding mode control.	L2 - Understand
CO5	Explain flatness-based control and its use in power converters.	L2 - Understand
	REFERENCE BOOKS:	
1.	Farzin Asadi, Kei Eguchi, "Dynamics and Control of DC-DC Converters", 1st Edition, Morgan & Claypool, 2018.	
2.	Andre Kislovski, "Dynamic Analysis of Switching-Mode DC/DC Converters", 1st Edition, Springer, 1991.	
3.	Ahmad Taher Azar, Quanmin Zhu, "Advances and Applications in Sliding Mode Control Systems", 1st Edition, Springer, 2015.	
4.	Jean Levine, "Analysis and Control of Nonlinear Systems: A Flatness-based Approach", 1st Edition, Springer, 2009.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		3	2	2	2
CO2	2		2	2	2	2
CO3	2		3	2	2	2
CO4	3		2	1	3	1
CO5	3		2	1	3	1
Avg.	2.4		2.4	1.6	2.4	1.6

1-Low,2-Medium,3-High

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ME23PE403	MODERN RECTIFIERS AND RESONANT CONVERTERS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:

1.	To understand power system harmonics and line-commutated rectifier operation.
2.	To learn the principles and performance of pulse width modulated rectifiers.
3.	To understand resonant converter concepts and soft switching techniques.
4.	To study dynamic modeling methods for switching converters.
5.	To learn PWM control techniques used in controlled rectifier systems.

UNIT–I	**POWER SYSTEM HARMONICS & LINE COMMUTATED RECTIFIERS**	**9**
	AC waveform Parameters - Power Factor - AC Line Current Harmonic Standards IEC 1000-IEEE 519 - Single-phase Full Wave Rectifier: Continuous Conduction Mode and Discontinuous Conduction Mode – Single-phase Rectifier's behavior for Large Value of Capacitance – Minimizing THD for Small Value of Capacitance – Three-phase Rectifiers: Continuous Conduction Mode and Discontinuous Conduction Mode - Introduction to Harmonic Trap Filters .	
UNIT–II	**PULSE WIDTH MODULATED RECTIFIERS**	**9**
	Properties of Ideal Rectifiers - Realization of Non-ideal Rectifier - Single Phase Converter System Incorporating Ideal Rectifiers - Modeling Losses and Efficiency in CCM – High Quality Rectifiers - Boost Rectifier - Expression for Controller Duty Cycle - Expression for DC Load Current - Solution for Converter Efficiency .	
UNIT–III	**RESONANT CONVERTERS**	**9**
	Review on Parallel and Series Resonant Switches - Soft Switching - Zero Current Switching – Zero Voltage Switching – Classification of Quasi Resonant Switches - Zero Current and Zero Voltage Switching of Quasi Resonant Buck Converter - Zero Current and Zero Voltage Switching of Quasi Resonant Boost Converter: Steady State Analysis .	
UNIT – IV	**DYNAMIC ANALYSIS OF SWITCHING CONVERTERS**	**9**
	Review of Linear System Analysis - State Space Averaging - Basic State Space Average Model - State Space Averaged Model for Buck Converter, Boost Converter, Buck Boost Converter and CUK Converter .	
UNIT – V	**CONTROL OF PWM RECTIFIERS**	**9**
	Pulse Width Modulation - Voltage Mode PWM Scheme and Current Mode PWM Scheme - Average Current Control -Current Programmed Control - Hysteresis Control - Nonlinear Carrier Control – Design of Controllers: PI Controller, Variable Structure Controller for Source Current Shaping of PWM Rectifiers .	
	Total(L)	**45 Periods**

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain harmonic effects and rectifier operation in power systems.	L2 - Understand
CO2	Explain the working and performance of PWM rectifiers.	L2 - Understand
CO3	Describe resonant converter operation and soft switching methods.	L2 - Understand
CO4	Apply state-space averaging to model basic switching converters.	L3 - Apply
CO5	Apply PWM control schemes for current and voltage control in rectifiers.	L3 - Apply
	REFERENCE BOOKS:	
1.	J. G. Kassakian, M. F. Schlecht, G. C. Verghese, "Principles of Power Electronics", 1st Edition, Pearson Education (India), 2010.	
2.	P. T. Krein, "Elements of Power Electronics", 1st Edition, Oxford University Press, 1998.	
3.	Ned Mohan, "Power Electronics: A First Course", 1st Edition, John Wiley & Sons, 2011.	
4.	I. Batarseh, A. Harb, "Power Electronics: Circuit Analysis and Design", 2nd Edition, Springer, 2018.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		2	2	3	2
CO2	2		2	2	2	2
CO3	3		3	2	2	2
CO4	3		2	2	2	2
CO5	3		2	2	2	2
Avg.	2.6		2.2	2	2	2

1-Low,2-Medium,3-High

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ME23PE404	HVDC AND FACTS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:		
1.	To understand basics of power transmission and the need for FACTS and HVDC systems.	
2.	To learn the operation and applications of thyristor-based FACTS controllers.	
3.	To understand LCC HVDC converter operation and control methods.	
4.	To study the working principles of VSI-based FACTS controllers.	
5.	To understand VSI-based HVDC systems and their control strategies.	
UNIT–I	**INTRODUCTION**	**9**
	Basics of Power Transmission Networks (L1) - Control of Power Flow in AC Lines - Uncompensated Line Analysis - Need and Types of FACTS Controllers - Need for HVDC System - LCC and VSC HVDC Basics - Monopolar, Bipolar, and Multi-terminal HVDC Configurations .	
UNIT–II	**THYRISTOR BASED FACTS CONTROLLERS**	**9**
	SVC: Configuration, Voltage Regulation, Modeling, Stability, Applications - TCSC: Operation, Modeling, Power Flow, and Stability Studies .	
UNIT–III	**ANALYSIS OF LCC HVDC CONVERTERS AND HVDC SYSTEM CONTROL**	**9**
	Converter Configurations - Graetz Circuit, Twelve-Pulse Converter - DC Link Control: Firing Angle, Current and Extinction Angle - Harmonic Filtering and Power Control - Controllers: POD, Frequency and Sub-synchronous Damping - Steady-state Modeling of LCC HVDC .	
UNIT – IV	**VSI BASED FACTS CONTROLLERS**	**9**
	STATCOM and SSSC: Operation, Modeling, Power Flow Control - UPFC and IPFC: Operation, Modeling, and Damping of Power Oscillations .	
UNIT – V	**VSI BASED HVDC SYSTEM AND CONTROLS**	**9**
	MMC-HVDC: Operating Principle, Dynamic Modeling, Current and Arm Balancing Control - Higher-level Control, Modulation, Submodule Energy Balancing - AC/DC Fault Management and Protection - AC-DC Power Flow Analysis (sequential and Simultaneous Methods) .	
	Total(L)	**45 Periods**

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain power flow control concepts and the role of FACTS and HVDC systems.	L2 - Understand
CO2	Describe the operation and applications of SVC and TCSC controllers.	L2 - Understand
CO3	Explain converter operation and basic control of LCC HVDC systems.	L2 - Understand
CO4	Describe the working and power flow control of VSI-based FACTS devices.	L2 - Understand
CO5	Explain the operation and control features of VSI-based HVDC systems.	L2 - Understand
	REFERENCE BOOKS:	
1.	R. Mohan Mathur, R. K. Varma, "Thyristor-Based FACTS Controllers for Electrical Transmission Systems", 1st Edition, IEEE Press & John Wiley, 2002.	
2.	K. R. Padiyar, "FACTS Controllers in Power Transmission and Distribution", 2nd Edition, New Age International, 2007.	
3.	K. R. Padiyar, "HVDC Power Transmission Systems", 2nd Edition, New Age International, 2010.	
4.	J. Arrillaga, V. K. Sood, "HVDC and FACTS Controllers: Applications of Static Converters in Power Systems", 1st Edition, Kluwer Academic Publishers, 2004.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		3	2	2	2
CO2	2		3	2	2	2
CO3	2		3	2	2	2
CO4	2		3	2	2	2
CO5	2		3	2	2	2
Avg.	2		3	2	2	2

1-Low,2-Medium,3-High

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ME23PE415	NONLINEAR DYNAMICS FOR POWER ELECTRONICS CIRCUITS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:

1.	To understand fundamental concepts of nonlinear dynamics and chaotic systems.
2.	To learn experimental and numerical techniques for investigating nonlinear behavior.
3.	To understand nonlinear phenomena occurring in DC–DC converters and inverters.
4.	To learn nonlinear dynamic behavior in electrical drive systems.
5.	To understand methods for controlling chaos in power electronic systems and drives.

UNIT–I	**BASICS OF NONLINEAR DYNAMICS**	**9**
	Basics of Nonlinear Dynamics: System, State and State Space Model, Vector Field - Modeling of Linear, Nonlinear and Linearized Systems - Attractors, Chaos, Poincare Map - Dynamics of Discrete Time System - Lyapunov Exponent, Bifurcations, Bifurcations of Smooth Map, Bifurcations in piece wise Smooth Maps - Border Crossing and Border Collision Bifurcation .	
UNIT–II	**TECHNIQUES FOR INVESTIGATION OF NONLINEAR PHENOMENA**	**9**
	Techniques for Experimental Investigation - Techniques for Numerical Investigation - Computation of Averages under Chaos - Computations of Spectral Peaks - Computation of the Bifurcation and Analyzing Stability .	
UNIT–III	**NONLINEAR PHENOMENA IN DC-DC CONVERTERS**	**9**
	Border Collision in the Current Mode Controlled Boost Converter - Bifurcation and Chaos in the Voltage Controlled Buck Converter with Latch - Bifurcation and Chaos in the Voltage Controlled Buck Converter without Latch - Bifurcation and chaos in CUK Converter - Nonlinear Phenomenon in the Inverter under Tolerance Band Control .	
UNIT – IV	**NONLINEAR PHENOMENA IN DRIVES**	**9**
	Nonlinear Phenomenon in Current Controlled and Voltage Controlled DC Drives - Nonlinear Phenomenon in PMSM Drives .	
UNIT – V	**CONTROL OF CHAOS**	**9**
	Hysteresis Control - Sliding Mode and Switching Surface Control - OGY Method - Pyragas Method - Time Delay Control - Application of the Techniques to the Power Electronics Circuit and Drives .	
	Total(L)	**45 Periods**

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain basic concepts of nonlinear dynamics, chaos, and bifurcations.	L2 - Understand
CO2	Describe numerical and experimental techniques for analyzing nonlinear systems.	L2 - Understand
CO3	Explain nonlinear phenomena observed in DC-DC converters and inverters.	L2 - Understand
CO4	Describe nonlinear behavior in DC and PMSM drive systems.	L2 - Understand
CO5	Apply basic chaos control methods to power electronic circuits and drives.	L3 - Apply
	REFERENCE BOOKS:	
1.	C. K. Tse, "Complex Behavior of Switching Power Converters", 1st Edition, CRC Press, 2003.	
2.	Alfredo Medio, Mauro Lines, "Nonlinear Dynamics: A Primer", 1st Edition, Cambridge University Press, 2003.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		3	3	-	1
CO2	2	1	3	3	-	1
CO3	2		3	3	-	2
CO4	2		3	3	-	2
CO5	2	1	3	3	-	1
Avg.	2	1	3	3	-	1.4

1-Low,2-Medium,3-High

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ME23PE416	ADVANCED POWER CONVERTERS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:		
1.	To understand the principles and operation of voltage-lift DC–DC converters.	
2.	To learn positive and negative output super-lift Luo converter configurations.	
3.	To understand ultra-lift and multi-quadrant operating Luo converters.	
4.	To learn the working principles of bidirectional dual active bridge DC–DC converters.	
5.	To understand impedance source converter topologies and modulation methods.	
UNIT–I	VOLTAGE-LIFT CONVERTERS	9
	Introduction (L1) - Self-lift and Reverse Self-lift Circuits - Cuk Converter, Luo Converter and SEPIC Converter - Continuous and Discontinuous Conduction Mode - Applications .	
UNIT–II	POSITIVE OUTPUT &NEGATIVE OUTPUT SUPER-LIFT LUO-CONVERTERS	9
	Main Series - Elementary Circuit - Re-Lift Circuit - Triple-Lift Circuit - Higher-Order Lift Circuit - Continuous and Discontinuous Conduction Modes - Applications .	
UNIT–III	ULTRA LIFT CONVERTERS AND MULTIPLE-QUADRANT OPERATING LUO-CONVERTERS	9
	Ultra-Lift Luo Converter - Operation - Continuous and Discontinuous Conduction Modes - Instantaneous Values - Multiple Quadrant Operating Luo Converters - Circuit Explanations - Modes of Operation - Applications	
UNIT – IV	BIDIRECTIONAL DUAL ACTIVE BRIDGE DC–DC CONVERTERS	9
	Classification of Bidirectional DC–DC Converter – Working Principle of Hybrid-Bridge-Based Dual Active Bridge Converter - Performance - Voltage Mode Control - Principle of Dual-Transformer based DAB Converter - Three-Level Bidirectional DC–DC Converter - Applications .	
UNIT – V	IMPEDANCE SOURCE CONVERTER	9
	Voltage-Fed Z-source Inverters – Topologies – Steady State and Dynamic Model - Current Fed Z-source Inverter –Topology – Modification and Operational Principles - Modulation Methods: Sine PWM, SVPWM and Pulse width Amplitude Modulation - Applications	
	Total(L)	45 Periods

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the operation and applications of voltage-lift DC–DC converters.	L2 - Understand
CO2	Describe positive and negative output super-lift Luo converter circuits.	L2 - Understand
CO3	Explain the working and operating modes of ultra-lift and multi-quadrant Luo converters.	L2 - Understand
CO4	Apply basic control concepts to analyze bidirectional dual active bridge DC–DC converters.	L3 - Apply
CO5	Explain impedance source converter operation and modulation techniques.	L2 - Understand
	REFERENCE BOOKS:	
1.	Fang Lin Luo, Hong Ye, "Essential DC/DC Converters", 1st Edition, CRC Press, 2005.	
2.	Fang Lin Luo, Hong Ye, "Power Electronics: Advanced Conversion Technologies", 2nd Edition, CRC Press, 2018.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		2	3	2	1
CO2	2		2	3	2	1
CO3	2		2	3	2	1
CO4	2		2	3	2	1
CO5	2		2	3	2	1
Avg.	**2**		**2**	**3**	**2**	**1**

1-Low,2-Medium,3-High

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ME23PE417	HIGH POWER CONVERTERS FOR AC DRIVES	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:	
1.	To understand the principles of high-power converters for AC drives.
2.	To learn the design and operation of three-phase and multilevel converters.
3.	To analyze control strategies for AC drives using power converters.
4.	To study advanced modulation techniques and PWM strategies.
5.	To understand protection, reliability, and industrial applications of AC drive converters.

UNIT–I	INTRODUCTION TO HIGH POWER AC DRIVE CONVERTERS	9
	Overview of AC drives – Fundamentals of high-power converters – Industrial applications – Features of high-power converters – Challenges in high-power conversion – Power quality issues – Harmonics in AC drives – Cooling methods – EMI and EMC issues – Reliability considerations .	
UNIT–II	THREE-PHASE CONVERTERS	9
	Three-phase diode and controlled converters – Performance parameters – Continuous and discontinuous conduction – Voltage and current waveforms – Effect of source inductance – Power factor and harmonic analysis .	
UNIT–III	PWM AND MODULATION TECHNIQUES	9
	Pulse Width Modulation (PWM) techniques – Sinusoidal PWM – Space Vector PWM – Selective harmonic elimination – Modulation for reduced harmonics .	
UNIT – IV	MULTILEVEL INVERTERS	9
	Neutral-point clamped (NPC) inverters – Cascaded H-bridge inverters – Flying capacitor inverters – Multilevel inverter applications in AC drives – Control strategies and voltage balancing .	
UNIT – V	ADVANCED CONTROL AND APPLICATIONS	9
	Vector control and direct torque control (DTC) of AC drives – Closed-loop control using high-power converters – Drive protection and reliability issues – Industrial applications of high-power AC drives – Case studies of high-power converter implementations .	
	Total(L)	45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	

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	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Understand the applications and challenges of high-power converters used in AC drive systems.	L2 - Understand
CO2	Understand the operation and performance of three-phase converters used in high-power AC drives.	L2 - Understand
CO3	Apply PWM and modulation techniques to control high-power converters for AC drive applications.	L3 - Apply
CO4	Understand multilevel inverter topologies and their characteristics for high-power AC drives.	L2 - Understand
CO5	Apply advanced control strategies and high-power converter techniques to practical AC drive applications.	L3 - Apply
	REFERENCE BOOKS:	
1.	Bimal N. Bose, Modern Power Electronics and AC Drives, Prentice Hall, 2002.	
2.	Ned Mohan, Power Electronics: Converters, Applications and Design, Wiley, 3rd Edition, 2003.	
3.	Bin Wu, High-Power Converters and AC Drives, IEEE Press & Wiley, 2006.	
4.	Muhammad H. Rashid, Power Electronics: Circuits, Devices and Applications, Pearson, 4th Edition, 2014.	
5.	Jose Rodriguez, Jih-Sheng Lai, Fang Zheng Peng, Multilevel Inverters: A Survey of Topologies, Controls and Applications, IEEE Transactions, 2002.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		2	3	2	1
CO2	2		2	3	2	1
CO3	2		2	3	2	1
CO4	2		2	3	2	1
CO5	2		2	3	2	1
Avg.	**2**		**2**	**3**	**2**	**1**

1–Low,2–Medium,3–High

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ME23PE405	**ENERGY STORAGE TECHNOLOGIES**		**CP**	**L**	**T**	**P**	**C**
			3	**3**	**0**	**0**	**3**
Programme & Branch	**M.E. – POWER ELECTRONICS AND DRIVES**		**Version: 1.0**				
Course Objectives:							
1.	To understand the need, types, and applications of energy storage systems.						
2.	To learn the concepts and modeling of thermal energy storage systems.						
3.	To understand battery technologies and electrical energy storage principles.						
4.	To learn the working principles and characteristics of fuel cells.						
5.	To understand alternate energy storage technologies and their applications.						
UNIT–I	**INTRODUCTION**	**9**					
	Necessity of Energy Storage – Types of Energy Storage –Energy Storage Technologies – Applications .						
UNIT–II	**THERMAL STORAGE SYSTEM**	**9**					
	Thermal Storage – Types – Modeling of Thermal Storage Units – Simple Water and Rock-bed Storage System – Pressurized Water Storage System – Modelling of Phase Change Storage System – Simple units, Packed Bed Storage Units - Modelling using Porous Medium Approach .						
UNIT–III	**ELECTRICAL ENERGY STORAGE**	**9**					
	Fundamental Concept of Batteries – Measuring of Battery Performance, Charging and Discharging of a Battery, Storage Density, Energy Density, and Safety Issues - Types of Batteries: Lead Acid, Nickel-Cadmium, Zinc-Manganese dioxide - Mathematical Modelling for Lead Acid Batteries – Flow Batteries .						
UNIT – IV	**FUEL CELL**	**9**					
	Fuel Cell – History of Fuel cell, Principles of Electrochemical Storage – Types: Hydrogen Oxygen Cells, Hydrogen Air Cell, Hydrocarbon Air Cell, Alkaline Fuel Cell - Detailed Analysis – Advantages and Disadvantages – Fuel Cell Thermodynamics .						
UNIT – V	**ALTERNATE ENERGY STORAGE TECHNOLOGIES**	**9**					
	Flywheel, Super capacitors, Principles & Methods – Compressed Air Energy Storage - Concept of Hybrid Storage – Pumped Hydro Storage – Applications .						
	Total(L)	**45 Periods**					

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the importance and applications of energy storage systems.	L2 - Understand
CO2	Describe thermal storage types and basic modeling approaches.	L2 - Understand
CO3	Explain battery operation, types, and performance parameters.	L2 - Understand
CO4	Describe fuel cell operation and basic thermodynamic concepts.	L2 - Understand
CO5	Explain the working principles of alternate energy storage technologies.	L2 - Understand
	REFERENCE BOOKS:	
1.	James Larminie, Andrew Dicks, "Fuel Cell Systems Explained", 3rd Edition, John Wiley & Sons, 2012.	
2.	V. J. Lunardini, "Heat Transfer in Cold Climates", 1st Edition, John Wiley & Sons, 1981.	
3.	Jiujun Zhang, Lei Zhang, Hansan Liu, Andy Sun, Ru-Shi Liu, "Electrochemical Technologies for Energy Storage and Conversion", 1st Edition (Two-Volume Set), John Wiley & Sons, 2012.	
4.	F. W. Schmidt, A. J. Willmott, "Thermal Storage and Regeneration", 1st Edition, Hemisphere Publishing Corporation, 1981.	
5.	Luisa F. Cabeza, "Advances in Thermal Energy Storage Systems: Methods and Applications", 2nd Edition, Woodhead Publishing, 2023.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	1		3	4	5	6
CO2					2	
CO3	2		2		3	
CO4	2		2		3	
CO5	3		3		3	3
Avg.	**2**		**2**	**2**	**2**	**3**

1-Low,2-Medium,3-High

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ME23PE406	WIND ENERGY CONVERSION SYSTEM	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:

1.	To understand the basic concepts and aerodynamics of wind energy conversion systems.
2.	To learn wind turbine types, design aspects, and power regulation methods.
3.	To understand fixed speed wind energy systems and generator selection.
4.	To learn the principles and operation of variable speed wind energy systems.
5.	To understand grid connection requirements and system impact of wind energy.

UNIT–I	**INTRODUCTION**	**9**
	Components of WECS - WECS Schemes - Power obtained from Wind - Simple Momentum Theory - Power Coefficient - Sabinin's theory - Aerodynamics of Wind turbine .	
UNIT–II	**WINDTURBINES**	**9**
	HAWT and VAWT – Power Developed - Thrust-Efficiency - Rotor Selection - Rotor Design Considerations - Tip Speed Ratio - No. of Blades - Blade Profile - Power Regulation - Yaw Control and Pitch Angle Control - Stall Control - Schemes for Maximum Power Extraction .	
UNIT–III	**FIXEDSPEEDSYSTEMS**	**9**
	Generating Systems - Constant Speed Constant Frequency Systems – Choice of Generators - Deciding Factors - Synchronous Generator - Squirrel Cage Induction Generator - Model of Wind Speed - Model Wind Turbine Rotor – Drive Train Model - Generator Model for Steady State and Transient Stability Analysis .	
UNIT – IV	**VARIABLESPEED SYSTEMS**	**9**
	Need of Variable Speed Systems - Power-wind Speed Characteristics - Variable Speed Constant Frequency Systems Synchronous Generator – DFIG - PMSG – Variable Speed Generators Modeling – Variable Speed Variable Frequency Schemes .	
UNIT – V	**GRIDCONNECTED SYSTEMS**	**9**
	Wind Interconnection Requirements - Low-voltage ride through (LVRT) , Ramp Rate Limitations and Supply of Ancillary Services for Frequency and Voltage Control - Current Practices and Industry Trends Wind Interconnection Impact on Steady-State and Dynamic Performance of the Power System including Modeling Issue .	
	Total(L)	**45 Periods**

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the working principles and power extraction of wind energy systems.	L2 - Understand
CO2	Describe wind turbine types, components, and control methods.	L2 - Understand
CO3	Apply basic models to study the performance of fixed speed wind energy systems.	L3 - Apply
CO4	Describe the operation of variable speed wind energy systems.	L2 - Understand
CO5	Explain grid connection requirements and power system impact of wind integration.	L2 - Understand
	REFERENCE BOOKS:	
1.	L. L. Freris, "Wind Energy Conversion Systems", 1st Edition, Prentice Hall, 1990.	
2.	S. N. Bhadra, D. Kastha, S. Banerjee, "Wind Electrical Systems", 1st Edition, Oxford University Press, 2005.	
3.	Ion Boldea, "Variable Speed Generators", 1st Edition, Taylor & Francis Group, 2006.	
4.	E. W. Golding, "The Generation of Electricity by Wind Power", 1st Edition, Redwood Burn Ltd., 1976.	
5.	N. Jenkins, "Wind Energy Technology", 1st Edition, John Wiley & Sons, 1997.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	1		3	2	3	1
CO2	3		3	2	3	1
CO3	3		3	2	3	1
CO4	3		3	2	3	1
CO5	3		3	2	3	1
Avg.	2.6		3	2	3	1

1-Low,2-Medium,3-High

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ME23PE407	**POWER ELECTRONICS FOR RENEWABLE ENERGY SYSTEMS**	**CP**	**L**	**T**	**P**	**C**
		3	**3**	**0**	**0**	**3**
Programme & Branch	**M.E. – POWER ELECTRONICS AND DRIVES**	**Version: 1.0**				

Course Objectives:		
1.	To understand renewable energy sources and their environmental impact.	
2.	To learn electrical machines used in wind energy conversion systems.	
3.	To understand power converters and basic analysis of solar PV systems.	
4.	To learn power converters and operational analysis of wind energy systems.	
5.	To understand hybrid renewable energy systems and their applications.	
UNIT–I	**INTRODUCTION TO RENEWABLE ENERGY SYSTEMS**	**9**
	Classification of Energy Sources (L1) – Importance of Non-conventional Energy Sources – Advantages and Disadvantages – Environmental Aspects of Energy – Qualitative Study of Renewable Energy Resources: Ocean Energy, Biomass Energy, Hydrogen Energy - Solar Photovoltaic (PV), Fuel Cells: Operating Principles and Characteristics - Wind Energy: Nature of Wind, Types, Control Strategy, Operating Area .	
UNIT–II	**ELECTRICAL MACHINES FOR WIND ENERGY CONVERSION SYSTEMS (WECS)**	**9**
	Review of Reference Theory Fundamentals – Construction, Principle of Operation and Analysis: Squirrel Cage Induction Generator (SCIG), Doubly Fed Induction Generator (DFIG) – Permanent Magnet Synchronous Generator (PMSG) .	
UNIT–III	**POWER CONVERTERS AND ANALYSIS OF SOLAR PV SYSTEMS**	**9**
	Power Converters: Line Commutated Converters (inversion-mode) – Boost and Buck-Boost converters - Selection of Inverter, Battery Sizing, Array Sizing - Block Diagram of the Solar PV Systems – Types of Solar PV Systems - Stand-alone PV Systems - Grid Integrated Solar PV Systems – Grid Connection Issues	
UNIT – IV	**POWER CONVERTERS AND ANALYSIS OF WIND SYSTEMS**	**9**
	Power Converters: Three-phase AC Voltage Controllers – AC-DC-AC Converters - Uncontrolled Rectifiers - PWM Inverters, Grid-Interactive Inverters – Matrix Converter . Analysis: Stand-alone Operation of Fixed and Variable Speed WECS - Grid Integrated SCIG and PMSG based WECS .	
UNIT – V	**HYBRID RENEWABLE ENERGY SYSTEMS**	**9**
	Need for Hybrid Systems - Range and Type of Hybrid Systems - Case Studies of Diesel-PV, Wind- PV, Microhydel-PV, Biomass-Diesel Systems – Maximum Power Point Tracking (MPPT) .	
	Total(L)	**45 Periods**

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain different renewable energy sources and their operating principles.	L2 - Understand
CO2	Describe the construction and operation of generators used in wind systems.	L2 - Understand
CO3	Apply basic methods to size and analyze solar PV system components.	L3 - Apply
CO4	Explain converter operation and system behavior of wind energy systems.	L2 - Understand
CO5	Describe hybrid renewable energy systems and MPPT concepts.	L2 - Understand
	REFERENCE BOOKS:	
1.	S. N. Bhadra, D. Kastha, S. Banerjee, "Wind Electrical Systems", 1st Edition, Oxford University Press, 2009.	
2.	M. H. Rashid, "Power Electronics Handbook", 3rd Edition, Academic Press (Elsevier), 2011.	
3.	G. D. Rai, "Non-Conventional Energy Sources", 1st Edition, Khanna Publishers, 2010.	
4.	G. D. Rai, "Solar Energy Utilization", 5th Edition, Khanna Publishers, 2008.	
5.	L. Gray, L. Johnson, "Wind Energy Systems", 1st Edition, Prentice Hall of India, 1995.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	1		3	2	3	2
CO2	2		3	2	2	2
CO3	2		3	2	2	2
CO4	1		3	2	2	2
CO5	1		3	2	2	2
Avg.	1.4		3	2	2.2	2

1-Low,2-Medium,3-High

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ME23PE418	RENEWABLE ENERGY TECHNOLOGY	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:

1.	To understand energy sources, emissions, and the importance of renewable energy.
2.	To learn the principles and characteristics of solar photovoltaic systems.
3.	To understand design aspects of photovoltaic power systems.
4.	To learn wind energy conversion principles and system configurations.
5.	To understand other renewable energy sources and their applications.

UNIT–I	**INTRODUCTION**	**9**
	Classification of Energy Sources (L1) – Co2 Emission - Features of Renewable Energy - Renewable Energy Scenario in India - Environmental Aspects of Electric Energy Conversion: Impacts of Renewable Energy Generation on Environment Per Capital Consumption - CO2 Emission - Importance of Renewable Energy Sources, Potentials – Achievements – Applications .	
UNIT–II	**SOLAR PHOTOVOLTAICS**	**9**
	Solar Energy: Sun and Earth - Basic Characteristics of Solar Radiation - Angle of Sunrays on Solar Collector - Estimating Solar Radiation Empirically - Equivalent Circuit of PV Cell - Photovoltaic Cell - Impact of Temperature and Insolation on I-V Characteristics - Shading Impacts on I-V Characteristics - Bypass Diode - Blocking Diode .	
UNIT–III	**PHOTOVOLTAIC SYSTEM DESIGN**	**9**
	Block Diagram of Solar PV System: Line Commutated Converters - Boost and Buck-Boost Converters - Selection of Inverter - PV Systems Classification - Standalone PV Systems - Grid Tied and Grid Interactive Inverters .	
UNIT – IV	**WIND ENERGY CONVERSION SYSTEMS**	**9**
	Origin of Winds: Global and Local Winds - Aerodynamics of Wind Turbine - Derivation of Betz's Limit - Power Available in Wind - Classification of Wind Turbine: Horizontal Axis Wind Turbine and Vertical Axis Wind Turbine - Aerodynamic Efficiency - Tip Speed Ratio - Solidity-Blade Count - Power Curve of Wind Turbine - Configurations of Wind Energy Conversion Systems: Type A, Type B, Type C and Type D Configurations - Grid Connection Issues - Grid Integrated SCIG and PMSG based WECS .	
UNIT – V	**OTHER RENEWABLE ENERGY SOURCES**	**9**
	Qualitative Study of Different Renewable Energy Resources: Ocean, Biomass, Hydrogen Energy Systems, Fuel Cells, Ocean Thermal Energy Conversion (OTEC), Tidal and Wave Energy, Geothermal Energy Resources .	
	Total(L)	**45 Periods**

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain renewable energy sources, emissions, and environmental impacts.	L2 - Understand
CO2	Describe the operation and characteristics of solar photovoltaic cells.	L2 - Understand
CO3	Apply basic methods to size and design photovoltaic system components.	L3 - Apply
CO4	Explain wind energy conversion principles and grid-connected systems.	L2 - Understand
CO5	Describe the working principles of other renewable energy sources.	L2 - Understand
	REFERENCE BOOKS:	
1.	S. N. Bhadra, D. Kastha, S. Banerjee, "Wind Electrical Systems", 1st Edition, Oxford University Press, 2009.	
2.	G. D. Rai, "Non-Conventional Energy Sources", 1st Edition, Khanna Publishers, 1993.	
3.	G. D. Rai, "Solar Energy Utilization", 1st Edition, Khanna Publishers, 1993.	
4.	B. H. Khan, "Non-Conventional Energy Sources", 2nd Edition, McGraw-Hill Education, 2009.	
5.	Fang Lin Luo, Hong Ye, "Renewable Energy Systems", 1st Edition, Taylor & Francis Group, 2013.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3		2	2	2	1
CO2	3		2	3	3	3
CO3	3		2	3	3	3
CO4	3		2	3	3	2
CO5	3		2	2	2	2
Avg.	3		2	2.6	2.6	2.2

1-Low,2-Medium,3-High

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ME23PE419	GRID INTEGRATION OF RENEWABLE ENERGY SOURCES	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:

1.	To understand concepts and challenges of renewable energy grid integration.
2.	To learn the influence of different generation types on power networks.
3.	To understand the dynamic performance of wind farms connected to power systems.
4.	To learn power system stabilizers and damping techniques for wind-integrated networks.
5.	To understand design and operation of standalone and grid-connected PV systems.

UNIT–I	**INTRODUCTION**	**9**
	Introduction to Renewable Energy Grid Integration (L1) - Concept of Mini/micro Grids and Smart Grids - Different types of Grid Interfaces - Issues Related to Grid Integration of Small and Large Scale of Synchronous Generator, Induction Generator, and Converter based Sources Together - Frequency Management - Influence of WECS on System Transient Response - Interconnection Standards and Grid Code Requirements for Integration .	
UNIT–II	**NETWORK INFLUENCE OF GENERATION TYPE**	**9**
	Starting – Network Voltage Management – Thermal/Active Power Management – Network Power Quality Management – Transient System Performance – Fault Level Issues – Protection .	
UNIT–III	**INFLUENCE OF WIND FARMS ON NETWORK DYNAMIC PERFORMANCE**	**9**
	Dynamic Stability and its Assessment – Dynamic Characteristics of Synchronous Generation - Damping power model of a Synchronous Generator – Influence of Automatic Voltage Regulator on Damping – Influence on Damping of Generator Operating Conditions – Influence of Turbine Governor on Generator Operation – Transient Stability – Voltage Stability – Influence of Generation Type on Network Dynamic Stability – Dynamic Interaction of Wind Farms with the Network – Influence of Wind Generation on Network Transient Performance .	
UNIT – IV	**POWER SYSTEM STABILIZERS AND NETWORK DAMPING CAPABILITY OF WIND**	**9**
	Power System Stabilizer for a Synchronous Generator - Power System Stabilizer for a DFIG - Power System Stabilizer for a FRC Wind Farm .	

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UNIT – V	STAND ALONE AND GRID CONNECTED PV SYSTEM	9
	Solar Modules – Storage Systems – Basics of Batteries – Batteries for PV Systems – Charge Controllers – MPPT and Inverters – Power Conditioning and Regulation – Protection – Types of Solar PV systems - Standalone PV Systems Design – Sizing – PV Systems in Buildings – Design Issues for Central Power Stations – Safety – Economic Aspect – Efficiency and Performance – International PV Programs .	
	Total(L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain grid integration concepts, standards, and power quality issues of renewable sources.	L2 - Understand
CO2	Describe network voltage, power, and protection issues due to generation types.	L2 - Understand
CO3	Explain dynamic stability and transient behavior of wind-integrated power systems.	L2 - Understand
CO4	Apply basic stabilizer concepts to improve damping in power systems with wind generation.	L3 - Apply
CO5	Apply basic methods to design and analyze standalone and grid-connected PV systems.	L3 - Apply
	REFERENCE BOOKS:	
1.	Stuart R. Wenham, Martin A. Green, Muriel E. Watt, Richard Corkish, "Applied Photovoltaics", 2nd Edition, Earthscan, 2007.	
2.	Joshua Earnest, "Wind Power Technology", 2nd Edition, PHI Learning, 2015.	
3.	Olimpo Anaya-Lara, Nick Jenkins, Janaka Ekanayake, Phill Cartwright, Mike Hughes, "Wind Generation: Modelling and Control", 1st Edition, John Wiley & Sons, 2009.	
4.	Brenden Fox, Damian Flynn, Leslie Bryans, "Wind Power Integration: Connection and System Operational Aspects", 1st Edition, Institution of Engineering and Technology, 2007.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3		2	3	3	1
CO2	3		2	3	3	1
CO3	3		2	3	3	1
CO4	3		2	3	3	1
CO5	3		2	3	3	1
Avg.	3		2	3	3	1

1–Low,2–Medium,3–High

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ME23PE420	DISTRIBUTED GENERATION AND MICRO GRID	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:

1.	To understand the concept, benefits, and impacts of distributed generation.
2.	To learn different types of distributed energy resources and their interfaces.
3.	To understand planning and protection issues in systems with distributed generation.
4.	To learn the concept, structure, and operation of micro grids.
5.	To understand technical and economic impacts of micro grids.

UNIT–I	**INTRODUCTION TO DISTRIBUTED GENERATION**	**9**
	DG Definition (L1) - Reasons for Distributed Generation (L1) - Benefits of Integration - Distributed Generation and the distribution System - Technical, Environmental and Economic Impacts of Distributed Generation on the Distribution System and Transmission System	
UNIT–II	**DISTRIBUTED ENERGY RESOURCES**	**9**
	Combined Heat and Power Systems - Wind Energy Conversion Systems - Solar Photovoltaic Systems - Small-scale Hydroelectric Power Generation - Other Renewable Energy Sources - Storage Devices - Inverter Interfaces	
UNIT–III	**DG PLANNING AND PROTECTION**	**9**
	Generation Capacity Adequacy in Conventional Thermal Generation Systems - Impact of Distributed Generation on Network Design - Protection of Distributed Generation - Protection of the Generation Equipment from Internal Faults - Impact of Distributed Generation on Existing Distribution System Protection .	
UNIT – IV	**CONCEPT OF MICROGRID**	**9**
	Microgrid Configuration - Functions of Micro Source Controller and Central Controller - Energy Management Module and Protection Co-ordination Module (PCM) - Modes of Operation - Grid Connected and Islanded Modes - Modelling of Microgrid, Microturbine Model and PV Solar Cell Model - Wind Turbine Model-Role of Microgrid in Power Market Competition .	
UNIT – V	**IMPACTS OF MICROGRID**	**9**
	Technical and Economical Advantages - Challenges and Disadvantages of Microgrid Development - Management and Operational issues - Impact on Heat Utilization, Process Optimization, Communication Standards and Protocols - Microgrids and Traditional Power System Economics - Potential Benefits of Microgrid Economics .	
	Total(L)	**45 Periods**

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the role and impacts of distributed generation in power systems.	L2 - Understand
CO2	Describe distributed energy resources and inverter-based interfaces.	L2 - Understand
CO3	Apply basic planning and protection concepts for distributed generation systems.	L3 - Apply
CO4	Explain microgrid configuration, control functions, and operating modes.	L2 - Understand
CO5	Describe technical and economic impacts of microgrids on power systems.	L2 - Understand
	REFERENCE BOOKS:	
1.	Nick Jenkins, Janaka Ekanayake, Goran Strbac, "Distributed Generation", 1st Edition, Institution of Engineering and Technology, 2010.	
2.	S. Chowdhury, S. P. Chowdhury, P. Crossley, "Microgrids and Active Distribution Networks", 1st Edition, Institution of Engineering and Technology, 2009.	
3.	Math H. J. Bollen, Fainan Hassan, "Integration of Distributed Generation in the Power System", 1st Edition, John Wiley & Sons, 2011.	
4.	Magdi S. Mahmoud, Fouad M. Al-Sunni, "Control and Optimization of Distributed Generation Systems", 1st Edition, Springer International Publishing, 2015.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	1		2	1	2	1
CO2	2		2	1	3	2
CO3	2		2	1	3	2
CO4	1		2	1	2	1
CO5	2		2	2	3	2
Avg.	**1.6**		**2**	**1.2**	**2.4**	**1.6**

1-Low,2-Medium,3-High

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ME23PE421	POWER QUALITY	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:

1.	To understand electric power quality issues and related standards.
2.	To learn analysis of single-phase and three-phase linear and non-linear systems.
3.	To understand conventional methods for load compensation and voltage regulation.
4.	To learn load compensation techniques using DSTATCOM.
5.	To understand series compensation methods for power distribution systems.

UNIT–I	**INTRODUCTION**	**9**
	Introduction (L1) – Characterization of Electric Power Quality: Transients, Short Duration and Long Duration Voltage Variations, Voltage Imbalance, Waveform Distortion, Voltage Fluctuations, Power Frequency Variation - Power Acceptability Curves – Power Quality Problems: Poor Load Power Factor, Non-linear and unbalanced Loads, DC offset in Loads, Notching in Load Voltage, Disturbance in Supply Voltage – Power Quality Standards .	
UNIT–II	**ANALYSIS OF SINGLE PHASE AND THREE PHASE SYSTEM**	**9**
	Single-phase Linear and Non-linear Loads – Single-phase Sinusoidal, Non-sinusoidal Source – Supplying Linear and Nonlinear Loads – Three-phase Balanced System and Unbalanced System – Three-phase Unbalanced and Distorted Source Supplying Non-linear Loads – Concept of Power Factor – Three-phase three wire – Three-phase four wire System .	
UNIT–III	**CONVENTIONAL LOAD COMPENSATION METHODS**	**9**
	Principle of Load Compensation and Voltage Regulation – classical load Balancing Problem: Open loop balancing – Closed Loop balancing, Current Balancing – Harmonic Reduction and Voltage Sag Reduction – Analysis of Unbalance – Instantaneous of Real and Reactive Powers – Extraction of Fundamental Sequence Component from Measured .	
UNIT – IV	**LOAD COMPENSATION USING DSTATCOM**	**9**
	Compensating Single-phase loads – Ideal Three-phase Shunt Compensator Structure – Generating Reference Currents using Instantaneous PQ Theory – Instantaneous Symmetrical Components Theory – Generating Reference Currents when the Source is Unbalanced – Realization and Control of DSTATCOM – DSTATCOM in Voltage Control Mode .	
UNIT – V	**SERIES COMPENSATION OF POWER DISTRIBUTION SYSTEM**	**9**
	Rectifier Supported DVR – DC Capacitor Supported DVR – DVR Structure – Voltage Restoration – Series Active Filter – Unified Power Quality Conditioner .	
	Total(L)	**45 Periods**

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain power quality problems, disturbances, and standards.	L2 - Understand
CO2	Describe power flow and power factor in single-phase and three-phase systems.	L2 - Understand
CO3	Explain conventional load compensation and harmonic reduction methods.	L2 - Understand
CO4	Apply basic control concepts to analyze load compensation using DSTATCOM.	L3 - Apply
CO5	Explain series compensation techniques such as DVR and UPQC.	L2 - Understand
	REFERENCE BOOKS:	
1.	R. C. Dugan, "Electrical Power Systems Quality", 3rd Edition, Tata McGraw-Hill, 2012.	
2.	J. Arrillaga, "Power System Harmonics", 2nd Edition, John Wiley & Sons, 2003.	
3.	Derek A. Paice, "Power Electronic Converter Harmonics", 1st Edition, IEEE Press, 1995.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3		3	3	3	2
CO2	3		3	3	3	2
CO3	3		3	3	3	2
CO4	3		3	3	3	2
CO5	3		3	3	3	2
Avg.	**3**		**3**	**3**	**3**	**2**

1-Low,2-Medium,3-High

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ME23PE422	ENERGY MANAGEMENT AND AUDITING	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:	
1.	To understand energy sources, policies, conservation acts, and energy management concepts.
2.	To learn energy cost analysis and load management techniques.
3.	To understand demand side management and energy conservation strategies.
4.	To learn the methodology and procedures of energy auditing.
5.	To understand energy efficient technologies and conservation methods.

UNIT–I	**ENERGY SCENARIO**	9
	Basics of Energy and its Various Forms (L1) - Conventional and Non-Conventional Sources - Energy Policy - Energy Conservation Act 2001, Amedments (India) in 2010 - Need for Energy Management - Designing and Starting an Energy Management Program - Energy Managers and Energy Auditors - Roles and Responsibilities of Energy Managers - Energy Labelling and Energy Standards .	
UNIT–II	**ENERGY COST AND LOAD MANAGEMENT**	9
	Important Concepts in an Economic Analysis (L1) - Economic Models - Time Value of Money-Utility rate Structures - Cost of Electricity - Loss Evaluation - Load management: Demand Control Techniques - Utility Monitoring and Control System -HVAC and Energy Management - Economic Justification .	
UNIT–III	**ENERGY MANAGEMENT**	9
	Demand Side Management – DSM Planning – DSM Techniques – Load Management as a DSM Strategy – Energy Conservation – Tariff Options for DSM .	
UNIT – IV	**ENERGY AUDITING**	9
	Energy audit methodology: audit preparation, execution and reporting – Financial analysis – Sensitivity analysis – Project financing options - Instruments for energy audit – Energy audit for generation, distribution and utilization systems – Economic analysis .	
UNIT – V	**ENERGY EFFICIENT TECHNOLOGIES**	9
	Energy Saving Opportunities in Electric Motors - Power factor Improvement Benefit and Techniques - Shunt Capacitor, Synchronous Condenser and Phase Advancer - Energy conservation in Industrial Drives, Electric Furnaces, Ovens and Boilers - Lighting Techniques: Natural, CFL, LED Lighting Sources and Fittings .	
	Total(L)	45 Periods

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain energy sources, policies, and the need for energy management.	L2 - Understand
CO2	Explain energy cost concepts and load management techniques.	L2 - Understand
CO3	Describe demand side management methods and tariff options.	L2 - Understand
CO4	Apply basic energy audit procedures and economic analysis methods.	L3 - Apply
CO5	Explain energy efficient technologies and power saving techniques.	L2 - Understand
	REFERENCE BOOKS:	
1.	Barney L. Capehart, Wayne C. Turner, William J. Kennedy, "Guide to Energy Management", 8th Edition, CRC Press (Taylor & Francis Group), 2016.	
2.	Government of India, "The Energy Conservation (Amendment) Bill", 1st Edition, Parliament of India, 2010.	
3.	T. D. Eastop, D. R. Croft, "Energy Efficiency for Engineers and Technologists", 1st Edition, Longman Scientific & Technical, 1990.	
4.	IEEE Standards Association, "IEEE Recommended Practice for Energy Management in Industrial and Commercial Facilities", 1st Edition, IEEE Press, 1996.	
5.	Amit K. Tyagi, "Handbook on Energy Audits and Management", 1st Edition, TERI Press, 2003.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		2		2	-
CO2	2		2	1	2	1
CO3	2		2	1	2	2
CO4	1		2	3		
CO5	3		2	3	3	3
Avg.	**2**		**2**	**2**	**2.25**	**2**

1-Low,2-Medium,3-High

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ME23PE408	SOFT COMPUTING TECHNIQUES	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:

1.	To understand the fundamentals of intelligent systems and neural networks.
2.	To learn neural network models and associative memory concepts.
3.	To understand fuzzy logic principles and control systems.
4.	To learn genetic algorithm concepts and optimization methods.
5.	To understand hybrid intelligent control techniques and their applications.

UNIT–I	**INTELLIGENT SYSTEMS AND NEURAL NETWORKS**	**9**
	Introduction to Intelligent systems – Soft computing techniques – Conventional vs swarm computing – Meta-heuristic methods – Swarm intelligence – Applications –Biological neuron – Artificial neuron & activation functions – Neural network types – McCulloch-Pitts neuron – Perceptron – Adaline & Madaline – Multilayer perceptron – Back-propagation learning – Learning rate – Training factors – Applications .	
UNIT–II	**ASSOCIATIVE MEMORY AND NEURAL NETWORK MODELS**	**9**
	Counter Propagation Network – Architecture & working – Hopfield & recurrent networks – Stability & associative memory – Features & limitations – Applications – Hopfield vs Boltzmann machine – Adaptive Resonance Theory – ART architecture & types – Training & associative memory .	
UNIT–III	**FUZZY LOGIC SYSTEMS**	**9**
	Introduction to Crisp & fuzzy sets – Fuzzy operations & reasoning – Fuzzy modeling & control – Fuzzification, inference & defuzzification – Fuzzy rule base – Fuzzy control for nonlinear systems – Self-organizing fuzzy control – Fuzzy control for time-delay systems .	
UNIT – IV	**GENETIC ALGORITHMS**	**9**
	Evolutionary programming – Genetic algorithms & programming – Genetic vs conventional optimization – Selection & genetic operators – Crossover & mutation – Optimization problems: discrete & continuous – Single & multi-objective optimization .	
UNIT – V	**HYBRID INTELLIGENT CONTROL**	**9**
	Neuro-fuzzy systems: ANN & fuzzification – ANFIS & fuzzy neuron – Optimize membership & rules using GA – Support Vector Machines – Evolutionary programming – Particle Swarm Optimization – Case study & toolbox: NN, FLC & ANFIS .	
	Total(L)	**45 Periods**

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the structure and learning process of artificial neural networks.	L2 - Understand
CO2	Describe associative memory models and recurrent neural networks.	L2 - Understand
CO3	Explain fuzzy logic concepts and fuzzy control systems.	L2 - Understand
CO4	Apply genetic algorithms to solve basic optimization problems.	L3 - Apply
CO5	Describe hybrid intelligent systems and their applications.	L2 - Understand
	REFERENCE BOOKS:	
1.	Laurene V. Fausett, "Fundamentals of Neural Networks: Architectures, Algorithms and Applications", 1st Edition, Pearson Education, 1994.	
2.	Timothy J. Ross, "Fuzzy Logic with Engineering Applications", 2nd Edition, Wiley India, 2008.	
3.	H. J. Zimmermann, "Fuzzy Set Theory and Its Applications", 4th Edition, Springer (International Edition), 2011.	
4.	David E. Goldberg, "Genetic Algorithms in Search, Optimization, and Machine Learning", 1st Edition, Pearson Education, 2009.	
5.	W. T. Miller, R. S. Sutton, P. J. Werbos, "Neural Networks for Control", 1st Edition, MIT Press, 1996.	

Mapping of COs with POs

COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3		2	2	1	1
CO2	3		2	2	1	1
CO3	3		2	2	1	1
CO4	3		2	2	1	1
CO5	2		2	2	1	1
Avg.	2.8		2	2	1	1

1-Low,2-Medium,3-High

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ME23PE409	MACHINE LEARNING AND DEEP LEARNING	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:		
1.	To understand different learning paradigms and types of learning problems.	
2.	To learn neural network models and learning algorithms.	
3.	To understand machine learning fundamentals, feature selection, and classification methods.	
4.	To learn the concepts and architecture of convolutional neural networks.	
5.	To understand recurrent networks, autoencoders, and generative models.	
UNIT–I	**LEARNING PROBLEMS AND ALGORITHMS**	**9**
	Learning paradigms – Supervised, Semi-supervised, and Unsupervised algorithms – Applications of learning problems .	
UNIT–II	**NEURAL NETWORKS**	**9**
	Neural network architecture and activation functions – Multilayer networks – Linear separability – Hebb Net, Perceptron, Adaline – Back-propagation training, Hebb rule and Delta rule – Hetero-associative and Auto-associative networks – Kohonen Self-Organizing Maps – Feature maps examples – Learning Vector Quantization – Gradient descent – Boltzmann Machine learning .	
UNIT–III	**MACHINE LEARNING**	**9**
	Confusion matrix, Accuracy, Precision, Recall, F1-score – Curse of dimensionality – Data splitting: training, testing, validation – Cross-validation – Overfitting, Underfitting, Early stopping, Regularization, Bias-Variance trade-off – Feature selection and normalization – Dimensionality reduction – Classifiers: KNN, SVM, Decision Trees, Naïve Bayes – Binary and multi-class classification – Clustering techniques .	
UNIT – IV	**DEEP LEARNING: CONVOLUTIONAL NEURAL NETWORKS**	**9**
	Feedforward networks – Activation functions – Backpropagation in CNN – Optimizers – Batch normalization – Convolution layers – Pooling layers – Fully connected layers – Dropout – CNN examples .	
UNIT – V	**DEEP LEARNING: RNNS, AUTOENCODERS AND GANS**	**9**
	RNNs: Cell structure, LSTM and GRU – Time-distributed layers – Sequence generation – Autoencoders: Convolutional, Denoising, Variational – GANs: Generator, Discriminator, DCGAN examples .	
	Total(L)	**45 Periods**

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain supervised, semi-supervised, and unsupervised learning approaches.	L2 - Understand
CO2	Describe neural network architectures and learning rules.	L2 - Understand
CO3	Implement machine learning techniques for classification, regression, and clustering tasks.	L3 - Apply
CO4	Explain the structure and working of convolutional neural networks.	L2 - Understand
CO5	Describe the operation of RNNs, autoencoders, and GANs.	L2 - Understand
	REFERENCE BOOKS:	
1.	J. S. R. Jang, C. T. Sun, E. Mizutani, "Neuro-Fuzzy and Soft Computing: A Computational Approach to Learning and Machine Intelligence", 1st Edition, PHI Learning, 2012.	
2.	Ian Goodfellow, Yoshua Bengio, Aaron Courville, "Deep Learning", 1st Edition, MIT Press, 2016.	
3.	Trevor Hastie, Robert Tibshirani, Jerome Friedman, "The Elements of Statistical Learning", 2nd Edition, Springer, 2009.	
4.	Christopher M. Bishop, "Pattern Recognition and Machine Learning", 1st Edition, Springer, 2006.	
5.	Shai Shalev-Shwartz, Shai Ben-David, "Understanding Machine Learning: From Theory to Algorithms", 1st Edition, Cambridge University Press, 2017.	

Mapping of COs with POs						
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	1		1			
CO2	2		2			
CO3	3		3		3	
CO4	2		3			
CO5	3		3		3	
Avg.	3		3		3	
1–Low,2–Medium,3–High						

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ME23PE410	OPTIMIZATION TECHNIQUES	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:	
1.	To understand the basics and classification of optimization problems.
2.	To learn linear programming concepts and solution techniques.
3.	To understand methods for solving nonlinear optimization problems.
4.	To learn dynamic programming concepts and solution approaches.
5.	To understand genetic algorithm principles and their applications.

UNIT–I	INTRODUCTION	9
	Definition (L1) - Classification of Optimization Problems - Classical Optimization Techniques - Single and Multiple Optimizations with and without Inequality constraints .	
UNIT–II	LINEAR PROGRAMMING (LP)	9
	Simplex Method of Solving LP - Revised Simplex Method , Duality - Constrained Optimization - Theorems and Procedure - Linear Programming - Mathematical Model - Solution Technique - Duality .	
UNIT–III	NON LINEAR PROGRAMMING	9
	Steepest Descent Method - Conjugates Gradient Method - Newton's Method - Sequential Quadratic Programming - Penalty Function Method - Augmented Lagrange Multiplier method .	
UNIT – IV	DYNAMIC PROGRAMMING (DP)	9
	Multistage Decision Processes - Concept of Sub-optimization and Principle of Optimality - Recursive Relations - Integer Linear Programming - Branch and Bound Algorithm .	
UNIT – V	GENETIC ALGORITHM (GA)	9
	Introduction to Genetic Algorithm - Working Principle, Coding of Variables, Fitness Function, GA Operators - Similarities and differences between GA's and Traditional Methods - Unconstrained and Constrained Optimization using Genetic Algorithm, Real Coded GA's, Advanced GA's, Global Optimization using GA - Applications to Power System .	
	Total(L)	45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	

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	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Explain different types of optimization problems and classical methods.	L2 - Understand
CO2	Apply linear programming techniques to solve constrained problems.	L3 - Apply
CO3	Apply numerical methods to solve non-linear programming problems.	L3 - Apply
CO4	Describe dynamic programming concepts and solution procedures.	L2 - Understand
CO5	Apply genetic algorithms to solve basic optimization problems.	L3 - Apply
	REFERENCE BOOKS:	
1.	S. S. Rao, "Engineering Optimization: Theory and Practice", 4th Edition, John Wiley & Sons, 2009.	
2.	Hamdy A. Taha, "Operations Research: An Introduction", 10th Edition, Pearson Education, 2016.	
3.	David G. Luenberger, "Introduction to Linear and Nonlinear Programming", 1st Edition, Addison-Wesley, 1973.	
4.	E. Polak, "Computational Methods in Optimization", 1st Edition, Academic Press, 1971.	
5.	D. A. Pierre, "Optimization Theory with Applications", 1st Edition, John Wiley & Sons, 1969.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3		3			1
CO2	3		3			1
CO3	3		3			1
CO4	3		3			1
CO5	3		3	3		1
Avg.	3		3	3		1

1-Low,2-Medium,3-High

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ME23PE423	PYTHON PROGRAMMING FOR MACHINE LEARNING	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:		
1.	To understand Python programming fundamentals for data analysis.	
2.	To learn Python libraries used in machine learning.	
3.	To apply data preprocessing and visualization techniques.	
4.	To implement basic machine learning algorithms using Python.	
5.	To use Python tools for simple machine learning applications.	
UNIT–I	**PYTHON BASICS**	**9**
	Introduction to Python – Python environment and IDEs – Variables and data types – Operators and expressions – Control statements – Functions and modules – Lists, tuples, sets, and dictionaries – File handling .	
UNIT–II	**NUMERICAL COMPUTING WITH PYTHON**	**9**
	Introduction to NumPy – Arrays and array operations – Matrix operations – Broadcasting and indexing – Introduction to SciPy – Mathematical and optimization functions .	
UNIT–III	**DATA HANDLING AND VISUALIZATION**	**9**
	Pandas for data analysis – Series and DataFrames – Data loading and cleaning – Handling missing data – Data preprocessing techniques – Data visualization using Matplotlib and Seaborn .	
UNIT – IV	**MACHINE LEARNING WITH PYTHON**	**9**
	Introduction to machine learning concepts – Scikit-learn overview – Data splitting: training and testing – Regression algorithms – Classification algorithms – Model evaluation metrics .	
UNIT – V	**ADVANCED ML TOOLS AND APPLICATIONS**	**9**
	Feature selection and dimensionality reduction – Clustering techniques – Introduction to deep learning libraries: TensorFlow and Keras – Simple neural network implementation – Case studies and applications in engineering systems .	
	Total(L)	**45 Periods**

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Understand Python programming fundamentals.	L2 - Understand
CO2	Understand numerical and data analysis libraries in Python.	L2 - Understand
CO3	Apply data preprocessing and visualization techniques using Python.	L3 - Apply
CO4	Implement basic machine learning algorithms using Python.	L3 - Apply
CO5	Apply Python tools to solve simple machine learning problems.	L3 - Apply
	REFERENCE BOOKS:	
1.	Jake VanderPlas, Python Data Science Handbook, O'Reilly Media, 2016.	
2.	Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly Media, 2019.	
3.	Wes McKinney, Python for Data Analysis, O'Reilly Media, 2018.	
4.	Sebastian Raschka, Python Machine Learning, Packt Publishing, 3rd Edition, 2019.	
5.	Ethem Alpaydin, Introduction to Machine Learning, MIT Press, 3rd Edition, 2014.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	-	-	2	3	3	-
CO2	3	1	3	-	3	1
CO3	2	1	2	-	3	3
CO4	3	2	3	3	3	3
CO5	-	-	-		3	-
Avg.	2.66	1.33	2.5	3	3	2.33

1-Low,2-Medium,3-High

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ME23PE424	AI FOR POWER ELECTRONICS		CP	L	T	P	C
			3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES		Version: 1.0				
Course Objectives:							
1.	To understand AI and data analytics concepts applied to power electronic systems.						
2.	To learn machine learning techniques for modeling and analysis of power converters.						
3.	To understand neural networks and deep learning methods for power electronics.						
4.	To learn AI-based control and fault diagnosis techniques in power electronic systems.						
5.	To understand advanced AI applications in power electronics and smart energy systems.						
UNIT–I	**INTRODUCTION**	**9**					
	Overview of Artificial Intelligence – Machine Learning Concepts – Supervised, Unsupervised and Reinforcement Learning – Role of Data in Power Electronics – Data Acquisition from Power Converters – Feature Extraction – Data Preprocessing – Applications of AI in Power Electronics .						
UNIT–II	**MACHINE LEARNING TECHNIQUES**	**9**					
	Linear and Nonlinear Regression – Decision Trees – Support Vector Machines – k-nearest Neighbor Algorithm – Performance Metrics – Training, Validation and Testing – Overfitting and Underfitting – Case Studies Related to Power Converter Modeling and Control .						
UNIT–III	**ARTIFICIAL NEURAL NETWORKS AND DEEP LEARNING**	**9**					
	Biological Neuron Model – Artificial Neural Networks – Feedforward and Recurrent Neural Networks – Training Algorithms – Deep Learning Fundamentals – Convolutional Neural Networks – Application of ANN for Power Converter Control and Parameter Estimation .						
UNIT – IV	**AI-BASED CONTROL AND FAULT DIAGNOSIS**	**9**					
	AI-based Controllers – Intelligent PID Tuning – Fuzzy Logic Control – Neuro-fuzzy Systems – AI-based Fault Detection and Diagnosis – Condition Monitoring of Power Electronic Devices – Thermal and Aging Analysis – Predictive Maintenance .						
UNIT – V	**ADVANCED AI APPLICATIONS IN POWER ELECTRONICS**	**9**					
	Reinforcement Learning for Power Converter Control – AI for Motor Drive Systems – Optimization of Converter Efficiency using AI – AI in Smart Grids and Renewable Energy Systems – Digital Twins – Emerging Trends and Future Scope of AI in Power Electronics .						
	Total(L)	**45 Periods**					

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	Bloom's Taxonomy Levels: Remember (L1), Understand , Apply , Analyze (L4), Evaluate (L5), Create (L6).	
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the role of AI and data processing in power electronic applications.	L2 - Understand
CO2	Apply basic machine learning techniques to analyze power electronic system data.	L3 - Apply
CO3	Explain neural network and deep learning concepts used in power electronics.	L2 - Understand
CO4	Apply AI-based methods for control and fault diagnosis of power electronic systems.	L3 - Apply
CO5	Explain advanced AI applications in power electronics and renewable energy systems.	L2 - Understand
	REFERENCE BOOKS:	
1.	Ian Goodfellow, Yoshua Bengio, Aaron Courville, "Deep Learning", 1st Edition, MIT Press, 2016.	
2.	Kevin P. Murphy, "Machine Learning: A Probabilistic Perspective", 1st Edition, MIT Press, 2012.	
3.	Simon Haykin, "Neural Networks and Learning Machines", 3rd Edition, Pearson Education, 2009.	
4.	B. K. Bose, "Artificial Intelligence Techniques in Power Electronics", 1st Edition, IEEE Press, 2007.	
5.	B. K. Bose, "Power Electronics and Motor Drives: Advances and Trends", 1st Edition, Academic Press, 2010.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3		3	2	2	2
CO2	3		3	2	2	2
CO3	3		3	2	2	2
CO4	3		3	2	2	2
CO5	3		3	2	2	2
Avg.	3		3	2	2	2

1-Low,2-Medium,3-High

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ME23PE425	INTELLIGENT CONTROL AND AUTOMATION	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:

1.	To apply intelligent control techniques to dynamic systems.
2.	To design ANN and fuzzy-based controllers for control applications.
3.	To implement hybrid intelligent control schemes.
4.	To understand automation architectures and modern automation tools.
5.	To apply intelligent controllers in industrial and energy systems.

UNIT–I	**INTELLIGENT CONTROL USING ANN AND FUZZY LOGIC**	**9**
	ANN-based system identification – ANN-based control structures – Adaptive Neuro-Controllers – Fuzzy logic controller design steps – Fuzzy modeling and system identification – Adaptive fuzzy control strategies – Comparison of ANN and fuzzy controllers .	
UNIT–II	**OPTIMIZATION TECHNIQUES FOR CONTROL APPLICATIONS**	**9**
	GA-based controller tuning – Hybrid genetic algorithms for control – Optimization of controller parameters – Tabu search for control optimization – Ant Colony Optimization for control problems – Particle Swarm Optimization for controller design – Comparative performance analysis .	
UNIT–III	**HYBRID INTELLIGENT CONTROL SYSTEMS**	**9**
	Neuro-fuzzy control architecture – ANFIS-based controller design – Optimization of membership functions – Rule base optimization using GA and PSO – Performance evaluation of hybrid controllers – Case studies in hybrid control systems .	
UNIT – IV	**AUTOMATION SYSTEMS AND TECHNOLOGIES**	**9**
	Automation concepts and objectives – Production automation systems – Elements of automated systems – Levels and strategies of automation – PLC-based automation – SCADA systems – Industrial communication and control networks – IoT-based automation – Industry 4.0 concepts .	
UNIT – V	**INTELLIGENT CONTROL FOR AUTOMATION APPLICATIONS**	**9**
	Intelligent control in industrial monitoring – Optimization and control of industrial processes – Smart appliances and embedded automation – Intelligent automation for electric vehicles – Intelligent controllers for power systems – Case studies in automation systems .	
	Total(L)	**45 Periods**

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Apply ANN and fuzzy logic techniques for intelligent control applications.	L3 - Apply
CO2	Apply optimization techniques for controller tuning and performance improvement.	L3 - Apply
CO3	Design hybrid intelligent control systems for dynamic processes.	L3 - Apply
CO4	Understand automation architectures and modern automation technologies.	L2 - Understand
CO5	Apply intelligent controllers to industrial, energy, and automation systems.	L3 - Apply
	REFERENCE BOOKS:	
1.	Bimal N. Bose, Modern Power Electronics and AC Drives, Prentice Hall, 2002.	
2.	Kevin M. Passino & Stephen Yurkovich, Fuzzy Control, Addison Wesley, 1998.	
3.	Frank L. Lewis, Draguna Vrabie, Optimal Control, Wiley, 2012.	
4.	S. N. Sivanandam, S. Sumathi, Introduction to Neural Networks using MATLAB, TMH, 2006.	
5.	M. P. Groover, Automation, Production Systems and Computer Integrated Manufacturing, Pearson, 2015.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3		2	2	2	2
CO2	3		2	2	2	2
CO3	3		2	2	2	2
CO4	3		2	2	2	2
CO5	3		2	2	2	2
Avg.	3		2	2	2	2

1-Low,2-Medium,3-High

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ME23PE411	SYSTEM DESIGN USING MICROCONTROLLER		CP	L	T	P	C
			3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0					

Course Objectives:	
1.	To understand the architecture and features of the 8051 microcontroller.
2.	To learn assembly level programming concepts of the 8051 microcontroller.
3.	To understand the architecture and programming of the PIC 16 microcontroller.
4.	To learn the operation of peripherals and interfacing techniques of PIC 16.
5.	To understand microcontroller based system design using practical case studies.

UNIT–I	**8051 ARCHITECTURE**	**9**
	Architecture – Memory Organization – Addressing Modes – Instruction Set – Timers – Interrupts – I/O Ports, Interfacing I/O Devices – Serial Communication .	
UNIT–II	**8051 PROGRAMMING**	**9**
	Assembly Language Programming – Arithmetic Instructions – Logical Instructions – Single Bit Instructions – Timer Counter Programming – Serial Communication Programming Interrupt Programming – LCD Digital Clock/thermometer - Introduction to IDE Based Assembler Programming .	
UNIT–III	**PIC 16 MICROCONTROLLER**	**9**
	Architecture – Memory Organization – Addressing Modes – Instruction Set – PIC Programming in Assembly & C – I/O Port, Data Conversion, RAM & ROM Allocation, Timer Programming, Practice in MP-LAB .	
UNIT – IV	**PERIPHERAL OF PIC 16 MICROCONTROLLER**	**9**
	Timers – Interrupts, I/O Ports - I2C Bus-A/D Converter – UART - CCP Modules – ADC, DAC and Sensor Interfacing – Flash and EEPRO Memories .	
UNIT – V	**SYSTEM DESIGN –CASE STUDY**	**9**
	Interfacing LCD Display – Keypad Interfacing – Generation of Gate Signals for Converters and Inverters – Motor Control – Controlling DC/AC Appliances – Measurement of Frequency – Stand-alone Data Acquisition System .	
	Total(L)	**45 Periods**

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the architecture and internal features of the 8051 microcontroller.	L2 - Understand
CO2	Write simple programs using 8051 instruction set and addressing modes.	L2 - Understand
CO3	Explain the architecture and programming features of the PIC 16 microcontroller.	L2 - Understand
CO4	Apply peripheral interfacing concepts using PIC 16 microcontroller.	L3 - Apply
CO5	Apply microcontroller concepts to develop simple embedded system applications.	L3 - Apply
	REFERENCE BOOKS:	
1.	Raj Kamal, "Microcontrollers: Architecture, Programming, Interfacing and System Design", 2nd Edition, Pearson Education, 2012.	
2.	Myke Predko, "Programming and Customizing the 8051 Microcontroller", 1st Edition, Tata McGraw-Hill, 2001.	
3.	Muhammad Ali Mazidi, Sarmad Naimi, Sepehr Naimi, "The AVR Microcontroller and Embedded Systems: Using Assembly and C", 1st Edition, Pearson Education, 2014.	
4.	Muhammad Ali Mazidi, Janice G. Mazidi, Rolin D. McKinlay, "The 8051 Microcontroller and Embedded Systems: Using Assembly and C", 2nd Edition, Prentice Hall, 2005.	
5.	John Iovine, "PIC Microcontroller Project Book", 1st Edition, McGraw-Hill, 2000.	

Mapping of COs with POs						
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	1		3	1	1	2
CO2	2		2	2	1	3
CO3	1		3	1	1	3
CO4	2		2	2	1	1
CO5	3		2	3	2	1
Avg.	1.8		2.4	1.8	1.2	2
1-Low,2-Medium,3-High						

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ME23PE412	DSP BASED SYSTEM DESIGN	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:	
1.	To understand representations and architectural requirements of DSP systems.
2.	To learn common DSP algorithms and their computational characteristics.
3.	To understand DSP processor architecture and system-level design concepts.
4.	To learn architecture analysis of DSP algorithms on programmable hardware.
5.	To understand system interfacing techniques for DSP applications.

UNIT–I	**REPRESENTATION OF DSP SYSTEM**	**9**
	Single Core and Multicore - Architectural Requirement of DSPs - High Throughput, Low Cost, Low Power, Small Code Size, Embedded Applications - Representation of Digital Signal Processing Systems - Block Diagrams, Signal Flow Graphs, Data-flow Graphs, Dependence Graphs - Techniques for Enhancing Computational Throughput - Parallelism and Pipelining .	
UNIT–II	**DSP ALGORITHMS**	**9**
	DSP Algorithms - Convolution, Correlation, FIR/IIR Filters, FFT, Adaptive Filters, Sampling Rate Converters, DCT, Decimator, Expander and Filter Banks - DSP Applications - Computational Characteristics of DSP Algorithms and Applications - Numerical Representation of Signals - Word Length Effect and its Impact - Carry Free Adders - Multiplier .	
UNIT–III	**SYSTEM ARCHITECTURE**	**9**
	Introduction (L1) - Basic Architectural Features - DSP Computational Building Blocks, Bus Architecture and Memory, Data Addressing Capabilities, Address Generation Unit, Programmability and Program Execution - Features for External Interfacing - VLIW Architecture - Basic Performance Issue in Pipelining - Simple Implementation of MIPS - Instruction Level Parallelism, Dynamic Scheduling, Dynamic Hardware Prediction, Memory Hierarchy - Study of Fixed Point and Floating Point DSP Architectures .	
UNIT – IV	**ARCHITECTURE ANALYSIS ON PROGRAMMABLE HARDWARE**	**9**
	Analysis of basic DSP Architectures on Programmable Hardwares - Algorithms for FIR , IIR, Lattice Filter Structures, Architectures for Real and Complex Fast Fourier Transforms - 1D/2D Convolutions - Winograd Minimal Filtering Algorithm - FPGA: Architecture, Different Sub-systems, Design Flow for DSP System Design - Mapping of DSP Algorithms onto FPGA .	

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UNIT – V	SYSTEM INTERFACING	9
	Examples of Digital Signal Processing Algorithms Suitable for Parallel Architectures such as GPUs and multiGPUs - Interfacing: Introduction, Synchronous Serial Interface CODE - A CODEC Interface Circuit - ADC Interface .	
	Total(L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain different representations and throughput enhancement techniques in DSP systems.	L2 - Understand
CO2	Explain the operation and applications of common DSP algorithms.	L2 - Understand
CO3	Describe DSP architectural features and execution mechanisms.	L2 - Understand
CO4	Apply basic DSP algorithm architectures on programmable hardware platforms.	L3 - Apply
CO5	Explain interfacing methods used in DSP-based systems.	L2 - Understand
	REFERENCE BOOKS:	
1.	Sen M. Kuo, Woon-Seng Gan, "Digital Signal Processors: Architectures, Implementations, and Applications", 1st Edition, Prentice Hall, 2005.	
2.	Rulph Chassaing, "Digital Signal Processing and Applications with the C6713 and C6416 DSK", 2nd Edition, John Wiley & Sons (Wiley-Interscience), 2005.	
3.	Peter Pirsch, "Architectures for Digital Signal Processing", 2nd Edition, John Wiley & Sons, 2007.	
4.	Phil Lapsley, Jeff Bier, Amit Shoham, Edward A. Lee, "DSP Processor Fundamentals: Architectures and Features", 1st Edition, John Wiley & Sons / IEEE Press, 1997.	
5.	K. K. Parhi, "VLSI Digital Signal Processing Systems: Design and Implementation", 1st Edition, John Wiley & Sons, 1999.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1						
CO2	3		3	2	3	2
CO3						
CO4	3		3	3	3	3
CO5	2		3	2	3	3
Avg.	2.67		3	2.33	3	2.67

1-Low,2-Medium,3-High

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ME23PE413	MEMS DESIGN: SENSORS AND ACTUATORS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:	
1.	To understand micro-fabrication processes, materials, and basic electro-mechanical concepts.
2.	To learn the principles and design of electrostatic sensing and actuation systems.
3.	To understand thermal sensing and actuation mechanisms and their applications.
4.	To learn piezoelectric sensing and actuation principles and material properties.
5.	To understand MEMS applications through practical case studies.

UNIT–I	**MICRO-FABRICATION, MATERIALS AND ELECTRO-MECHANICAL CONEPTS**	**9**
	Overview of Micro Fabrication (L1) – Silicon and Other Material based Fabrication Processes – Concepts: Conductivity of Semiconductors-Crystal Planes and Orientation-Stress - Strain-Flexural Beam Bending Analysis - Torsional Deflections -Intrinsic Stress - Resonant Frequency and Quality Factor .	
UNIT–II	**ELECTROSTATIC SENSORS AND ACTUATION**	**9**
	Principle - Material - Design and Fabrication of Parallel Plate Capacitors as Electrostatic Sensors and Actuators - Applications .	
UNIT–III	**THERMAL SENSING AND ACTUATION**	**9**
	Principle - Material - Design and Fabrication of Thermal Couples - Thermal Bimorph Sensors - Thermal Resistor Sensors - Applications .	
UNIT – IV	**PIEZOELECTRIC SENSING AND ACTUATION**	**9**
	Piezoelectric Effect - Cantilever Piezo-electric Actuator Model - Properties of Piezoelectric Materials Applications .	
UNIT – V	**CASE STUDIES**	**9**
	Piezo-resistive Sensors - Magnetic Actuation - Micro Fluidics Applications - Medical Applications - Optical MEMS - NEMS Devices .	
	Total(L)	**45 Periods**

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain micro-fabrication techniques and electro-mechanical concepts used in MEMS.	L2 - Understand
CO2	Describe the working principles of electrostatic sensors and actuators.	L2 - Understand
CO3	Explain thermal sensing and actuation methods and their applications.	L2 - Understand
CO4	Describe piezoelectric sensing and actuation mechanisms.	L2 - Understand
CO5	Apply MEMS concepts to understand selected engineering and medical applications.	L3 - Apply
	REFERENCE BOOKS:	
1.	Chang Liu, "Foundations of MEMS", 1st Edition, Pearson (International Edition), 2006.	
2.	Marc J. Madou, "Fundamentals of Microfabrication", 2nd Edition, CRC Press, 2002.	
3.	Gregory T. A. Kovacs, "Micromachined Transducers Sourcebook", 1st Edition, McGraw-Hill, 1998.	
4.	M. H. Bao, "Micromechanical Transducers: Pressure Sensors, Accelerometers and Gyroscopes", 1st Edition, Elsevier, 2000.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3		3	3	3	3
CO2	3		2		3	
CO3	3		2		3	
CO4	3		2		3	
CO5	3		2		3	
Avg.	3		2.2	3	3	3

1-Low,2-Medium,3-High

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ME23PE414	IoT FOR SMART SYSTEMS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:	
1.	To study about Internet of Things technologies and its role in real time applications.
2.	To introduce the infrastructure required for IoT
3.	To familiarize the accessories and communication techniques for IoT.
4.	To provide insight about the embedded processor and sensors required for IoT
5.	To familiarize the different platforms and Attributes for IoT

UNIT–I	INTRODUCTION TO INTERNET OF THINGS	9
	Overview (L1) - Hardware and Software Requirements for IOT - Sensor and Actuators - Technology Drivers - Business Drivers - Typical IoT Applications - Trends and Implications .	
UNIT–II	IOT ARCHITECTURE	9
	IoT Reference Model and Architecture - Node Structure - Sensing - Processing - Communication - Powering, Networking - Topologies, Layer/Stack Architecture - IoT Standards - Cloud Computing for IoT, Bluetooth - Bluetooth Low Energy Beacons .	
UNIT–III	PROTOCOLS	9
	NFC, SCADA and RFID, Zigbee MIPI, M-PHY, UniPro, SPMI, SPI, M-PCIe GSM, CDMA, LTE, GPRS, small cell - Wireless Technologies for IoT : WiFi (IEEE 802.11) - Bluetooth/Bluetooth Smart - ZigBee/ZigBee Smart - UWB (IEEE 802.15.4) - LoWPAN, Proprietary Systems - Recent Trends .	
UNIT – IV	IOT PROCESSORS	9
	Services/Attributes: Big-Data Analytics for IOT, Dependability, Interoperability, Security, Maintainability - Embedded processors for IOT - Introduction to Python programming - Building IOT with RasperryPI and Arduino	
UNIT – V	CASE STUDIES	9
	Industrial IoT - Home Automation - Smart Cities - Smart Grid - Connected Vehicles - electric Vehicle Charging - Agriculture - IOT Defense	
	Total(L)	45 Periods

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the basic concepts, drivers, and applications of IoT systems.	L2 - Understand
CO2	Describe IoT architecture, node structure, and networking models.	L2 - Understand
CO3	Explain IoT protocols and wireless communication technologies.	L2 - Understand
CO4	Apply basic programming and embedded platforms to build simple IoT solutions.	L3 - Apply
CO5	Explain IoT use cases in industrial, smart city, and energy applications.	L2 - Understand
	REFERENCE BOOKS:	
1.	Arshdeep Bahga, Vijai Madisetti, "Internet of Things: A Hands-on Approach", 1st Edition, Universities Press, 2015.	
2.	Olivier Hersent, David Boswarthick, Omar Elloumi, "The Internet of Things: Key Applications and Protocols", 2nd Edition, John Wiley & Sons, 2016.	
3.	Samuel Greengard, "The Internet of Things", 1st Edition, MIT Press, 2015.	
4.	Adrian McEwen, Hakim Cassimally, "Designing the Internet of Things", 1st Edition, John Wiley & Sons, 2014.	
5.	Jean-Philippe Vasseur, Adam Dunkels, "Interconnecting Smart Objects with IP: The Next Internet", 1st Edition, Morgan Kaufmann Publishers, 2010.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	1		1			
CO2						
CO3	1			1	3	
CO4	2		3	3	3	3
CO5	3		3	3	3	3
Avg.	1.75		2.33	2.33	3	2

1-Low,2-Medium,3-High

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ME23PE426	AUTOMOTIVE EMBEDDED SYSTEM	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:		
1.	To understand basic electronic engine control systems and automotive standards.	
2.	To learn sensors and actuators used in automotive electronic systems.	
3.	To understand vehicle management systems and electronic control functions.	
4.	To learn onboard diagnostics, vehicle communication, and telematics concepts.	
5.	To understand electric vehicle systems and emerging automotive technologies.	
UNIT–I	**BASIC ELECTRONIC ENGINE CONTROL SYSTEMS**	**9**
	Overview of Automotive Systems - Fuel Economy - Air-Fuel Ratio - Emission Limits - Vehicle Performance - Automotive Microcontrollers - Electronic Control Unit - Hardware & Software Selection - Requirements for Automotive Applications - Open Source ECU - RTOS - Concept for Engine Management - Introduction to AUTOSAR and Introduction to Society SAE - Functional Safety ISO 26262 .	
UNIT–II	**SENSORS AND ACTUATORS FOR AUTOMOTIVES**	**9**
	Review of Sensors - Sensors Interface to the ECU, Conventional Sensors and Actuators, Modern Sensors and Actuators - LIDAR Sensor - Smart Sensors - MEMS/NEMS Sensors and Actuators for Automotive Applications .	
UNIT–III	**VEHICLE MANAGEMENT SYSTEMS**	**9**
	Electronic Engine Control - Engine Mapping - Spark Timing control Strategy - Electronic Ignition - Adaptive Cruise Control - Anti-locking Braking System - Electronic Suspension - Electronic Steering - Body Control System - Vehicle System Schematic for Interfacing with EMS, ECU - Adaptive Lighting System - Safety and Collision Avoidance .	
UNIT – IV	**ONBOARD DIAGNOSTICS AND TELEMATICS**	**9**
	Onboard Diagnosis of Vehicles - System Diagnostic Standards and Regulation Requirements Vehicle Communication Protocols Bluetooth, CAN, LIN, FlexRay, MOST, KWP2000 - Navigation - Connected Cars Technology - Tracking - Security for Data Communication - Dashboard Display and Virtual Instrumentation, Multimedia Electronics - Role of IOT in Automotive Systems .	
UNIT – V	**ELECTRIC VEHICLES**	**9**
	Electric Vehicles - Components - Plug-in Electrical Vehicle - Charging Station - Battery Management System - Power Management System - Aggregator - Autonomous Vehicles .	
	Total(L)	**45 Periods**

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain electronic engine control concepts and automotive safety standards.	L2 - Understand
CO2	Describe automotive sensors, actuators, and their interfacing with ECU.	L2 - Understand
CO3	Apply basic electronic control concepts in vehicle management systems.	L3 - Apply
CO4	Explain onboard diagnostics, vehicle communication protocols, and telematics.	L2 - Understand
CO5	Describe electric vehicle components and advanced automotive technologies.	L2 - Understand
	REFERENCE BOOKS:	
1.	William B. Ribbens, "Understanding Automotive Electronics", 4th Edition, Elsevier, 2012.	
2.	Ali Emadi, Mehrdad Ehsani, John M. Miller, "Vehicular Electric Power Systems: Land, Sea, Air, and Space Vehicles", 1st Edition, Marcel Dekker, 2004.	
3.	Ljubo Vlacic, Michel Parent, Fumio Harashima, "Intelligent Vehicle Technologies", 1st Edition, SAE International, 2001.	
4.	Jack Erjavec, Jeff Arias, "Alternative Fuel Technology: Electric, Hybrid, and Fuel Cell Vehicles", 2nd Edition, Cengage Learning, 2012.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1			1	1		2
CO2	2		2	2	2	3
CO3	3		3	3	3	2
CO4	3		3	3	3	2
CO5	3		3	3	3	2
Avg.	2.75		2.4	2.4	2.75	2.2
1-Low,2-Medium,3-High						

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ME23PE427	EMBEDDED CONTROL SYSTEMS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:

1.	To understand embedded control system concepts and real-time requirements.
2.	To learn microcontrollers and DSPs used for control applications.
3.	To understand real-time operating systems and embedded software design.
4.	To learn embedded control techniques for power electronic converters and drives.
5.	To understand advanced embedded control applications and communication methods.

UNIT–I	**INTRODUCTION TO EMBEDDED CONTROL SYSTEMS**	**9**
	Embedded Systems Overview – Embedded Control Systems in Power Electronics and Drives – Characteristics of Real-time Systems – Hard and Soft real-time Constraints – Embedded System Design flow – Hardware–Software co-design – Sensors and Actuators for control Applications – Signal Conditioning – Data Acquisition Systems .	
UNIT–II	**MICROCONTROLLERS AND DSPs FOR CONTROL APPLICATIONS**	**9**
	Architecture of Microcontrollers – Memory and I/O Organization – Timers, Counters and Interrupts – PWM Generation Techniques – ADC and DAC Interfacing – Digital Signal Processors: Architecture and Features – Fixed and Floating point DSPs – Comparison of Microcontrollers and DSPs – Selection Criteria for Control Applications .	
UNIT–III	**REAL-TIME OPERATING SYSTEMS AND EMBEDDED SOFTWARE DESIGN**	**9**
	Embedded C Programming Concepts – Interrupt-driven Programming – Real-Time Operating Systems (RTOS) – Task Scheduling and Context Switching – Inter-task Communication – Semaphores and Mutexes – Timing Analysis – Embedded Software Debugging and Testing – Reliability and Safety Considerations .	
UNIT – IV	**EMBEDDED CONTROL OF POWER ELECTRONIC CONVERTERS AND DRIVES**	**9**
	Digital Control of DC–DC Converters – Sampling and Discretization – Digital PWM Techniques – Current and Voltage Control Loops – Embedded Control of DC Motor Drives – Embedded Control of Induction Motor Drives – Vector Control Implementation – Speed and Torque Control – Fault Detection and Protection Schemes .	
UNIT – V	**ADVANCED EMBEDDED CONTROL APPLICATIONS**	**9**
	Embedded Control of PMSM and BLDC Drives – Sensor less Control Techniques – Embedded Implementation of Control Algorithms – Communication Protocols (CAN, SPI, I2C, UART) – Industrial Embedded Controllers – Case Studies in Power Electronics and drive Systems – Recent Trends in Embedded Control Systems .	
	Total(L)	**45 Periods**

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the fundamentals of embedded control systems and real-time constraints.	L2 - Understand
CO2	Describe the architecture and features of microcontrollers and DSPs for control.	L2 - Understand
CO3	Apply basic RTOS concepts in embedded software development.	L3 - Apply
CO4	Apply embedded control methods for power converters and motor drives.	L3 - Apply
CO5	Explain advanced embedded control applications and industrial communication protocols.	L2 - Understand
	REFERENCE BOOKS:	
1.	Raj Kamal, "Embedded Systems: Architecture, Programming and Design", 3rd Edition, McGraw Hill Education, 2013.	
2.	Wayne Wolf, "Computers as Components: Principles of Embedded Computing System Design", 3rd Edition, Morgan Kaufmann, 2012.	
3.	B. K. Bose, "Power Electronics and Motor Drives: Advances and Trends", 1st Edition, Academic Press, 2010.	
4.	R. Krishnan, "Electric Motor Drives: Modeling, Analysis and Control", 1st Edition, Pearson Education, 2010.	
5.	Steve Furber, "ARM System-on-Chip Architecture", 2nd Edition, Addison-Wesley, 2000.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3		3	2	2	2
CO2	3		3	2	2	2
CO3	3		3	2	2	2
CO4	3		3	2	2	2
CO5	3		3	2	2	2
Avg.	3		3	2	2	2

1-Low,2-Medium,3-High

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ME23PE428	FPGA BASED CONTROLLERS FOR POWER CONVERTERS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. – POWER ELECTRONICS AND DRIVES	Version: 1.0				

Course Objectives:	
1.	To understand digital control concepts and the role of FPGA in power electronics.
2.	To learn FPGA architecture and hardware description language basics.
3.	To understand digital control and PWM implementation using FPGA.
4.	To learn FPGA-based control of power electronic converters.
5.	To understand advanced FPGA applications and emerging trends in power electronics.

UNIT–I	INTRODUCTION TO FPGA AND DIGITAL CONTROL	9
	Overview of Digital Control in Power Electronics – Limitations of Microcontroller and DSP Based Control – Advantages of FPGA - Based Control – FPGA Architecture: Logic Blocks, Interconnects and I/O Blocks – CPLD vs FPGA – FPGA Development Flow – Introduction to VHDL/Verilog – Applications of FPGA in Power Converters .	
UNIT–II	FPGA ARCHITECTURE AND HARDWARE DESCRIPTION LANGUAGES	9
	FPGA Internal Architecture – Configurable Logic Blocks – Look-up tables – Flip-flops – Clock Management – I/O Standards – Introduction to VHDL/Verilog syntax – Concurrent and Sequential Statements – Modeling Styles – Simulation and Synthesis – Timing Constraints .	
UNIT–III	DIGITAL CONTROL AND PWM IMPLEMENTATION USING FPGA	9
	Sampling and Discretization – Digital Control Algorithms – FPGA-based Implementation of PI and PID Controllers – Digital PWM Techniques – Carrier-Based PWM – Space Vector PWM – Resolution and Timing Issues – Dead-time Generation – Multi-Channel PWM Generation .	
UNIT – IV	FPGA CONTROL OF POWER ELECTRONIC CONVERTERS	9
	FPGA-based Control of DC–DC Converters – Buck, Boost and Buck–Boost Converters – FPGA Control of AC–DC Converters – Inverter Control using FPGA – Current and Voltage Feedback Implementation – Protection Schemes – Fault Detection and Isolation .	
UNIT – V	ADVANCED FPGA APPLICATIONS IN POWER ELECTRONICS	9
	FPGA-based Control of Motor Drive Converters – High-Speed Control and Parallel Processing – FPGA and SoC Platforms – Communication Interfaces (SPI, I2C, UART, CAN) – Hardware–Software Co-design – Case Studies in FPGA-based Power Converter Control – Emerging Trends .	
	Total(L)	45 Periods

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain digital control requirements and advantages of FPGA-based control.	L2 - Understand
CO2	Describe FPGA architecture and basic VHDL/Verilog modeling concepts.	L2 - Understand
CO3	Apply FPGA-based techniques to implement digital control and PWM schemes.	L3 - Apply
CO4	Apply FPGA control methods for DC–DC and AC–DC power converters.	L3 - Apply
CO5	Explain advanced FPGA applications and communication interfaces in power electronics.	L2 - Understand
	REFERENCE BOOKS:	
1.	Wayne Wolf, "FPGA-Based System Design", 1st Edition, Prentice Hall, 2004.	
2.	Clive Maxfield, "The Design Warrior's Guide to FPGAs", 1st Edition, Elsevier, 2004.	
3.	Pong P. Chu, "FPGA Prototyping by Verilog Examples", 1st Edition, John Wiley & Sons, 2008.	
4.	B. K. Bose, "Power Electronics and Motor Drives: Advances and Trends", 1st Edition, Academic Press, 2010.	
5.	Robert W. Erickson, Dragan Maksimović, "Fundamentals of Power Electronics", 2nd Edition, Springer, 2001.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3		3	2	2	2
CO2	3		3	2	2	2
CO3	3		3	2	2	2
CO4	3		3	2	2	2
CO5	3		3	2	2	2
Avg.	3		3	2	2	2

1–Low,2–Medium,3–High

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ME23IS501	ENVIRONMENTAL SAFETY	CP 3	L 3	T 0	P 0	C 3
Type of Course	**Open Elective**					
Offering Dept.	**Mechanical Engineering**					
Programme & Branch	**Common to M.E. (AE, PED and SE)**	**Version: 1.0**				
Course Objectives:						
1.	To provide in depth knowledge in Principles of Environmental safety and its applications in various fields.					
2.	To give understanding of air and water pollution and their control.					
3.	To expose the students to the basis in hazardous waste management.					
4.	To provide knowledge on pollution monitoring and control devices.					
5.	To design emission measurement devices.					
UNIT–I	**AIR POLLUTION**	**9**				
	Classification and properties of air pollutants -Pollution sources -Effects of air pollutants on human beings, Animals, Plants, and Materials -Automobile pollution -Hazards of air pollution -Concept of clean coal combustion technology -Ultra violet radiation, infrared radiation, radiation from the sun-Hazards due to depletion of ozone-Deforestation, ozone holes, automobile exhausts, chemical factory stack emissions, CFC.					
UNIT–II	**WATER POLLUTION**	**9**				
	Classification of water pollutants-Health hazards-Sampling and analysis of water-Water treatment -Different industrial effluents and their treatment and disposal-Advanced wastewater treatment-Effluent quality standards and laws-Chemical industries, tannery, textile effluents -Common treatment.					
UNIT–III	**HAZARDOUS WASTE MANAGEMENT**	**9**				
	Hazardous waste management in India-Waste identification, characterization, and classification-Technological options for collection, treatment and disposal of hazardous waste, Selection charts for the treatment of different hazardous wastes-Methods of collection and disposal of solid wastes-Health hazards -Toxic and radioactive wastes-Incineration and vitrification-Hazards due to bio-process, dilution, standards, and restrictions-Recycling and reuse.					
UNIT–IV	**ENVIRONMENTAL MEASUREMENT AND CONTROL**	**9**				
	Sampling and analysis-Dust monitor -Gas analyzer- particle size analyzer-Lux meter- pH meter-Gas chromatograph-Atomic absorption spectrometer-Gravitational settling chambers, cyclone separators, scrubbers-Electrostatic precipitator, bag filter, maintenance-Control of gaseous emission by adsorption, absorption, and combustion methods-Pollution Control Board, laws.					

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UNIT–V	**POLLUTION CONTROL IN PROCESS INDUSTRIES**	**9**
	Pollution control in process industries -Cement, paper, petroleum, petroleum products, textile-Tanneries, thermal power plants-Dyeing and pigment industries -Eco-friendly energy.	
	Total(L)	**45 Periods**
	OPEN-ENDEDPROBLEMS/QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Illustrate and familiarize the basic concepts scope of environmental safety.	L2-Understand
CO2	Interpret the standards of professional conduct that are published by professional safety organizations and/or certification bodies.	L2- Understand
CO3	Explain the ways in which environmental health problems have arisen due to air and water pollution.	L2- Understand
CO4	Examine the role of hazardous waste management and use of critical thinking to identify and assess environmental health risks.	L4 - Analyze
CO5	Apply concepts of emission measurement and design emission measurement devices.	L3 - Apply

S.No.	**REFERENCE BOOKS:**
1.	E C Wolfe, "Race to Save to Save Planet", Wadsworth Publishing Co., Belmont, 2006.
2.	G T Miller, "Environmental Science: Working with the Earth", Wadsworth Publishing Co., Belmont, 11th Edition, 2006.
3.	M J Hammer and M J Hammer, "Water and Wastewater Technology", Pearson Prentice Hall, 2006
4.	Rao C S, "Environmental pollution Engineering", Wiley Eastern Limited, New Delhi, 2018.
5.	S P Mahajan, "Pollution control in process industries", Tata McGraw Hill Publishing Company, New Delhi, 2006.
6.	Varma and Braner, "Air pollution Equipment", Springer Publishers, Second Edition.

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ME23IS502	ELECTRICAL SAFETY	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	Mechanical Engineering					
Programme & Branch	Common to M.E. (AE, PED and SE)	Version: 1.0				

Course Objectives:

1.	To impart knowledge on fundamental electrical concepts, equipment principles, and comply with safety regulations, including basic first aid.
2.	To familiarize students with primary electrical hazards, insulation, and lightning protection measures.
3.	To provide an in depth knowledge on functioning of fuses, circuit breakers, and safety measures against electrical faults.
4.	To provide knowledge on equipment selection, safety features, and maintenance for electrical tools.
5.	To familiarize students with hazardous zone classification, safe equipment, and safety measures in different environments.

UNIT–I	**CONCEPTS AND STATUTORY REQUIREMENTS**	**9**
	Introduction – electrostatics, electro magnetism, stored energy, energy radiation and electromagnetic interference– Working principles of electrical equipment-Indian electricity act and rules-statutory requirements from electrical inspectorate-international standards on electrical safety– first aid-Cardio Pulmonary Resuscitation(CPR).	
UNIT–II	**ELECTRICAL HAZARDS**	**9**
	Primary and secondary hazards-shocks, burns, scalds, falls-human safety in the use of electricity. Energy leakage-clearances and insulation-classes of insulation-voltage classifications-excess energy current surges-Safety in handling of war equipment's-over current and short circuit current-heating effects of current-electromagnetic forces-corona effect-static electricity–definition, sources, hazardous conditions, control, electrical causes of fire and explosion-ionization, spark and arc ignition energy-national electrical safety code ANSI. Lightning, hazards, lightning arrestor, installation – earthing, specifications, earth resistance, earth pit maintenance.	
UNIT–III	**PROTECTION SYSTEMS**	**9**
	Fuse, circuit breakers and overload relays– protection against over voltage and under voltage– safe limits of amperage – voltage –safe distance from lines-capacity and protection of conductor-joints-and connections, overload and short circuit protection-no load protection-earth fault protection. FRLS insulation-insulation and continuity test-system grounding-equipment grounding-earth leakage circuit breaker (ELCB) -cable wires-maintenance of ground-ground fault circuit interrupter-use of low voltage-electrical guards-Personal protective equipment–safety in handling hand held electrical appliances tools and medical equipment's.	

UNIT–IV	**SELECTION, INSTALLATION, OPERATION AND MAINTENANCE**	**9**
	Role of environment in selection-safety aspects in application - protection and interlock-self diagnostic features and fail safe concepts-lock out and work permit system-discharge rod and earthing devices safety in the use of portable tools-cabling and cable joints-preventive maintenance.	
UNIT–V	**HAZARDOUS ZONES**	**9**
	Classification of hazardous zones-intrinsically safe and explosion proof electrical apparatus-increase safe equipment-their selection for different zones-temperature classification-grouping of gases-use of barriers and isolators-equipment certifying agencies.	
	Total(L)	**45 Periods**
	OPEN-ENDEDPROBLEMS/QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Demonstrate understanding of electrical concepts and legal compliance for safe operation, within regulatory constraints.	L2 - Understand
CO2	Identify and mitigate electrical hazards, ensuring safety adherence to protocols and guidelines.	L3 - Apply
CO3	Utilize protection systems effectively, ensuring electrical safety within specified standards.	L3 - Apply
CO4	Apply a safe and efficient process for selecting, installing, operating, and maintaining electrical equipment, adhering to industry regulations.	L3 - Apply
CO5	Develop expertise in managing hazardous zones safely, within the constraints of applicable safety standards.	L3 - Apply

S.No.	**REFERENCE BOOKS:**
1.	" Accident prevention manual for industrial operations", N.S.C., Chicago, 1982.
2.	Indian Electricity Act and Rules, Government of India.
3.	Power Engineers – Handbook of TNEB, Chennai, 1989.
4.	Martin Glov, "Electrostatic Hazards in powder handling", Research Studies Pvt. Ltd., England, 1988.
5.	Fordham Cooper W, "Electrical Safety Engineering", Butterworth and Company, London, 1986.

ME23IS503	SAFETY IN ENGINEERING INDUSTRY	CP 3	L 3	T 0	P 0	C 3
Type of Course	Open Elective					
Offering Dept.	Mechanical Engineering					
Programme & Branch	Common to M.E. (AE, PED and SE)	Version: 1.0				

Course Objectives:

1.	To know the safety rules and regulations, standards and codes.
2.	To study various mechanical machines and their safety importance.
3.	To understand the principles of machine guarding and operation of protective devices.
4.	To know the working principle of mechanical engineering processes such as metal forming and joining process and their safety risks.
5.	To impart knowledge on finishing, inspection and testing operations in engineering industry.

UNIT–I	**SAFETY IN METAL WORKING MACHINERY AND WOOD WORKING MACHINES**	**9**
	General safety rules, principles, maintenance, Inspections of turning machines, boring machines, milling machine, planning machine and grinding machines, CNC machines. Wood working machinery, types, safety principles, electrical guards, work area, material handling, inspection, standards and codes- saws, types, hazards.	
UNIT–II	**PRINCIPLES OF MACHINE GUARDING**	**9**
	Guarding during maintenance, Zero Mechanical State (ZMS), Definition, Policy for ZMS– guarding of hazards- point of operation protective devices, machine guarding, types, fixed guard, interlock guard, automatic guard, trip guard, electron eye, positional control guard, fixed guard fencing- guard construction- guard opening. Selection and suitability: lathe-drilling-boring-milling-grinding-shaping-sawing-shearing-presses-forge hammer -flywheels-shafts-couplings-gears-sprockets wheels and chains-pulleys and belts-authorized entry to hazardous installations-benefits of good guarding systems.	
UNIT–III	**SAFETY IN WELDING AND GAS CUTTING**	**9**
	Gas welding and oxygen cutting, resistances welding, arc welding and cutting, common hazards, personal protective equipment, training, safety precautions in brazing, soldering and metalizing– explosive welding, selection, care and maintenance of the associated equipment and instruments – safety in generation, distribution and handling of industrial gases-colour coding– flashback arrestor– leak detection-pipe line safety-storage and handling of gas cylinders.	

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UNIT–IV	**SAFETY IN COLD FARMING AND HOT WORKING OF METALS**	**9**
	Cold working, power presses, point of operation safe guarding, auxiliary mechanisms, feeding and cutting mechanism, hand or foot-operated presses, power press electric controls, power press set up and die removal, inspection and maintenance - metal sheers-press brakes. Hot working safety in forging, hot rolling mill operation, safe guards in hot rolling mills – hot bending of pipes, hazards and control measures. Safety in gas furnace operation, cupola, crucibles, ovens-foundry health hazards, work environment, material handling in foundries, foundry production cleaning and finishing foundry processes.	
UNIT–V	**SAFETY IN FINISHING, INSPECTION AND TESTING**	**9**
	Heat treatment operations, electro plating, paint shops, sand and shot blasting, safety in inspection and testing, dynamic balancing, hydro testing, valves, boiler drums and headers, pressure vessels, air leak test, steam testing, safety in radiography, personal monitoring devices, radiation hazards, engineering and administrative controls, Indian Boilers Regulation. Health and welfare measures in engineering industry-pollution control in engineering industry- industrial waste disposal.	
	Total(L)	**45 Periods**
	OPEN-ENDEDPROBLEMS/QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Apply safety rules for maintaining and inspecting metal and wood working machines, ensuring industry standards.	L3 - Apply
CO2	Apply effective design strategies for machine guarding systems, emphasizing zero mechanical state (ZMS) during maintenance.	L3 - Apply
CO3	Demonstrate proficiency in safe welding and cutting, ensuring proper equipment selection, care, and maintenance.	L2 - Understand
CO4	Make use of safety measures in cold and hot metalworking, ensuring proper equipment setup, inspection, and maintenance.	L3 - Apply
CO5	Apply safety protocols in finishing, inspection, and testing, adhering to regulations and considering health and pollution control in engineering.	L3 - Apply
S.No.	**REFERENCE BOOKS:**	
1.	"Accident Prevention Manual", NSC, Chicago, 1982.	
2.	"Occupational safety Manual", BHEL, Trichy, 1988.	

3.	John V. Grimaldi and Rollin H. Simonds, "Safety Management", All India Travelers Book seller, New Delhi, 1989.
4.	Krishnan N V, "Safety in Industry", Jaico Publishery House, 1996.
5.	Indian Boiler acts and Regulations, Government of India.
6.	"Safety in the use of wood working machines", HMSO, UK, 1992.
7.	"Health and Safety in welding and Allied processes", welding Institute, UK, High Tech. Publishing Ltd., London, 1989.

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ME23IS504	DESIGN OF EXPERIMENTS	CP 3	L 3	T 0	P 0	C 3
Type of Course	**Open Elective**					
Offering Dept.	**Mechanical Engineering**					
Programme & Branch	**Common to M.E. (AE, PED and SE)**	**Version: 1.0**				

Course Objectives:

1.	To impart knowledge on principles and steps in designing a statistically designed experiment.
2.	To build foundation in analysing the data in single factor experiments and to perform post hoc tests.
3.	To provide knowledge on analysing the data in factorial experiments.
4.	To educate on analysing the data analysis in special experimental designs and Response Surface Methods.
5.	To impart knowledge in designing and analysing the data in Taguchi's Design of Experiments to improve Process/Product quality.

UNIT–I	**EXPERIMENTAL DESIGN FUNDAMENTALS**	**9**
	Importance of experiments, experimental strategies, basic principles of design, terminology, ANOVA, steps in experimentation, sample size, normal probability plot, linear regression models.	
UNIT–II	**SINGLE FACTOR EXPERIMENTS**	**9**
	Completely randomized design, Randomized block design, Latin square design, Statistical analysis, estimation of model parameters, model adequacy checking, pair wise comparison tests.	
UNIT–III	**MULTIFACTOR EXPERIMENTS**	**9**
	Two and three factor full factorial experiments, Randomized block factorial design, Experiments with random factors, rules for expected mean squares, approximate F-tests. 2^K factorial Experiments.	
UNIT–IV	**SPECIAL EXPERIMENTAL DESIGNS**	**9**
	Blocking and confounding in 2^K designs. Two level Fractional factorial design, nested designs, Split plot design, Introduction to Response Surface Methods.	
UNIT–V	**TAGUCHI METHODS**	**9**
	Steps in experimentation, design using Orthogonal Arrays, data analysis, Robust design,- control and noise factors, S/N ratios, parameter design, Multi-level experiments, Multi-response optimization, Introduction to Shainin DOE.	
	Total(L)	**45 Periods**

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	OPEN-ENDEDPROBLEMS/QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Interpret the Design of Experiments principles, strategizing experiment design within practical resource considerations and goals.	L2 - Understand
CO2	Analyze single-factor experiment data, focusing on randomization and pair-wise comparison tests.	L4 - Analyze
CO3	Analyze multifactor experiment data, applying rules for expected mean squares and approximate F-tests.	L4 - Analyze
CO4	Apply special experimental designs, minimize confounding effects, optimize data collection, and introduce Response Surface Methods with practical considerations.	L3 - Apply
CO5	Apply Taguchi-based approaches for quality evaluation, emphasizing practical experimentation with orthogonal arrays and multi-response optimization.	L3 - Apply

S.No.	REFERENCE BOOKS:
1.	Krishnaiah K and Shahabudeen P, "Applied Design of Experiments and Taguchi Methods", PHI learning private Ltd., 2012.
2.	Montgomery D C, "Design and Analysis of Experiments", John Wiley and Sons, Eighth edition, 2012.
3.	NicoloBelavendram, "Quality by Design: Taguchi techniques for industrial Experimentation", Prentice Hall, 1995.
4.	Phillip J Rose, "Taguchi techniques for quality Engineering", McGraw Hill, 1996.

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ME23IS505	CIRCULAR ECONOMY	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	Mechanical Engineering					
Programme & Branch	Common to M.E. (AE, PED and SE)	Version: 1.0				

Course Objectives:	
1.	To equip graduates with circularity expertise for diverse national and international job opportunities.
2.	To develop skilled manpower and foster entrepreneurship in Circular Economy.
3.	To facilitate student-professional interactions for real-world exposure in technology, research, innovation, and circular business models.
4.	To inspire students to address circularity business needs and pursue Research and Development (R&D) and entrepreneurship.
5.	To cultivate environmentally conscious entrepreneurs through core competencies in environmental education and collaborative university-industry partnerships.

UNIT–I	INTRODUCTION TO CIRCULAR ECONOMY	9
	Linear Economy and its emergence, Economic and Ecological disadvantages of linear economy, Replacing Linear economy by Circular Economy, Development of Concept of Circular Economy, A differential - Linear Vs Circular Economy.	
UNIT–II	CHARACTERISTICS OF CIRCULAR ECONOMY	9
	Material recovery, Waste Reduction, reducing negative externalities, Explaining Butterfly diagram, Concept of Loops.	
UNIT–III	CIRCULAR DESIGN, INNOVATION AND ASSESSMENT	9
	Zero waste: Waste Management in context of Circular Economy, Circular design, Research and innovation, LCA, Circular Business.	
UNIT–IV	CASE STUDIES	9
	Business models, Solid Waste Management / Wastewater, Plastics: A case study, EPR: polluters pay principle, Industrial symbiosis/ Eco-parks.	
UNIT–V	LEGAL AND POLICY FRAMEWORK	9
	Role of governments and networks, Sharing best practices, Universal circular economy policy goals, India and CE strategy, ESG.	
	Total(L)	45 Periods
	OPEN-ENDEDPROBLEMS/QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.	

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	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Differentiate Circular Economy from Linear Economy and showcase its practical application.	L2 - Understand
CO2	Apply Circular Economy principles, incorporating material recovery and waste reduction to illustrate the Butterfly diagram and emphasize the loops within the circular system.	L3 - Apply
CO3	Apply circular design and innovation principles, assess sustainability in Circular Economy, and examine circular business models	L3 - Apply
CO4	Analyze case studies on circular economy from different fields and connect these cases to Circular Economy concepts professionally.	L4 - Analyze
CO5	Infer government roles, share best practices, and articulate Circular Economy policy goals, demonstrating expertise in legal frameworks with an ESG focus, especially in India.	L2 - Understand

S.No.	REFERENCE BOOKS:
1.	María-Laura Franco-García, Jorge Carlos Carpio-Aguilar, Hans Bressers, "Towards Zero Waste: Circular Economy Boost, Waste to Resources", Springer International Publishing, 2019.
2.	Marcello Tonelli and Nicolo Cristoni, "Strategic Management and the Circular Economy", Routledge, 2018.
3.	Sadhan Kumar Ghosh, "Circular Economy: Global Perspective", Springer, 2020.
4.	Stahel and Walter R, "The Circular Economy: A User's Guide", Routledge, 2019.
5.	Lerwen Liu, Seeram Ramakrishna, "An Introduction to Circular Economy", Springer Singapore, 2021.

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ME23AE501	FUNDAMENTALS OF AUTOMOTIVE ELECTRONICS	CP	L	T	P	C
		3	3	0	0	3

Type of Course	**Open Elective**
Offering Dept.	**M.E. AUTOMOTIVE ELECTRONICS**
Programme & Branch	**Common to M.E (SE, PED, ISE)** — **Version: 1.0**

Course Objectives:

1.	To learn the basics of automotive electronics and system architecture.
2.	To understand different sensors, actuators, and control methods in vehicles.
3.	To study engine and powertrain electronics, including emission and hybrid systems.
4.	To learn about automotive communication networks and diagnostic systems.
5.	To explore safety systems, comfort features, and new technologies in automobiles.

UNIT–I	**INTRODUCTION TO AUTOMOTIVE ELECTRONICS**	**9**
	Evolution of automotive Electronics-Modern automotive electrical system Overview-Vehicle architecture: Body electronics- Powertrain electronics- Chassis electronics-Automotive sensors and actuators: characteristics, Applications -Introduction to ECUs: architecture- microcontrollers-memory-Automotive Wiring Harness- Connectors- Grounding-Shielding- Overview of automotive standards: ISO-SAE-AIS.	
UNIT–II	**SENSORS, ACTUATORS & CONTROL SYSTEMS**	**9**
	Position sensors: TPS-Hall sensors-crank/cam Sensors-Speed sensors-pressure sensors-temperature Sensors-Oxygen/λ sensor-MAF-MAP-knock Sensor-Actuators: fuel injectors-ignition coils, stepper motors, BLDC motors, relays. Engine control system basics: closed-loop control, feedback, PID-Electronic throttle control system-Signal conditioning and data acquisition in vehicles.	
UNIT–III	**POWERTRAIN AND ENGINE MANAGEMENT ELECTRONICS**	**9**
	Electronic Engine Control (EEC) architecture-Fuel injection systems: MPFI, GDI, CRDI-Ignition systems: electronic ignition, spark timing control-Air–Air-fuel ratio control-emission control electronics-Exhaust after-treatment systems (EGR, SCR, DPF sensors & control)-Transmission control electronics: TCU, sensors & actuation-Hybrid and electric vehicle basics: motor controllers, inverters, battery electronics.	
UNIT–IV	**VEHICLE COMMUNICATION NETWORKS**	**9**
	Automotive communication basics: need for networking protocol: frame structure- arbitration- error handling-LIN- Flex Ray- MOST – applications and differences- Automotive Ethernet --Network architecture in modern vehicles.	
UNIT– V	**SAFETY, COMFORT SYSTEMS**	**9**
	Electronic stability control (ESC)- traction control (TCS)-Anti-Lock Braking System (ABS): sensors- electronic control-Airbag	

	systems: crash sensors- deployment logic-Body control electronics: lighting- windows- HVAC- Immobilizers-ADAS overview: radar- camera- ultrasonic systems-Cybersecurity challenges in automotive electronics-Emerging trends: Software defined vehicles-Autonomous driving electronics.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the basic structure and components of automotive electronic systems.	L2 – Understand
CO2	Identify common sensors, actuators, and their functions in vehicles.	L2 – Understand
CO3	Describe electronic control of engine, fuel, and emission systems.	L2 – Understand
CO4	Understand communication networks and diagnostic procedures in vehicles.	L2 – Understand
CO5	Outline important safety features and modern trends in automotive electronics.	L2 – Understand
	REFERENCE BOOKS:	
1.	William B. Ribbens, "Understanding Automotive Electronics", Newnes, 8th Edition, 2017.	
2.	Tom Denton, "Automotive Electronics", Routledge, 3rd Edition, 2021.	
3.	Robert Bosch GmbH, "Bosch Automotive Handbook", Bentley Publishers, 10th Edition, 2018.	
4.	Robert Bosch GmbH, "Bosch Automotive Electrics and Automotive Electronics: Systems and Components, Networking and Hybrid Drive", Springer, 6th Edition, 2014.	
5.	James D. Halderman, "Automotive Electricity and Electronics", Pearson, 5th Edition, 2016.	

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ME23AE502	VEHICLE SENSORS AND ELECTRONIC CONTROL SYSTEMS	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	M.E. AUTOMOTIVE ELECTRONICS					
Programme & Branch	Common to M.E (SE, PED, ISE)	Version: 1.0				

Course Objectives:

1.	To understand the role, types, and characteristics of sensors used in modern vehicles.
2.	To learn powertrain sensors and their use in engine and transmission control.
3.	To study chassis, dynamics, and safety-related sensors and their integration.
4.	To explore body electronics sensors used for comfort, environment, and security systems.
5.	To gain knowledge of ADAS sensors, sensor fusion, diagnostics, safety, and emerging technologies.

UNIT–I	INTRODUCTION TO VEHICLE SENSORS & MEASUREMENT TECHNOLOGY	9
	Need for sensors in automotive systems-Classification of sensors: mechanical- electrical- magnetic- thermal- optical-chemical-Sensor characteristics: sensitivity- accuracy- drift-hysteresis- linearity- bandwidth-Signal conditioning: filtering-amplification- analog–digital conversion-Smart sensors and micro-electromechanical systems (MEMS)-Automotive standards: ISO- SAE- AEC-Q100 for sensor qualification.	
UNIT–II	POWERTRAIN SENSORS AND ELECTRONIC CONTROL	9
	Engine sensors: -Crankshaft & camshaft position Sensors-Manifold Absolute Pressure (MAP) Sensor-Mass Air Flow (MAF) Sensor-Throttle Position Sensor (TPS)-Knock sensor- oxygen (λ) Sensor-Coolant temperature and intake air temperature Sensors-Fuel injection- ignition timing- and combustion control using Sensors-Transmission sensors: speed- gear-position-clutch pressure sensors--Electronic Engine Management System (EEMS) Architecture-Closed-loop and open-loop control strategies.	
UNIT–III	CHASSIS, DYNAMICS & SAFETY SENSORS	9
	Wheel speed sensors for ABS-Yaw rate- lateral acceleration-and gyro Sensors-Steering angle sensors and suspension height Sensors-Brake pressure sensors- tire pressure monitoring systems (TPMS)-Airbag sensors: crash accelerometers- impact detection Systems-Electronic Stability Control (ESC)-Traction Control System (TCS) sensors and integration.	
UNIT–IV	BODY ELECTRONICS, ENVIRONMENT & COMFORT SYSTEM SENSORS	9
	Climate control sensors: humidity- cabin temperature- sunlight sensors-Rain sensors- wiper control- interior air quality sensors-Parking sensors: ultrasonic- radar-based-Interior sensors: occupant detection- seat-belt sensors- weight classification sensors-Door- window- anti-pinch and proximity sensors-Immobilizer and smart key system sensors-Comfort system control units and communication interfaces.	

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UNIT– V	ADVANCED DRIVER ASSISTANCE SYSTEM (ADAS) & ELECTRONIC CONTROL	9
	ADAS sensor technologies: - Radar- LiDAR- cameras- ultrasonic sensors-Sensor fusion and perception systems-Electronic control units for ADAS-Autonomous vehicle sensing architecture- Diagnostics: OBD-II- UDS- fault codes- sensor health monitoring-Functional safety: ISO 26262 considerations for sensor integration-Cybersecurity for sensor networks-Future trends: solid-state LiDAR- high-resolution radar- V2X sensors- self-calibrating sensors.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain different vehicle sensors and their basic characteristics.	L2 - Understand
CO2	Describe powertrain sensors and their role in fuel, ignition, and transmission control.	L2 - Understand
CO3	Interpret sensor functions for chassis control, vehicle dynamics, and safety systems.	L2 - Understand
CO4	Identify body electronics sensors used for comfort, environment, and occupant safety.	L3 - Apply
CO5	Evaluate ADAS sensor technologies, diagnostic methods, safety standards, and future trends.	L3 - Apply
	REFERENCE BOOKS:	
1.	Tan Sinclair, "Sensors and Transducers", Newnes, 3rd Edition, 2001.	
2.	Konrad Reif (Ed.), "Automotive Sensors", Springer, 1st Edition, 2014.	
3.	William B. Ribbens, "Understanding Automotive Electronics", Newnes, 8th Edition, 2017.	
4.	Robert Bosch GmbH, "Bosch Automotive Handbook", Bentley Publishers, 10th Edition, 2018.	
5.	Dale R. Patrick and Stephen W. Fardo, "Understanding Sensors", Prentice Hall, 1999.	

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ME23AE503	AUTOMOTIVE EMBEDDED SYSTEMS	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	M.E. AUTOMOTIVE ELECTRONICS					
Programme & Branch	Common to M.E (SE, PED, ISE)	Version: 1.0				

Course Objectives:

1.	To understand the fundamentals, architecture, and standards of embedded systems used in modern vehicles.
2.	To study automotive sensors, actuators, and interfacing techniques for real-time control.
3.	To learn ECU hardware/software architecture, AUTOSAR framework, and functional safety concepts.
4.	To explore automotive communication networks, diagnostic protocols, and network security.
5.	To analyze embedded systems used in advanced vehicle functions, EVs, ADAS, and emerging trends.

UNIT–I	INTRODUCTION TO EMBEDDED SYSTEMS IN AUTOMOBILES	9
	Overview of embedded systems: definition- characteristics-real-time constraints-Evolution and role of embedded electronics in modern vehicles-Embedded system architecture: hardware- firmware- software - stack-Microcontrollers for automotive applications: ARM- Infineon Aurix- Renesas- NXP-Memory systems: Flash- EEPROM- RAM- external memory-Introduction to real-time operating systems (RTOS)-Overview of automotive standards (ISO- SAE- AUTOSAR- AEC-Q100).	
UNIT–II	AUTOMOTIVE SENSORS, ACTUATORS & INTERFACING	9
	Sensors: position- speed- pressure- temperature- oxygen-radar- Camera-Actuators: motors- solenoids- injectors- relays-Valves-Signal conditioning & sensor interfaces-A/D- D/A converters- PWM- timers and Counters-Hardware-software interfacing- GPIO-Diagnostics and fault detection at sensor/actuator Level-Case study: Electronic Throttle Control Module.	
UNIT–III	ELECTRONIC CONTROL UNITS (ECUS) AND SOFTWARE ARCHITECTURE	9
	ECU architecture: CPU- peripherals- communication modules-power supply-Types of ECUs: Engine Control Unit- TCU- BCU-ABS/ESC- ADAS ECUs-Embedded software development lifecycle: V-model- ASPICE-Bootloaders and ECU flashing-Introduction to AUTOSAR architecture: - Classic Platform: BSW- RTE- SWC-Adaptive Platform: POSIX- middleware-services-Model-based development (MATLAB/Simulink)-Functional safety concepts: ISO 26262- HARA- safety goals-ASIL levels.	
UNIT–IV	AUTOMOTIVE COMMUNICATION NETWORKS & PROTOCOLS	9
	Need for in-vehicle networking-CAN: frame formats-arbitration- error handling- CAN FD-LIN: master-slave concept-scheduling-FlexRay: time-triggered communication-MOST and Automotive Ethernet-UDS diagnostics- OBD-II- DTCs-Gateway	

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	ECUs and network security-Time-synchronized networking and deterministic communication.	
UNIT– V	**EMBEDDED SYSTEMS FOR ADVANCED VEHICLE FUNCTIONS**	**9**
	Powertrain embedded systems: engine control- transmission control-Chassis embedded systems: ABS- ESC- TPMS- steering systems-Body electronics: lighting- HVAC- seat control-keyless entry-ADAS control systems: path planning-perception- sensor fusion basics-Embedded systems in Electric Vehicles (EV): - BMS- motor controllers- DC-DC converters-On-board chargers- inverter control-Embedded system testing: HiL- SiL- MiL-Cybersecurity in automotive embedded systems.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the structure, features, and standards of automotive embedded systems.	L2 - Understand
CO2	Describe the operation of sensors, actuators, and interface modules used in vehicles.	L2 - Understand
CO3	Interpret ECU architectures, AUTOSAR platforms, safety methods, and software development processes.	L2 - Understand
CO4	Understand vehicle communication protocols, diagnostics, and network protection mechanisms.	L2 - Understand
CO5	Evaluate embedded systems applied to powertrain, chassis, EVs, ADAS, and future vehicle technologies.	L3 - Apply
	REFERENCE BOOKS:	
1.	Robert Bosch GmbH, "Automotive Electrics and Automotive Electronics", Springer, 6th Edition, 2014.	
2.	Nicolas Navet & François Simonot-Lion, "Automotive Embedded Systems Handbook", CRC Press, 1st Edition, 2008.	
3.	William B. Ribbens, "Understanding Automotive Electronics", Newnes, 8th Edition, 2017.	
4.	Matthias Klauda, "AUTOSAR – A Practical Approach", Springer, 1st Edition, 2020.	
5.	John Wiley & Sons, "Embedded Systems Handbook", CRC Press, 2nd Edition, 2009.	

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ME23AE504	AUTOMOTIVE COMMUNICATION NETWORKS	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	M.E. AUTOMOTIVE ELECTRONICS					
Programme & Branch	Common to M.E (SE, PED, ISE)	Version: 1.0				

Course Objectives:	
1.	To understand the need, evolution, and architecture of in-vehicle communication networks.
2.	To study CAN and CAN FD protocols, frame formats, error handling, and applications.
3.	To learn LIN, FlexRay, MOST networks and their roles in vehicle communication.
4.	To explore automotive Ethernet, diagnostic protocols, and ECU flashing methods.
5.	To analyze network integration, security mechanisms, testing approaches, and future trends

UNIT–I	**INTRODUCTION TO IN-VEHICLE NETWORKING**	**9**
	Need for networking in modern vehicles-Evolution of automotive electronics and distributed Communication requirements: speed- latency- reliability- Determinism-Network topologies: bus- star- ring- hybrid-OSI model and relevance to automotive Networks-Classification of automotive networks: low-speed- high-speed- multimedia- Safety-Critical-Overview of standards: ISO- SAE- AEC- AUTOSAR networking concepts.	
UNIT–II	**CAN AND CAN FD NETWORKS**	**9**
	Controller Area Network (CAN): features- architecture-CAN frame structure: base/extended frames- ID- DLC- CRC-Bit stuffing- arbitration- error detection & handling-CAN protocols: Classical CAN vs CAN FD-Physical layer requirements and Transceivers-Higher-layer protocols: CANopen- J1939- XCP-Use cases: powertrain- chassis- body systems-CAN bus diagnostics and troubleshooting.	
UNIT–III	**LIN, FLEXRAY AND MOST NETWORKS**	**9**
	LIN (Local Interconnect Network)-Master–slave architecture-Frames- schedule tables- clusters-Applications: body electronics- door modules- HVAC- FlexRay- Motivation for FlexRay: high-speed- deterministic communication-Static and dynamic segments- TDMA concept-Fault tolerance and redundancy-Applications: safety-critical (braking- steering-chassis)-MOST (Media Oriented Systems Transport)-Multimedia & infotainment communication-Network layers-synchronous/asynchronous channels-Applications: audio-video- telematics.	
UNIT–IV	**AUTOMOTIVE ETHERNET & DIAGNOSTICS COMMUNICATION**	**9**
	Automotive Ethernet-Introduction and need for high-bandwidth communication-100BASE-T1/1000BASE-T1 automotive standards -TSN (Time-Sensitive Networking) for deterministic performance-Applications: ADAS, autonomous	

	driving, camera & sensor networks - **Diagnostics & Flashing**- On-board diagnostics (OBD-II): modes- PIDs-Unified Diagnostic Services (UDS): request/response- service IDs-KWP2000 basics-ECU flashing and reprogramming methods-DTCs (Diagnostic Trouble Codes) and fault memory.	
UNIT– V	**NETWORK INTEGRATION, SECURITY**	**9**
	Network management: sleep/wake-up- bus-off recovery-Gateway ECUs: routing- filtering- protocol translation-AUTOSAR communication stack: COM- PDUR- CANIF- ETHIF-Cybersecurity for in-vehicle networks: -Threats- attack surfaces-Secure Onboard Communication (SecOC)-Intrusion detection systems - Testing and validation: HiL- SiL- bus analyzers- network simulators.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the purpose, architecture, and types of automotive communication networks.	L2 - Understand
CO2	Describe CAN, CAN FD protocols, frame structure, and their use in vehicle subsystems.	L2 - Understand
CO3	Interpret communication principles of LIN, FlexRay, and MOST networks and their applications.	L3 - Apply
CO4	Understand automotive Ethernet, diagnostic services, and ECU programming techniques.	L3 - Apply
CO5	Apply basic network integration ideas and identify common security issues and new trends.	L3 - Apply
	REFERENCE BOOKS:	
1.	Kirsten Matheus and Thomas Königseder,"Automotive Ethernet", Springer, 2nd Edition, 2021.	
2.	Pawel Gaj and Josep Aguilar,"In-Vehicle Communication Networks", Wiley, 1st Edition, 2016.	
3.	Konrad Reif (Ed.),"Automotive Mechatronics: Automotive Networking, Driving Stability Systems, Electronics", Springer, 2nd Edition, 2015.	
4.	P.R. Kumar & P.R. Sundararajan, "Automotive Communication Protocols", McGraw Hill, 1st Edition, 2019.	
5.	Florian Schäfer, "CAN System Engineering: From Theory to Practical Applications", Springer, 1st Edition, 2016.	

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ME23AE505	INTRODUCTION TO ELECTRIC AND HYBRID VEHICLE ELECTRONICS	CP	L	T	P	C
		3	3	0	0	3
Type of Course	**Open Elective**					
Offering Dept.	**M.E. AUTOMOTIVE ELECTRONICS**					
Programme & Branch	**Common to M.E (SE, PED, ISE)**	**Version: 1.0**				

Course Objectives:

1.	To provide basic knowledge of electric and hybrid vehicle types and architectures.
2.	To explain electric motors, power electronic converters and control techniques used in EVs.
3.	To describe battery technologies, BMS functions and energy management systems.
4.	To illustrate hybrid powertrain configurations and control strategies.
5.	To introduce EV control systems, communication, safety and future technologies.

UNIT–I	**FUNDAMENTALS OF ELECTRIC & HYBRID VEHICLES**	**9**
	Evolution of vehicle electrification-Classification of EVs: HEV, PHEV, BEV, FCEV-Vehicle system architecture: conventional vs hybrid vs electric-Energy flow in hybrid and electric powertrains-Drive cycles: NEDC, WLTP, FTP-Overview of EV subsystems: traction motor, inverter, battery pack, charger-Automotive standards & regulations related to EVs (ISO, AIS, UN ECE).	
UNIT–II	**ELECTRIC MACHINES & POWER ELECTRONICS**	**9**
	Types of electric motors for EVs: BLDC, PMSM, Induction Motor, Switched Reluctance Motor-Torque–speed characteristics & efficiency maps-Motor control techniques: FOC, DTC, PWM-Power electronics converters: - DC–DC (buck, boost, bidirectional)-Inverters (single-phase, three-phase)-On-board chargers (AC/DC)-Power semiconductor devices: MOSFET, IGBT, SiC, GaN -Protection circuits, EMI/EMC considerations.	
UNIT–III	**BATTERIES, BMS & ENERGY MANAGEMENT**	**9**
	Battery basics: electrochemistry, Li-ion chemistries (NMC, LFP, NCA), temperature effects-Battery pack design: cell configuration, thermal management-Battery parameters: SOC, SOH, SOP, SOF-Battery Management System (BMS) architecture: - Centralized, distributed, modular topologies-Cell balancing: passive and active-Charging methods: CC–CV, fast charging, regenerative charging-Charging standards: CCS, CHAdeMO, GB/T, IEC 61851-Supercapacitors and hybrid energy storage systems.	
UNIT–IV	**HYBRID POWERTRAIN ARCHITECTURE & CONTROL**	**9**
	Hybrid configurations: - Series, parallel, series–parallel, power-split hybrids-E-Drive unit (EDU) and e-Axle design-Transmission systems for EVs-Energy management strategies:-Rule-based, optimization-based-Torque split and power flow control-Regenerative braking system: electronics,	

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	control & safety-Thermal management for powertrain electronics	
UNIT– V	**ELECTRONIC CONTROL SYSTEMS, COMMUNICATION & FUTURE TRENDS**	**9**
	Vehicle Control Unit (VCU) architecture-Powertrain ECUs: inverter controller, motor controller, BMS controller-Vehicle communication networks: CAN, CAN FD, LIN, Automotive Ethernet-Diagnostics: OBD for EV/HEV, UDS, DTC handling-Functional safety: ISO 26262 for EV systems-Cybersecurity in EV electronics-Emerging technologies: - Solid-state batteries-Wireless charging systems-V2G/V2H/V2X integration-Software-defined EV architectures-AI-based energy management-Autonomous/ADAS considerations for EVs	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Identify different types of electric and hybrid vehicles and their subsystems.	L2 - Understand
CO2	Explain motor types, converters and basic control methods used in EVs.	L2 - Understand
CO3	Describe batteries, BMS functions, charging methods and storage systems.	L3 - Apply
CO4	Design hybrid architectures and summarize energy control strategies.	L3 - Apply
CO5	Apply basic EV control and diagnostic concepts in real systems.	L3 - Apply
	REFERENCE BOOKS:	
1.	Iqbal Husain, "Electric and Hybrid Vehicles: Design Fundamentals", CRC Press, 3rd Edition, 2021.	
2.	James Larminie & John Lowry, "Electric Vehicle Technology Explained", Wiley, 2nd Edition, 2012.	
3.	Sandeep Dhameja, "Electric Vehicle Battery Systems", Newnes, 1st Edition, 2001	
4.	S. M. Mousazadeh, "Hybrid Electric Vehicles, Springer", 1st Edition, 2023.	
5.	Gonzalez & Ali Emadi, "Advanced Electric Drive Vehicles", CRC Press, 1st Edition, 2017.	

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ME23SE501	BLOCKCHAIN TECHNOLOGIES	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	Computer Science and Engineering					
Programme & Branch	Common to M.E. (AE,PED,ISE)	Version: 1.0				

Course Objectives:

1.	To understand the fundamental concepts and working principles of Blockchain technology.
2.	To explore various components, mechanisms, and architectures used in Blockchain systems.
3.	To examine real-world applications of Blockchain across different domains.
4.	To gain practical knowledge of private and public Blockchain networks.
5.	To learn the basics of smart contracts and understand how they are implemented in Blockchain platforms.

UNIT-I	**INTRODUCTION OF CRYPTOGRAPHY AND BLOCKCHAIN**	**9**
	Introduction to Blockchain - Blockchain Technology Mechanisms & Networks - Blockchain Origins - Objective of Blockchain - Blockchain Challenges - Transactions and Blocks - P2P Systems - Keys as Identity - Digital Signatures - Hashing - and public key cryptosystems - private vs. public Blockchain.	
UNIT-II	**INTRODUCTION TO ETHEREUM**	**9**
	Introduction to Bitcoin - The Bitcoin Network - The Bitcoin Mining Process - Mining Developments - Bitcoin Wallets - Decentralization and Hard Forks - Ethereum Virtual Machine (EVM) - Merkle Tree - Double-Spend Problem - Blockchain and Digital Currency - Transactional Blocks - Impact of Blockchain Technology on Cryptocurrency.	
UNIT-III	**BUSINESS FORECASTING**	**9**
	Introduction to Ethereum - Consensus Mechanisms - Metamask Setup - Ethereum Accounts - Transactions - Receiving Ethers - Smart Contracts.	
UNIT-IV	**INTRODUCTION TO HYPERLEDGER AND SOLIDITY PROGRAMMING**	**9**
	Introduction to Hyperledger - Distributed Ledger Technology & its Challenges - Hyperledger & Distributed Ledger Technology - Hyperledger Fabric - Hyperledger Composer. Solidity - Language of Smart Contracts - Installing Solidity & Ethereum Wallet - Basics of	

	Solidity - Layout of a Solidity Source File & Structure of Smart Contracts - General Value Types.	
UNIT–V	**BLOCKCHAIN APPLICATIONS**	**9**
	Internet of Things - Medical Record Management System - Domain Name Service and Future of Blockchain - Alt Coins.	
	Total (L)	**45 Periods**
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Understand and explore the working of blockchain technology.	L2 – Understand
CO2	Implement the bitcoin mining process.	L3 – Apply
CO3	Develop and deploy simple smart contracts.	L3 – Apply
CO4	Apply the concept of solidity to build de-centralized apps on ethereum.	L3 – Apply
CO5	Develop applications on blockchain.	L3 – Apply
	REFERENCE BOOKS:	
1.	Imran Bashir, "Mastering Blockchain: Distributed Ledger Technology, Decentralization, and Smart Contracts Explained", Second Edition, Packt Publishing, 2018.	
2.	Narayanan, J. Bonneau, E. Felten, A. Miller, S. Goldfeder, "Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction", Princeton University Press, 2016	
3.	Antonopoulos, "Mastering Bitcoin", O'Reilly Publishing, 2014.	
4.	Antonopoulos and G. Wood, "Mastering Ethereum: Building Smart Contracts and Dapps", O'Reilly Publishing, 2018.	
5.	D. Drescher, "Blockchain Basics", Apress, 2017.	

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ME23SE502	DEEP LEARNING	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	Computer Science and Engineering					
Programme & Branch	Common to M.E. (AE,PED,ISE)	Version: 1.0				

Course Objectives:

1.	To understand Deep Neural Networks for solving complex learning tasks.
2.	To implement CNN-based models such as R-CNN, Fast R-CNN, Faster R-CNN, and Mask R-CNN for object detection and recognition.
3.	To build and train Recurrent Neural Networks and apply NLP techniques using word embedding.
4.	To understand the internal structure, working mechanisms, and differences between LSTM and GRU units.
5.	To apply Autoencoders for image processing, feature extraction, and representation learning.

UNIT–I	DEEP LEARNING CONCEPTS	9
	Fundamentals about Deep Learning - Perception Learning Algorithms - Probabilistic modelling - Early Neural Networks - How Deep Learning different from Machine Learning – Scalars - Vectors – Matrixes - Higher Dimensional Tensors - Manipulating Tensors- Vector Data- Time Series Data - Image Data - Video Data.	
UNIT–II	NEURAL NETWORKS	9
	About Neural Network - Building Blocks of Neural Network. Optimizers - Activation Functions - Loss Functions - Data Pre - processing for neural networks - Feature Engineering - Overfitting and Underfitting - Hyperparameters.	
UNIT–III	CONVOLUTIONAL NEURAL NETWORK	9
	CNN Basics: Linear Time Invariant, Image Processing & Filtering. Network Architecture: Input, Convolution, Pooling, Dense, Dropout, Batch Normalization Layers; Activation Functions; Optimizers; Backpropagation. Pretrained Models & Transfer Learning: LeNet, AlexNet, VGG16, ResNet, Inception, Google Inception, Microsoft ResNet. Object Detection: R-CNN, Fast R-CNN, Faster R-CNN, Mask R-CNN, YOLO.	
UNIT–IV	NATURAL LANGUAGE PROCESSING USING RNN	9
	About NLP & its Toolkits - Language Modeling - Vector Space Model (VSM) - Continuous Bag of Words (CBOW) - Skip-Gram Model for Word Embedding - Part of Speech (PoS) Global Co-	

	occurrence Statistics – based Word Vectors - Transfer Learning - Word2Vec - Global Vectors for Word Representation GloVe - Backpropagation Through Time - Bidirectional RNNs (BRNN) - Long Short Term Memory (LSTM) - Bi-directional LSTM - Sequence-to-Sequence Models (Seq2Seq) - Gated recurrent unit GRU.	
UNIT–V	**DEEP REINFORCEMENT & UNSUPERVISED LEARNING**	**9**
	About Deep Reinforcement Learning. Q-Learning - Deep Q-Network (DQN). Policy Gradient Methods - Actor-Critic Algorithm - About Autoencoding - Convolutional Auto Encoding - Variational Auto Encoding. Generative Adversarial Networks - Autoencoders for Feature Extraction - Auto Encoders for Classification - Denoising Autoencoders - Sparse Autoencoders.	
	Total (L)	**45 Periods**
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Apply feature extraction techniques to derive meaningful representations from image and video datasets.	L3 – Apply
CO2	Apply neural network preprocessing and tuning techniques to build effective learning models.	L3 – Apply
CO3	Implement image recognition and image classification using a pretrained network.	L3 – Apply
CO4	Apply text mining and analytics techniques on Twitter data.	L3 – Apply
CO5	Implement autoencoders for classification and feature extraction.	L3 – Apply
	REFERENCE BOOKS:	
1.	Josh Patterson and Adam Gibson, "Deep Learning A Practitioner's Approach", O'Reilly Media, Inc.2017.	
2.	Jojo Moolayil ,"Learn Keras for Deep Neural Networks", Apress,2018.	
3.	Vinita Silaparasetty, "Deep Learning Projects Using TensorFlow 2", Apress, 2020.	
4.	Francois Chollet, "Deep Learning with Python", Manning Shelter Island, 2017.	
5.	Santanu Pattanayak, "Pro Deep Learning with TensorFlow", Apress, 2017.	

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ME23SE503	FULL STACK WEB APPLICATION DEVELOPMENT	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	Computer Science and Engineering					
Programme & Branch	Common to M.E. (AE,PED,ISE)	Version: 1.0				

Course Objectives:	
1.	To develop typescript application.
2.	To develop single page application.
3.	To enable to communicate with a server over the HTTP protocol.
4.	To learn all the tools need to start building applications with Node.js.
5.	To implement the full stack development using MEAN Stack.

UNIT–I	FUNDAMENTALS & TYPESCRIPT LANGUAGE	9
	Server-Side Web Applications - Client-Side Web Applications - Single Page Application - About TypeScript - Creating TypeScript Projects - TypeScript Data Types – Variables - Expression and Operators – Functions - OOP in Typescript – Interfaces – Generics – Modules – Enums – Decorators – Enums – Iterators - Generators.	
UNIT–II	ANGULAR	9
	Angular Overview: Angular CLI - Project Creation - Components & Interaction - Dynamic Components - Angular Elements. Forms: Template - driven & Reactive Forms - Property/Style/Class/Event Binding - Two-way Binding – FormGroup – FormControl - Routing: Router Configuration – Navigation – RouterLink - Query Parameters - URL Matching Strategies. Services & HTTP: Dependency Injection – HttpClient - CRUD Operations - Http Headers - Request/Response.	
UNIT–III	NODE.Js	9
	Node.js Overview: Environment Setup – NPM – Modules - Asynchronous Programming - Callbacks & Error Handling - Call Stack & Event Loop. File System & I/O: Synchronous vs Asynchronous I/O - File Handling - Path/Directory Operations - Timers & Timer Promises API. Events & Buffers: EventEmitter - Event API - Buffers & TypedArrays - Binary Data Handling - Flowing vs Non-Flowing Streams. Data Handling: JSON Processing.	

UNIT–IV	**EXPRESS.Js**	**9**
	Introduction to Express.js - Configuring Express.js App Settings - Defining Routes - Starting the App - Express.js Application Structure – Configuration - Settings – Middleware - body-parser - cookie-parser - express-session - response-time - Template Engine – Jade – EJS – Parameters – Routing - router.route(path) - Router Class - Request Object - Response Object - Error Handling - RESTful.	
UNIT–V	**MONGODB**	**9**
	Introduction to MongoDB – Documents – Collections-Subcollections – Database - Data Types – Dates- Arrays - Embedded Documents - CRUD Operations - Batch Insert - Insert Validation - Querying The Documents – Cursors – Indexing - Unique Indexes - Sparse Indexes - Special Index and Collection Types - Full-Text Indexes- Geospatial Indexing - Aggregation framework.	
	Total (L)	**45 Periods**
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Develop basic programming skills using javascript.	L3 – Apply
CO2	Implement a front-end web application using Angular.	L3 – Apply
CO3	Apply Node.js environment setup and NPM to create and manage modular javascript applications.	L3 – Apply
CO4	Apply Express.js RESTful principles to develop server-side web applications.	L3 – Apply
CO5	Implement the aggregation framework to process and analyze complex datasets in MongoDB.	L3 – Apply

	REFERENCE BOOKS:
1.	Adam Freeman, "Essential TypeScript", Apress, 2019.
2.	Mark Clow, "Angular Projects", Apress, 2018.
3.	Alex R. Young, Marc Harter,"Node.js in Practice", Manning Publication, 2014.
4.	Azat Mardan, "Pro Express.js", Apress, 2015.
5.	Kyle Banker, Peter Bakkum, Shaun Verch, Douglas Garrett, Tim Hawkins, "MongoDB in Action", Manning Publication, Second edition, 2016.

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ME23SE504	PRINCIPLES OF MULTIMEDIA	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	Computer Science and Engineering					
Programme & Branch	Common to M.E. (AE,PED,ISE)	Version: 1.0				

Course Objectives:	
1.	To understand the gamut of multimedia and its significance.
2.	To understand the components and characteristics of multimedia systems.
3.	To gain familiarity with multimedia tools and authoring environments.
4.	To develop skills in designing and implementing multimedia applications.
5.	To explore the latest trends and technologies in multimedia.

UNIT–I	INTRODUCTION	9
	Introduction to Multimedia – Characteristics of Multimedia Presentation – Multimedia Components – Promotion of Multimedia Based Components – Digital Representation – Media and Data Streams – Multimedia Architecture – Multimedia Documents, Multimedia Tasks and Concerns, Production, sharing and distribution, Hypermedia, WWW and Internet, Authoring, Multimedia over wireless and mobile networks.	
UNIT–II	ELEMENTS OF MULTIMEDIA	9
	Text-Types, Font, Unicode Standard, File Formats, Graphics and Image data representations – data types, file formats, color models; video – color models in video, analog video, digital video, file formats, video display interfaces, 3D video and TV: Audio – Digitization, SNR, SQNR, quantization, audio quality, file formats, MIDI; Animation- Key Frames and Tweening - other Techniques, 2D and 3D Animation.	
UNIT–III	MULTIMEDIA TOOLS	9
	Authoring Tools – Features and Types – Card and Page Based Tools – Icon and Object Based Tools – Time Based Tools – Cross Platform Authoring Tools – Editing Tools – Painting and Drawing Tools – 3D Modeling and Animation Tools – Image Editing Tools – Sound Editing Tools – Digital Movie Tools.	
UNIT–IV	MULTIMEDIA SYSTEMS	9
	Compression Types and Techniques: CODEC, Text Compression: GIF Coding Standards, JPEG standard – JPEG 2000, basic audio compression – ADPCM, MPEG Psychoacoustics, basic Video	

	compression techniques - MPEG, H.26X - Multimedia Database System - User Interfaces - OS Multimedia Support - Hardware Support - Real Time Protocols - Play Back Architectures - Synchronization - Document Architecture - Hypermedia Concepts: Hypermedia Design - Digital Copyrights, Content analysis.	
UNIT–V	**MULTIMEDIA APPLICATIONS FOR THE WEB AND MOBILE PLATFORMS**	**9**
	ADDIE Model - Conceptualization - Content Collection - Storyboard–Script Authoring Metaphors - Testing - Report Writing - Documentation - Multimedia for the web and mobile platforms. Virtual Reality - Internet multimedia content distribution - Multimedia Information sharing - social media sharing - cloud computing for multimedia services, interactive cloud gaming - Multimedia information retrieval.	
	Total (L)	**45 Periods**
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Illustrate multimedia architectures and their role in hypermedia and mobile networks.	L2 - Understand
CO2	Explain the key concepts and techniques used in multimedia applications.	L2 - Understand
CO3	Develop effective strategies to deliver quality of experience in multimedia applications.	L3 - Apply
CO4	Design and implement algorithms and techniques applied to multimedia objects.	L3 - Apply
CO5	Apply the ADDIE model and multimedia technologies for web, mobile, and cloud-based platforms.	L3 - Apply
	REFERENCE BOOKS:	
1.	Li, Ze-Nian, Drew, Mark, Liu, Jiangchuan, "Fundamentals of Multimedia", Third Edition, Springer, 2021.	
2.	Prabhat K.Andleigh, Kiran Thakrar, "Multimedia Systems Design", Pearson Education, 2015.	
3.	Gerald Friedland, Ramesh Jain, "Multimedia Computing", Cambridge University Press, 2018. (digital book)	
4.	Ranjan Parekh, "Principles of Multimedia", Second Edition, McGraw-Hill Education, 2017	

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ME23SE505	CYBER PHYSICAL SYSTEMS	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	Computer Science and Engineering					
Programme & Branch	Common to M.E. (AE,PED,ISE)	Version: 1.0				

Course Objectives:

1.	To understand the design of cyber-physical systems advanced techniques for modeling and analysis.
2.	To explore cyber-physical systems from attacks by understanding and improving their security.
3.	To design and analyze synchronization techniques for reliable coordination in distributed cyber-physical systems.
4.	To apply real-time scheduling methods that handle fixed timing, memory effects, multicore systems, and system variability in cyber-physical systems.
5.	To learn techniques for modeling, integrating, and formalizing semantics of cyber-physical systems using domain-specific languages.

UNIT-I	**SYMBOLIC SYNTHESIS FOR CYBER-PHYSICAL SYSTEMS**	**9**
	Introduction and Motivation, Basic Techniques - Preliminaries, Problem Definition, Solving the Synthesis Problem, Construction of Symbolic Models, Advanced Techniques: Construction of Symbolic Models, Continuous-Time Controllers, Software Tools.	
UNIT-II	**SECURITY OF CYBER-PHYSICAL SYSTEMS**	**9**
	Introduction and Motivation - Basic Techniques - Cyber Security Requirements - Attack Model – Countermeasures - Advanced Techniques: System Theoretic Approaches.	
UNIT-III	**SYNCHRONIZATION IN DISTRIBUTED CYBER-PHYSICAL SYSTEMS**	9
	Challenges in Cyber-Physical Systems - A Complexity-Reducing Technique for Synchronization - Formal Software Engineering - Distributed Consensus Algorithms - Synchronous Lockstep Executions - Time-Triggered Architecture - Related Technology - Advanced Techniques.	
UNIT-IV	**REAL-TIME SCHEDULING FOR CYBER-PHYSICAL SYSTEMS**	**9**
	Introduction and Motivation - Basic Techniques - Scheduling with Fixed Timing Parameters - Memory Effects - Multiprocessor/Multicore Scheduling - Accommodating Variability and Uncertainty.	

UNIT–V	MODEL INTEGRATION IN CYBER-PHYSICAL SYSTEMS	9
	Introduction and Motivation – Causality - Semantic Domains for Time - Interaction Models for Computational Processes - Semantics of CPS DSMLs - Advanced Techniques – ForSpec - The Syntax of CyPhyML - Formalization of Semantics - Formalization of Language Integration.	
	Total (L)	**45 Periods**
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Understand the core principles behind Cyber physical systems.	L2 – Understand
CO2	Identify Security mechanisms of Cyber physical systems.	L2 – Understand
CO3	Understand Synchronization in Distributed Cyber-Physical Systems.	L2 – Understand
CO4	Apply basic scheduling techniques with fixed timing parameters.	L3 -Apply
CO5	Apply the semantics of Cyber physical Domain-Specific Modeling Languages to model system behavior accurately.	L3 -Apply
	REFERENCE BOOKS:	
1.	Raj Rajkumar, Dionisio De Niz, and Mark Klein, "Cyber-Physical Systems", 1st edition, Addison-Wesley Professional, 2017.	
2.	Rajeev Alur, "Principles of Cyber-Physical Systems", MIT Press, 2015.	

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UX23AC701	INDUSTRIAL SAFETY ENGINEERING	CP	L	T	P	C
		2	2	0	0	0
Type of Course	**Audit Course**					
Offering Dept.	**Mechanical Engineering**					
Programme & Branch	**Common to M.E. (AE, ISE, PED and SE)**	**Version: 1.0**				

Course Objectives:

1.	To provide foundational knowledge on the principles, history and global evolution of industrial safety.	
2.	To give understanding of air and water pollution and their control.	
3.	To understand the causes and effects of fire.	
4.	To understand the causes of accidents due to electrical hazards.	
5.	To enable students to have knowledge on improving road safety in relation with the driver.	
UNIT–I	**INTRODUCTION TO INDUSTRIAL SAFETY**	**6**
	Fundamentals of Safety and it's Management- Safety Planning and Risk Management - Accident Investigation and Prevention- Safety Performance Monitoring- Safety Education - Training and Employee Participation- Data Analysis for safety- Industrial Safety the Factories Act - Environmental Legislations and Waste Management Rules- International Standards.	
UNIT–II	**ENVIRONMENTAL SAFETY**	**6**
	Air Pollution- Air Quality Standards - Prevention and Control Measures - Water Pollution - Types of Water Pollutants (Organic, Inorganic, Microbial) - Treatment and Control Methods - Hazardous Waste Management - Treatment and Disposal Methods- Recycling and Recovery Techniques - Environmental Measurement and Control - Pollution Control in Process Industries.	
UNIT–III	**FIRE SAFETY AND CHEMICAL SAFETY**	**6**
	Fire Safety - Physics and Chemistry of Fire- Fire Prevention and Protection- Industrial Fire Protection Systems - Building Fire Safety - Explosion Protecting Systems. Chemical Safety - Hazard Identification (GHS / MSDS) -Acute and Chronic effects - Chemical Storage and Segregation –Handling, Transfer and PPE Requirements–Toxic Exposure and First Aid Measures.	

UNIT – IV	**ELECTRICAL SAFETY**	**6**
	Concepts and Statutory Requirements - Electrical Hazards - Types of Electrical Hazards - Electric Shock and Burns - Protection Systems - Fuses and Circuit Breakers - Earth Fault Protection-Overload and Short-Circuit Protection-Selection and Installation-Operation and Maintenance - Preventive and Breakdown Maintenance - Lockout/Tagout Procedures - Hazardous zones.	
UNIT–V	**TRANSPORT SAFETY**	**6**
	Transportation of Hazardous Goods - Types of Hazardous Goods - Documentation, Regulatory Compliance- Road Transport - Driver Safety - Road Safety - Speed Control and Safe Distance - Accident Prevention Methods - Shop Floor and Repair Shop Safety.	
	Total (L)	**30 Periods**

OPEN-ENDED PROBLEMS / QUESTIONS

Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.

	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the safety education and training.	L2-Understand
CO2	Illustrate and familiarize the basic concepts scope of environmental safety.	L2-Understand
CO3	Apply explosion protection techniques and their significances to suit the Industrial requirement.	L3 – Apply
CO4	Choose various protection systems from different electrical operations.	L3 – Apply
CO5	Apply road safety principles, including accident prevention, Motor Vehicles Act, insurance, and safety surveys to improver transport safety.	L3 – Apply

	REFERENCE BOOKS:
1.	Hughes. P & Ferrett. E, "Introduction to health and safety at work", Routledge.
2.	The Factories Act, 1948, Madras Book Agency.
3.	T. Miller, "Environmental Science: Working with the Earth", 11th Edition, Wadsworth Publishing Co., Belmont, CA, 2006.

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4.	"Fire Prevention and firefighting", Loss prevention Association, India.
5.	Indian Electricity Act and Rules, Government of India.
6.	K.W.Ogden, "Safer Roads – A guide to Road Safety Engineering".

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UX23AC702	SUSTAINABILITY ENGINEERING	CP	L	T	P	C
		2	2	0	0	0
Type of Course	**Audit Course**					
Offering Dept.	**Mechanical Engineering**					
Programme & Branch	**Common to M.E (AE, ISE, PED and SE)**	**Version: 1.0**				

Course Objectives:

1.	To develop an understanding of sustainability principles and their application in engineering design and practice.
2.	To understand the impacts of greenhouse gas emissions on global warming and climate change.
3.	To understand and evaluate strategies for enhancing water conservation and minimizing water wastage.
4.	To understand the relationship between air quality and human health and the environmental impacts of air pollution.
5.	To examine governance challenges associated with effective supplier and partner engagement.

UNIT–I	INTRODUCTION TO SUSTAINABILITY ENGINEERING	6
	Definition and principles of sustainability–History and evolution of sustainability–Growth into the mainstream – Sustainability in engineering practice – Sustainable Development Goals (SDGs)–Challenges and opportunities in sustainability engineering. * **Experiential Learning:** Zero Waste Campus Challenge. **Idea:** Develop a full segregation and composting system. Introduce "Green Points" for departments reducing single-use plastics.	
UNIT–II	**GREENHOUSE GAS EMISSION AND CLIMATE CHANGE**	**6**
	Introduction of Green House Gas (GHG) - Major GHGs- GHG Effects-Realization of climate action: BIRTH OF United Nations Framework Convention on Climate Change (UNFCCC) - Carbon Border Adjustment Mechanism (CBAM) - ISO 14064 series– Scope 1,2&3– GHG Protocol - GHG accounting and reporting principles – Emission calculation – Global Warming Potential (GWP)–Reporting procedure–Mandatory requirement. * **Experiential Learning:** CO_2 Capture through Green Walls and Algae Bio-Reactors. **Idea:** Design vertical gardens and small algae ponds near high-traffic areas (canteen, generator room) to absorb CO_2 and improve air quality.	

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UNIT–III	**ENVIRONMENTAL STEWARDSHIP: WATER, AIR AND LAND**	**6**
	Water Stewardship: Managing freshwater - Groundwater- Oceans through rainwater harvesting - Preventing pollution. **Air Stewardship:** Reducing air pollution - Improving air quality. **Land Stewardship:** Responsible land use - Preventing erosion- Improving soil fertility - Managing land for multiple uses: Biodiversity, Community and Economy. *** Experiential Learning:** Campus Water Neutrality Project. **Idea:** Measure total water consumption vs. recharge. Implement rainwater harvesting, greywater reuse (from hostels/labs), and smart metering.	
UNIT – IV	**CORPORATE SOCIAL RESPONSIBILITY**	**6**
	Concept of Corporate Social Responsibility (CSR) - Principles of CSR: Accountability, Transparency, Ethical behavior - Respect for stakeholder interests - Respect for rule of law - Respect for international norms - Respect for human rights - Guidelines on CSR - Impact of CSR-CSR community partnership-ISO 26000-ESG reporting - key features. *** Experiential Learning:** ESG Student Ambassador Program. **Idea:** Form a student-led ESG club responsible for awareness, campaigns, and behavior-change initiatives like "bike-to-cycle".	
UNIT–V	**GOVERNANCE, RISK MANAGEMENT AND ETHICS**	**6**
	Conceptual framework of corporate governance - Board effectiveness - Corporate policies & disclosures - Shareholders' rights- Risk identification, mitigation and audit - Compliance management - Ethics & Business - Indian and contemporary laws relating to anti-bribery-Global report. ***Experiential Learning:** Green Procurement and Ethical Supply Chain Policy. **Idea:** Develop campus policy for sustainable purchases - Eco-stationery, energy-efficient equipment and recycled materials.	
	Total (L)	**30 Periods**
	*** INTERNAL EVALUATION ONLY**	
	OPEN-ENDED PROBLEMS / QUESTIONS	

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	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the challenges and opportunities involved in sustainability engineering practices.	L2-Understand
CO2	Apply GHG accounting and reporting boundaries and to prepare a basic GHG inventory.	L3 – Apply
CO3	Discuss principles of land stewardship, including responsible land use, erosion control, and improvement of soil fertility.	L3 – Apply
CO4	Analyze the impact of CSR initiatives on business performance, reputation, and sustainable development outcomes.	L3 – Apply
CO5	Outline processes for risk identification, mitigation, internal control, audit, and compliance management within a corporate governance system.	L3 – Apply
	REFERENCE BOOKS:	
1.	Wahidul K. Biswas, Michele John, "Engineering for Sustainable Development: Theory and Practice", Wiley Publishers, 2022.	
2.	R. L. Rag and Lekshmi Dinachandran Remesh, "Introduction to Sustainable Engineering", 2nd Edition, PHI Learning Pvt. Ltd., 2016.	
3.	Bhavik R. Bakshi, "Sustainable Engineering: Principles and Practice", Cambridge University Press, 2019.	
4.	A.S. Bradley, A. O. Adebayo, P.Maria, "Engineering applications in sustainable design and development", 1st Edition, Cengage learning, 2016.	

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UX23AC703	ENERGY AUDIT	CP	L	T	P	C
		2	2	0	0	0
Type of Course	Audit Course					
Offering Dept.	Electrical and Electronics Engineering					
Programme & Branch	Common to M.E (ISE, AE, PED and SE)	Version: 1.0				

Course Objectives:

1.	To understand India's energy scenario, energy audits, and energy policies.	
2.	To learn audit types, methods, cost analysis, and audit tools.	
3.	To learn energy management programs, audit reports, and energy savings.	
4.	To understand thermal energy saving methods in industrial and building systems.	
5.	To learn efficient electrical energy use and conservation techniques.	
UNIT–I	**INTRODUCTION**	**6**
	Review of Energy Scenario in India - General Philosophy and Need of Energy Audit and Management - Basic Elements and Measurements - Mass and Energy Balances – Scope of Energy Auditing Industries - Evaluation of Energy Conserving Opportunities - Energy Performance Contracts, Need for Energy Policy for Industries, National & State level Energy policies.	
UNIT–II	**ENERGY AUDIT METHODOLOGIES**	**6**
	Types of Energy Audit – Energy Management (audit) Approach - Understanding Energy Costs - Benchmarking – Matching Energy use to Requirement - Optimizing the input Energy Requirements - Duties and Responsibilities of Energy Auditors - Energy Audit Instruments.	
UNIT–III	**ENERGY MANAGEMENT**	**6**
	Design of Energy Management Programmes - Development of Energy Management Systems – Importance - Duties of Energy Manager - Energy Audit Reports - Case Study and Potential Energy Savings. * **Experiential Learning**: Preparation and presentation of Energy Audit Reports.	
UNIT – IV	**THERMAL ENERGY MANAGEMENT**	**6**
	Energy Conservation in Boilers - Steam Turbines and Industrial Heating systems - Cogeneration and Waste heat recovery - Thermal Insulation - Heat Exchangers and Heat Pumps – Building Energy Management.	
UNIT – V	**ELECTRICAL ENERGY MANAGEMENT**	**6**
	Methods to Minimize Supply-demand Gap - Renovation and Modernization of Power Plants - Reactive Power Management – Conservation in Motors, Pumps and Fan Systems – Energy Efficient Motors.	
	Total(L)	**30 Periods**

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	*** INTERNAL EVALUATION ONLY**	
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain India's energy scenario, energy audits, and energy policies.	L2 - Understand
CO2	Apply energy audit methods and benchmarking techniques.	L3 - Apply
CO3	Develop basic energy management programs and audit reports.	L3 - Apply
CO4	Identify thermal energy-saving opportunities in industrial systems.	L2 - Understand
CO5	Apply electrical energy conservation techniques in power systems and equipment.	L3 - Apply
	REFERENCE BOOKS:	
1.	De, B. K., "Energy Management, Audit & Conservation", 2nd Edition, Vrinda Publication, 2010	
2.	Murphy, W. R., "Energy Management", 1st Edition, Elsevier India Private Limited, 2007	
3.	Turner, W. C., Doty, S., Turner, W. C., "Energy Management Handbook", 7th Edition, Fairmont Press, 2009	
4.	Patrick, F., "Energy Conservation Guidebook", 1st Edition, Prentice Hall, 1993	

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UX23AC704	DESIGN THINKING	CP	L	T	P	C
		2	2	0	0	0
Type of Course	Audit Course					
Offering Dept.	Computer Science and Engineering					
Programme & Branch	Common to M.E. (ISE, AE, PED, SE)	Version: 1.0				

Course Objectives:	
1.	To learn design thinking concepts and principles.
2.	To use design thinking methods in every stage of the problem.
3.	To learn the different phases of design thinking.
4.	To develop a prototype and perform testing.
5.	To understand the character and quality of an entrepreneur.

UNIT–I	INTRODUCTION	6
	Need for Design - Four Questions - Ten Tools - Principles of Design Thinking - The process of Design Thinking.	
UNIT–II	UNDERSTAND, OBSERVE AND DEFINE THE PROBLEM	6
	Understanding of the problem - Problem analysis - Reformulation of the problem – Observation Phase - Empathetic design - Characterization of the target group - Description of customer needs.	
UNIT–III	IDEATION AND PROTOTYPING	6
	Ideate Phase - Evaluation of ideas - Prototype Phase - Lean Startup Method for Prototype Development - Visualization and presentation techniques.	
UNIT–IV	TESTING AND IMPLEMENTATION	6
	Test Phase - Tips for interviews - Tips for surveys - Kano Model - Desirability Testing - Conducting workshops - Requirements for the space.	
UNIT–V	ENTREPRENEURSHIP	6
	Entrepreneurship – Character - Quality of Entrepreneur – Opportunity - Entrepreneurial design thinking – The New Social Contract – Design Activism – Designing tomorrow.	
	Total (L)	30 Periods

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	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Describe key concepts of design thinking.	L2 – Understand
CO2	Discuss the phases of design thinking process.	L2 – Understand
CO3	Illustrate design thinking in all stages of problem solving.	L3 – Apply
CO4	Apply testing methodologies to validate the prototype.	L3 – Apply
CO5	Determine the role of an entrepreneur.	L3 – Apply
	REFERENCE BOOKS:	
1.	Christian Mueller-Rotenberg, "Handbook of Design Thinking - Tips & Tools for how to design thinking", 2018.	
2.	Jeanne Liedtka and TimOgilvie, "Designing for Growth: A Design Thinking Tool Kit for Managers", Columbia University Press, 2011	
3.	Tim Brown, "Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation", HarperCollins e-books, 2009.	

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UX23AC705	BUSINESS ANALYTICS	CP	L	T	P	C
		2	2	0	0	0
Type of Course	Audit Course					
Offering Dept.	Computer Science and Engineering					
Programme & Branch	Common to M.E. (ISE, AE, PED, SE)	Version: 1.0				

Course Objectives:

1.	To understand the analytics life cycle.
2.	To comprehend the process of acquiring business intelligence.
3.	To understand various types of analytics for business forecasting.
4.	To model the supply chain management for analytics.
5.	To apply analytics for different functions of a business.

UNIT–I	**INTRODUCTION TO BUSINESS ANALYTICS**	6
	Analytics and Data Science – Analytics Life Cycle – Types of Analytics – Business Problem Definition – Data Collection – Data Preparation – Hypothesis Generation – Modeling – Validation and Evaluation – Interpretation – Deployment and Iteration.	
UNIT–II	**BUSINESS INTELLIGENCE**	6
	Data Warehouses and Data Mart - Knowledge Management –Types of Decisions - Decision Making Process - Decision Support Systems – Business Intelligence – OLAP – Analytic functions.	
UNIT–III	**BUSINESS FORECASTING**	6
	Introduction to Business Forecasting and Predictive analytics - Logic and Data Driven Models – Data Mining and Predictive Analysis Modelling – Machine Learning for Predictive analytics.	
UNIT–IV	**HR & SUPPLY CHAIN ANALYTICS**	6
	Human Resources – Planning and Recruitment – Training and Development - Supply chain network - Planning Demand, Inventory and Supply – Logistics – Analytics applications in HR & Supply Chain - Applying HR Analytics to make a prediction of the demand for hourly employees for a year.	
UNIT–V	**MARKETING & SALES ANALYTICS**	6
	Marketing Strategy - Marketing Mix - Customer Behaviour – selling Process – Sales Planning – Analytics applications in Marketing and Sales - predictive analytics for customers' behaviour in marketing and sales.	
	Total (L)	**30 Periods**

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	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Explain the real world business problems and model with analytical solutions.	L2 – Understand
CO2	Identify the business processes for extracting Business Intelligence	L2 – Understand
CO3	Apply predictive analytics for business fore-casting	L3 – Apply
CO4	Apply analytics for supply chain and logistics management	L3 – Apply
CO5	Use analytics for marketing and sales.	L3 – Apply
	REFERENCE BOOKS:	
1.	R. Evans James, "Business Analytics", 2nd Edition, Pearson, 2017.	
2.	R.N. Prasad, Seema Acharya, "Fundamentals of Business Analytics", 2nd Edition, Wiley, 2016.	
3.	Philip Kotler and Kevin Keller, "Marketing Management", 15th edition, PHI, 2016.	
4.	VSP RAO, "Human Resource Management", 3rd Edition, Excel Books, 2010.	
5.	Mahadevan B, "Operations Management -Theory and Practice",3rd Edition, Pearson Education, 2018.	

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